ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Marine Technician (MAR TECH) 00379-03

USQ-S1 Electrical Specialist
(ALUT)

MITE Course IDs
Residential: 122020

RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Commanding Officer of the Naval Training Development Centre (NTDC). The approval or revision date should appear at the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J “Course History.”

Revision	Approval Date	Authority
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FOREWORD

1. This training document is developed under the authority of the Commander Naval Training and Personnel Group (CNPTG) and in accordance with the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-002 Non-Commissioned Member General Specification 5555-00379-MAR-TECH dated 15 Dec 16;

c. A-PD-055-002/PP-001 Marine Technician dated 28 Apr 17; and

d. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA, who will forward to NPTG for approval, as appropriate.

5. The designated TE(s) for this training is indicated below:

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	122020	ALUT
NFS(A)	All	122020	ALUT

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Marine Technician Electrical Specialist. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22- Qualification Standard and Plan, Annex B para 3.0.

Allocation of ATT	**Minutes NFS(A)**	**Minutes NFS(P)**
Academics (Taught thru Institution. Camosun's course schedule is completed in 32 weeks)	59650	32 weeks
Applications	18475	18475
Enabling Checks (EC)	1700	1700
Performance Checks (PC)	3460	3460
Actual Training Time (ATT) Sub-total	23635 (+ 59650)	23635 (+ 32 weeks)
Allocation of BAT, TAT, TTT, PT IAW NPTORD 4500-22, Annex B Table 3.1	Based on NFS(A) App Phase	Based on NFS(P) App Phase
Base Administrative Time (BAT)	840	840
Training Administrative Time (TAT)	840	840
Training Travel Time (TTT)	300	300
Physical Fitness (PT)	2160	2160
Ceremonial Divisions	240	240
Sub-total	4380	4380
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	28015 (+ 59650)	28015 (+ 32 weeks)
For Scheduling duration total mins/400 = # days	220 Days	71 Days (+32 weeks)
XX days/20 days per month =	11	3.6 (+32 weeks)

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO / EO
Academic Instructor	Civilian	The minimum civilian instructor qualification is a technologist diploma or Degree in Engineering in the required speciality.	4	EO001.01 EO 002.01 EO 002.02 EO 003.01 EO 003.02 EO 003.06 EO 004.01 EO 004.02 EO 007.01 EO 007.02 EO 008.02
IPMS Instructor	PO2	IPMS Course	1	PO 003 PO 008
Vibration Analysis	MS-PO2	VA Level 1 Minimum	1	EO 010.01
Thermal Imaging Instructor	Civilian	FMF Engineering (EHM)	1	EO 010.02
DRMIS Instructor	Civilian	DRMIS SME	1	EO 001.02
Certified Gas Free Instructor	Civilian/PO2 Mar-Tech Qualified Gas-Free Instructor	FMF Safety (50 minutes required)	1	EO 002.03
Course Instructor	PO2/MS	PLQ, QL5A ETech Journeyman Electrician (legacy) / 00379-03	4	PO 001 PO 002 PO 003 PO 004 PO 005 PO 006 PO 007 PO 008 PO 009 PO 010
Senior Instructor	PO1	QL6B ETech Journeyman Electrician (legacy) / 00379-01	1	

Critical Training Resources	No of Days
As detailed in Annex L	

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in NPTORD 4500-40 – Quantity Control Process for the NTS.

a. Maximum – 12;

b. Minimum – 8; and

c. Optimum – 12.

Note: When determining Max, Min and Capacity, the QSPWB is to consult with the Naval Fleet Schools (NFS) regarding the number of trainees per course, which is based on resource limitations (money, time, instructors to teach and supervise)), equipment limitations (boats, tools, workstations), classroom limitations (desks and chairs), safety concerns (RSO, and Range coaches) and instructional techniques (syndicate size). Operational considerations can be captured if required under Optimum Capacity (4-man attack team, bridge team, RAS team). Camosun College is capable of accommodating 12 learners to a maximum learning effect.

MITE

5. Loading. The authority to load members onto this course lies with the NFS. NPTG is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW NPTORD 4500-1 - Individual Training and Education Policy and NPTORD 4500-40 Quantity Control Management for the Naval Training System (NTS); in MITE by the NFS or Training Site (TS) conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

7. Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. USQ-S2 Machinery Maintainer (ALMI);

b. 180 days from completion of the USQ-S2 Machinery Maintainer in a MAR TECH billet; and

c. Chain-of-Command Confirmation that member has sufficient on-job experience to consolidate tasks trained on the USQ-S2 Machinery Maintainer(ALMI).

8. Qualification. Successful completion of this course constitutes the primary means of attaining the MAR TECH Electrical Specialist (ALUT). Additionally, the following MITE Course IDs will be granted:

a. DRMIS Maintenance Supervisor – Module 2 – (MITE Course ID 119097);

b. Vibration Analysis – Vibration Monitor (MITE ID 108503);

c. Halon (MITE ID 109149); and

d. Naval Technical Communication Module B, (MITE ID 118129).

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed DAOD 3031-9 – Course Reporting and Certificates and NPTORD 4500-51- Course Reports and Certification.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. TBD – Formal NTS Validation Instructions will be developed.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with NAVORD 4500-27 - PLAR, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. Nil.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

PO 001 PERFORM TECHNICAL ADMINISTRATION

1. Performance. Complete Technical Administration.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;

      (2) PPE;

      (3) Testing equipment and applicable tools;

      (4) Access to DRMIS;

      (5) References; and

      (6) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Marine Technician Electrical Specialist shall perform technical administration IAW Ref by:

   a. Reporting technical information; and

   b. Producing naval maintenance documentation with DRMIS.

4. References. Specified in applicable EOs.

5. Task Numbers. AMT0160, AFT0470, AFT1000.

6. Training Limitations. NFS may not have all the required equipment/trainer.

7. Remarks. Nil.

PO 002 SERVICE MOTOR CONTROLLERS

1. Performance. Service Motor Controllers.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) LOTO equipment;
      (5) ISSC;
      (6) References; and
      (7) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Marine Technician Electrical Specialist shall service motor controllers IAW Ref by:

   a. Analyzing electrical circuits;

   b. Performing mathematical calculations;

   c. Troubleshooting Motors and Motor Controllers; and

   d. Maintaining Motors and Motor Controllers.

4. References. Specified in applicable EOs.

5. Task Numbers. AUT4060, ALT0650, ALT2110, AKT0660, AHT0570, AUT0930, AUT0890, ALT2160, AUT0920, AUT2480, AUT2140, AUT2390, AUT2310, AUT1270, ALT1030, AUT0220, AUT2120, AOT0120, AUT2130, AKT2480, AIT0630, AUT2630, AUT3250, AUT3140, AUT0210, AUT3530, AUT3870, AUT0910, AUT3130, AUT0270, AUT0580, AUT3950, AUT3160, ALT1740, AOT1750, ART0120, AOT0270, AWT0210, AUT0470, AUT0640, AUT3120, ALT1720, ART1060, ART1030, ANT0260, BT0220, AOT1950, ART0380, ART1310, AWT0060, ART1270, AIT0960, BT0260, AUT4080, AUT1180, ART1340, AIT0980, HT0160, ART0600, AIT0830, AUT2360, AUT0250, AKT2540, AKT1450, ART0730, ART0020, AIT0700, AUT2200, AUT2320, ART0510, AUT2490, ART0550, ART0530, ART0650, AOT0070, AKT1460, AUT3390, AOT1800, AOT0290.

6. Training Limitations. NFS may not have all the required equipment/trainer.

7. Remarks. Interpreting electrical drawings and requirements encompasses task (AMT0160), Interpreting drawings/schematics.

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

1. Performance. Service Electronic Control Systems.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;

      (2) PPE;

      (3) Testing equipment and applicable tools;

      (4) LOTO equipment;

      (5) References; and

      (6) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Marine Technician Electrical Specialist shall service electronic control systems independently IAW Ref by:

   a. Troubleshooting Control Systems;

   b. Repairing Control Systems; and

   c. Repairing Engine Control Systems.

4. References. Specified in applicable EOs.

5. Task Numbers. CT0420, ALT2180, ALT1900, ALT1820, ALT0480, ALT2190, ANT1610, ALT1860, AVT1390, ALT1930, ALT1800, ALT0650, AUT1730, AOT0450, AOT1840, AIT0910, AWT0220, ALT2140, BAT0130, BAT0380, BAT0060, ALT1920, ALT1990, ALT1950, ALT0660, AUT0970, ALT1840, BT0210, ALT1980, AUT1360, AUT1370, BAT0390, AFT0970, ABT0300, ALT2010, ALT2210, ALT1970, ALT2000, ALT2160, ALT1830, AOT1960, ALT0620, AUT0700, AUT4010, BAT0100, AYT0420, AVT1140, AUT2010, ALT0960, ALT0400, ALT0380, ALT0570, ALT0790, AUT4100, ALT1060,BAT 0220, AUT0230, AUT1850, AUT2570, ALT0950, ALT1340, ANT0060, ALT0080, AUT2580,CT0070, AVT1140, AYT0290, BAT0180, AVT1230, AUT0560, ALT1750, AUT3030, AUT0510, HT0170, ALT1770, ALT0910, ALT1630, ALT1640, ALT1760, ALT1520, ALT1700, ALT1780, AUT1920, AUT3590, AOT1760, AOT1580, AUT3010, AUT3580, AUT0510, AUT0570, AUT0440, ALT1540, ALT1680, ALT1710, ALT1690, ALT1650, ALT1670, ALT1500, ANT1300, ALT1490, ALT1790, ALT1470, ANT0250, AYT0060, AUT3220, CT0070, AUT3590, AUT2580, AUT1920, AUT1370.

6. Training Limitations. NFS may not have all the required equipment/trainer.

7. Remarks. Nil.

PO 004 MAINTAIN FIELD DEVICES

1. Performance. Maintain Field Devices in support of preventative and corrective maintenance.

2. Conditions.

a. Given:

(1) Ship and applicable equipment and/or system;

(2) PPE;

(3) Testing equipment and applicable tools;

(4) Lock-out/tag-out equipment;

(5) References; and

(6) Supervision;

b. Denied: Nil;

c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The marine technician electrical specialist shall maintain field devices ISO preventative and corrective maintenance IAW Ref by:

a. Troubleshooting field devices;

b. Repairing Field Devices; and

c. Testing Field Devices.

4. References. Specified in applicable EOs.

5. Task Numbers. ALT1850, AYT0390, AT0100, AYT0410, AHT0570, ALT1870, ANT1670, ANT1370, AKT0650, IT0270, ANT1690, ANT0470, AKT1520, AUT2290, AUT2790, ATT0270, AUT2840, AUT2680, ATT0980, AOT1360, AWT0130, ATT0060, BBT0390, AUT2830, BAT0270, AKT2500, ATT0180, ATT0790, AUT2750, AVT0970, ALT1320, AKT2080, BAT0210, ATT0730, AKT1890, ATT0820, AUT2780, ANT0960, AST0080, AUT2820, ATT0840, AUT2280, AUT1830, AUT2600, AOT0620, AUT2610, AVT0970, AUT2530, ATT0720, ATT0250, ATT0170, ATT0830, ANT0960, BAT0270, BAT0290, AYT0260, AYT0210, ANT0860, ANT0930, ANT0090, AOT0240, ALT1510, AUT3640, AUT3910, AUT3620,AUT3550, AUT3520, AUT3860, AUT3820, AUT3790, AOT1640, AUT3780, AUT3720, AUT0600, AOT1700, AOT0280, AUT3630, AT0240, AUT3680, AUT3810, ANT1510, ANT0160, ANT1340, ANT1290.

6. Training Limitations. NFS may not have all the required equipment/trainer.

7. Remarks. Nil.

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

1. Performance: Maintain Primary Power Generation Systems.

2. Conditions:

   a. Given:

      (1) Power Generation System(Virtual/Real);

      (2) PPE;

      (3) Testing equipment and applicable tools;

      (4) Lock-out/tag-out equipment;

      (5) References; and

      (6) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard: The Marine Technician Electrical Specialist shall maintain primary power generation systems IAW Ref by:

   a. Troubleshooting Primary Power Generation Systems; and

   b. Repairing Primary Power Generation Systems.

4. References: Specified in applicable EOs.

5. Task Numbers: CT0440, AUT4050, AUT4030, AUT0800, AUT2740, AUT2000, AUT0380, AUT1980, AUT0100, AUT2990, AUT3380, AUT3670, AUT1910, AUT1900 AUT3980, AUT1610, AUT1640, AUT0770, AUT1410, AUT0410, AUT2850, AJT0360, AUT0400, AUT2590, AUT2880, AUT1990, AUT2340, AUT2690, AUT3890, AUT3660, AUT3600, AUT0590, AUT3880, AUT0630, AUT0810.

6. Training Limitations: NFS may not have all the required equipment/trainer.

7. Remarks. Nil.

PO 006 MAINTAIN ELECTRICAL DISTRIBUTION SYSTEMS

1. Performance. Maintain electrical distribution systems.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;

      (2) PPE;

      (3) Testing equipment and applicable tools;

      (4) LOTO equipment;

      (5) References; and

      (6) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The marine technician electrical specialist shall maintain electrical distribution systems IAW Ref by:

   a. Troubleshooting Electrical Distribution Systems; and

   b. Repairing Electrical Distribution Systems.

4. References. Specified in applicable EOs.

5. Task Numbers. ANT1390, ANT1420, ANT0320, ANT0320, ANT1400, CT0440, ALT1910, AUT0850, AUT1080, AUT0760, AUT1490, AUT1510, AUT1020, AUT1060, AUT1500, AUT0810, AUT0750, AUT2040, AUT2730, AUT2710, AUT2010, AUT2720, AUT2270, AUT0080, AUT0260, AUT0170, AUT0130, AUT3060, AUT3300, AUT3050, AUT0550, AUT3030, AUT3770, AUT3750, AUT3260, AUT0520, AUT0430, AUT3760, AUT0580, AUT1980, AUT2990.

6. Training Limitations. NFS may not have all required equipment/ trainer.

7. Remarks. Nil.

PO 007 SERVICE FLUID POWER SYSTEMS

1. Performance. Service fluid power systems.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The marine technician electrical specialist shall service fluid power systems IAW Ref by:

   a. Troubleshooting Electro-Hydraulic systems; and

   b. Troubleshooting Electro-Pneumatic systems.

4. References. Specified in applicable EOs.

5. Task Numbers. AQT0030, ATT0630, BAT0140, AOT0630, ALT2200, ATT0370, AOT1940, BAT0240, AKT1840, ATT0970, AOT0080, AKT2110, AKT2150, AOT0720, AKT2550, AKT0180, ATT0440, BAT0350, ATT1170, ALT1660, AOT1790.

6. Training Limitations. NFS may not have all the required equipment/trainer.

7. Remarks. Nil.

PO 008 MAINTAIN CONTROL NETWORKS

1. Performance. Maintain Control Networks.

2. Conditions.

   a. Given:

      (1) Ship and applicable equipment and/or system;

      (2) PPE;

      (3) Testing equipment and applicable tools;

      (4) References; and

      (5) Supervision;

   b. Denied: Nil;

   c. Environmental: Any conditions, day or night, at sea or alongside.

3. Standard. The Marine Technician Electrical Specialist shall maintain control networks, IAW Ref by:

   a. Troubleshooting control networks;

   b. Repairing fiber optic networks; and

   c. Repairing control networks.

4. References. Specified in applicable EOs.

5. Task Numbers. ALT0400, ALT0960, ALT1640, ALT1490, ALT0660, ALT1830, ALT1840, ALT1860, ALT1500, ALT1930.

6. Training Limitations. NFS may not have all the required equipment/trainer.

7. Remarks. Nil.

PO 009 MAINTAIN HALON 1301 EXTINGUISHING SYSTEM

1. Performance. Maintain Halon 1301 Extinguishing System

2. Conditions.
   a. Given:
      (1) Operational Halon 1301 extinguishing system;
      (2) Maintenance schedule;
      (3) Appropriate tools, equipment and spare parts;
      (4) References; and
      (5) Assistance;
   b. Denied: Nil;
   c. Environmental: At sea or alongside, in all weather conditions.

3. Standard.The Marine Technician Electrical Specialist shall maintain the Halon 1301 extinguishing system by:
   a. Completing a preventive maintenance routine on Halon bottles;
   b. Preparing Halon bottles for replacement; and
   c. Correcting faults in the Halon 1301 extinguishing system.

4. References. A1, A2, A3, A4, C1, C2, C3.

5. Task Number ET043, ET044, ET045, ET046, ET147, ET191.

6. Training Limitations. The non-availability of the following will impact the ability of personnel to properly train for the qualification:
   a. Training platform;
   b. Damage control console; and
   c. Compressor.

7. Remarks. Nil.

PO 010 PERFORM SCHEDULED MOTOR MONITORING ROUTINE

1. Performance. Perform scheduled motor monitoring routine

2. Conditions.

   a. Given:

      (1) Specialty tools;

      (2) Appropriate tools, equipment and spare parts;

      (3) References; and

      (4) Assistance;

   b. Denied: Nil;

   c. Environment: At sea or alongside in any weather conditions.

3. Standard. The Marine Technician Electrical Specialist shall perform a scheduled motor monitoring routine using specialty tools IAW refs to include:

   a. Testing electric motors by Operating vibration monitor suite (HT0160); and

   b. Testing electrical systems with thermal imaging equipment.

4. References. Specified in applicable EOs.

5. Task Numbers. HT0160

6. Training Limitations. Requires VA lab and Testable equipment.

7. Remarks. Nil.

PO 011 PERFORM HIGH RELIABILITY SOLDERING AND TERMINATION TECHNIQUES

1. Performance. Perform high reliability soldering and termination techniques

2. Conditions.

   a. Given:

      (1) PPE
      (2) Specialty tools;
      (3) Appropriate tools, equipment and spare parts;
      (4) References; and
      (5) Assistance;
      (6) Supervision.

   b. Denied: Nil;

   c. Environment: At sea or alongside in any weather conditions.

3. Standard. The Marine Technician Electrical Specialist shall perform soldering and termination techniques using specialty tools IAW refs to include:

   a. Perform high reliability soldering techniques; and

   b. Perform connector terminations and wire wrapping techniques .

4. References. TBD by contractors.

5. Task Numbers.

6. Training Limitations. Requires soldering stations and various connector types.

7. Remarks. Nil.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of (PRACTICAL or WRITTEN) examinations, and the trainee must obtain a pass in all POs to complete this course successfully.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee, and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning so that it will be better retained for more extended periods. Formative tests are designed to give both the instructor(s) and trainees feedback. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the PO's conditions (where possible) and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation

5. The instructor provides quality control by administering and scoring all Performance Checks. Any PC not administered or invigilated by standards staff will be administered by a non-involved instructor designated by the instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required in this course to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the TRB; and

d. Information to assist in the completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled about the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. CEC/PC failures and supplemental CEC/PC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental CECs and PCs shall not be given prior to 24 hours having elapsed since the trainee was informed of their CEC or PC failure.

10. Record of trainee counselling is to be maintained by the training establishment.

TRAINING REVIEW BOARD

11. TRBs shall be convened and conducted IAW with NPTORD 4500-54 - Training Review Board Process.

COURSE REPORTS

12. Course reports (CF 377) shall be prepared and distributed IAW DAOD, 5031-9 - Course Reporting and Certificates and amplified in NPTORD 4500-51 – Course Reporting and Certificates.

13. Trainees, at the minimum, will be graded pass or fail.

14. Where necessary for Rank Qualifications, the narrative will indicate the trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

15. Training limitations and/or incomplete training must also be included in the course report.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTRE

PO 001 PERFORM TECHNICAL ADMINISTRATION

EO 001.01 Report Technical Information

1. Performance. The Enabling Objectives associated with EO 001.01 – Report Technical Information and associated tasks shall be trained IAW Naval Technical Communication Module B QSP, MITE ID 118129.

2. Time Required.

MODULE B:

Total Minutes (Instruction, Data Collection & Assessment)	1375 minutes
Actual Training Days (ATT) (based on 400 minutes per day)	3.50 days

3. Test Details. The Performance Objective will be assessed IAW Naval Technical Communication Module B QSP.

4. References. C1, C2, C3, and C4.

5. Remarks. Any considerations or special instructions (i.e.: Instructor requirements, Advance scheduling, etc.) are IAW Naval Technical Communication Module B QSP.

PO 001 PERFORM TECHNICAL ADMINISTRATION

EO 001.02 Produce Naval Maintenance Documentation within DRMIS

1. Performance. The Enabling Objectives associated with EO 001.02 – Produce Naval Maintenance Documentation within DRMIS, and associated tasks, shall be trained IAW DRMIS Maintenance Supervisor Module 2 QSP, MITE ID 119097.

2. Time Required.

MODULE 2:

Total Minutes (Instruction, Data Collection & Assessment)	285 minutes
Actual Training Days (ATT) (based on 400 minutes per day)	1 days

3. Test Details. The Performance Objective will be assessed IAW DRMIS Maintenance Supervisor Module 2 QSP.

4. References. C5.

Remarks. Any considerations or special instructions (i.e. Instructor requirements, Advance scheduling, etc.) are IAW DRMIS Maintenance Supervisor Module 2 QSP.

PO 002 SERVICE MOTOR CONTROLLERS

EO 002.01 Analyze Electrical Circuits

1. Performance: Analyze electrical circuits.

2. Conditions:

   a. Given:

      (1) Supervision;
      (2) DND Learn account;
      (3) Internet/intranet capable computer; and
      (4) References;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Laboratory.

3. Standard: The Marine Technician Electrical Specialist shall analyze electric circuits, IAW refs to include, but not limited to:

   a. Identifying electrical circuits to include:

      (1) Series resistive circuits;
      (2) Parallel resistive circuits;
      (3) Series-parallel circuits;
      (4) Capacitance;
      (5) Inductance;
      (6) Inductance and capacitance in DC circuits;
      (7) Alternating current and voltage;
      (8) Inductance and capacitance in AC circuits;
      (9) Series-parallel AC circuits;
      (10) Power in AC circuits; and
      (11) AC circuit analysis;

   b. Utilizing electrical concepts to include, but not limited to:

      (1) Resonance;
      (2) Filters;
      (3) Electromechanical energy conversion;

(4) DC generator characteristics;
(5) DC motor characteristics;
(6) Dynamo efficiency;
(7) Single phase transformers;
(8) Poly-phase transformer connections;
(9) Alternators;
(10) Parallel operation of alternators;
(11) Poly-phase induction motor; and
(12) Single phase motors.

4. Time Required.

a. EO – 12800 minutes; and
b. CEC – 500 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. (TBD by Academic instructors).

7. Test Details. EO 002.01 will be tested in CEC 002.01 administered by academic instructors IAW Annex K.

8. References. C8, C9, C10, C11, C12, and (TBD by Academic instructors).

9. Remarks. Laboratory experiments to support knowledge. The CEC will be a two-part written assessment. Trainee will need a minimum of 60% on each CEC to be successful.

5. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Reviewing Fundamentals of electrical circuits to include:							
a. Series resistive circuits to include: 1) Circuit analysis to include: i. Current; ii. Voltage drops; and							

iii. KVL and Analysis procedure for a series circuit. 2) Voltage divider i. Voltage divider theorem; ii. Voltage divider design; and iii. Potentiometer as a voltage divider. 3) Power in series circuits; 4) Voltage-dropping and current limiting resistors; and 5) Open-circuit and short-circuit in series circuits.						
b. Parallel resistive circuits to include: 1) Voltage and current in a parallel circuit to include: i. Kirchhoff's current law; ii. Parallel equivalent circuit; iii. Analysis procedure for a parallel circuit; iv. Conductance in parallel; v. Current divider; vi. Power in parallel circuits; and vii. Open-circuit and short-circuit in parallel circuits.						
c. Series-parallel circuits to include: 1) Equivalent circuit; 2) Current and voltage droops in a series-parallel circuits; 3) Open-circuit and short-circuit in series-parallel circuits; and 4) Analysis of a series parallel circuit.						Laboratory experiments to support knowledge
d. Capacitance to include: 1) Reading assignment; 2) Electric circuit storage; i. Electric field; ii. Capacitance and capacitor dimensions; and iii. Capacitor types and characteristics. 3) Parallel and series capacitors; 4) Energy stored in a capacitor; and 5) Stray capacitance.						Laboratory experiments to support knowledge

e.	Inductance to include: 1) Electro-magnetic induction; 2) Induced EMF and current; 3) Self-inductance; 4) Mutual-inductance; 5) Types of inductors; 6) Energy stored in an inductive circuit; 7) Inductors in series and parallel; and 8) Stray inductance.						Laboratory experiments to support knowledge
f.	Inductance and capacitance in DC circuits to include: 1) RL circuit operation 2) Instantaneous current and voltage in LR circuits 3) Open circuiting and inductive circuits 4) RL circuit waveforms 5) RC circuit operation 6) Instantaneous current and voltage in a RC circuit 7) Discharging a capacitor 8) RC circuit waveforms						
g.	Alternating current and voltage to include: 1) Resistive load with AC supply; 2) Peak, average, and RMS values; and 3) Oscilloscope control.						
h.	Inductance and capacitance in AC circuits by: 1) Alternating current and voltage in an inductive circuit; 2) Inductive reactance (XL); 3) Alternating current and voltage in a capacitive circuit; 4) Capacitive reactance (XC); 5) Series RL and RC circuits; and 6) Series and parallel RLC circuits.						Laboratory experiments to support knowledge
i.	Series-parallel AC circuits to include: 1) Series connected impedances; 2) AC voltage divider; 3) Impedance in parallel; 4) AC current divider; and 5) Series-parallel impedances.						Laboratory experiments to support knowledge

j. Power in AC circuits to include: 1) Power dissipated in resistance; 2) Power in an inductor; 3) Power in a capacitor; 4) True power and reactive power; 5) Power in RL and RC circuits; and 6) Power factor.						Laboratory experiments to support knowledge
k. AC circuit analysis to include: 1) Loop equations; 2) Nodal analysis; 3) Superposition; 4) Thevenin's and Norton's; 5) Maximum power transfer; and 6) Delta-wye transformer.						
TP 2 Applying the concepts of circuit analysis to include, but not limited to						
a. Resonance to include: 1) Series resonance: i. Q factor; and ii. Bandwidth. 2) Parallel resonance; i. Q factor; and ii. Resistance damping.						
b. Filters to include: 1) Low pass; 2) High pass; and 3) Resonance filters.						Laboratory experiments to support knowledge
c. Electromechanical energy conversion to include: 1) Voltage induced in a conductor; 2) Direction of induced voltage; 3) Lenz's law; 4) Induced voltage in series; and 5) Commutation.						
d. DC generator characteristics to include: 1) Induced voltage, magnetisation curve; 2) Field resistance and the build-up process; 3) Critical resistance and the failure of the generator to build up; 4) Shunt-generator characteristics; 5) Cumulative-compound generator characteristics;						

	6) Differential-compound generator characteristics; and 7) Series generator.						
e.	DC motor characteristics to include: 1) Force on a conductor-magnitude and direction; 2) Generation of torque; 3) Counter EMF; 4) Power conversion; 5) Motor speed characteristics; 6) Effects of armature reaction; 7) Shunt motor characteristics; 8) Series motor characteristics; and 9) Compound motor characteristics.						Laboratory experiments to support knowledge
f.	Dynamo efficiency to include: 1) Dynamo losses-coppers tray power and stray load; 2) Determining the stray power losses; 3) Method of duplicating flux and speed; and 4) Dynamo efficiency.						
g.	Single phase transformers to include: 1) Transformer fundamentals; 2) No-load voltage relations; 3) Effects of frequency relations; 4) No-load phasor diagrams; 5) Current relations; 6) Phasor diagrams with load; 7) Equivalent transformer circuit; 8) Full load phasor diagram; 9) Short circuit test; 10) Full load copper loss; 11) Transformer regulation; and 12) Transformer efficiency.						
h.	Poly-phase transformer connections to include: 1) The Y-Y transformation; 2) The Δ-Δ transformation; 3) The Y-Δ transformation; 4) The Δ- Y transformation; 5) The open Δ connection; and 6) Power supplied in the open Δ transformation.						
i.	Alternators to include: 1) Generated EMF;						

	2) Distributed winding and distribution factor; 3) Pitch factor; 4) Alternator characteristics; 5) Armature reaction i. Lagging power factor ii. Leading power factor 6) Alternator regulation 7) Alternator efficiency 8) Alternator ratings						
j.	Parallel operation of alternators to include: 1) Requirements for parallel operation; 2) Synchronising procedure; 3) Variation of field current of alternators in parallel; 4) Dividing load of parallel alternators; 5) Hunting; 6) Torque at standstill; 7) Slip; 8) Rotor EMF, frequency and current; 9) Running torques and maximum torque; 10) Description of operation; 11) Wound rotor induction motor; 12) Current and power factor; 13) Losses and efficiency (theory); and 14) Equivalent circuit and slip.						Laboratory experiments to support knowledge
k.	Poly-phase induction motor to include: 1) Introduction; 2) Effect of load; 3) Increased load; 4) Power and torque relationships; 5) Variation of field current; 6) Calculation of generated voltage and torque angle from load conditions; 7) Synchronous motor armature reaction; 8) V curves; 9) Motor ratings and speed control; and 10) Power factor correction.						Laboratory experiments to support knowledge
l.	Single phase motors to include:						

1) Direct current motors operated on alternating current; 2) Universal motor characteristics; 3) Single phase induction motor-cross field theory; 4) Split phase induction motor; 5) Capacitor start motor characteristics; 6) Permanent split capacitor motor; and 7) Two value capacitor motor.					

Critical Enabling Check – CEC 002.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitate Instructions

1. Trainer.

Arrangements and Procedures

2. CEC 002.01 will be a written assessment in 2 parts based on content from EO 002.01.
3. Pass/Fail IAW Annex K.
4. Time required: 500 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 002.01 will be a written assessment based on content from EO 002.01.
2. During this written assessment, you will be required to apply detailed knowledge of circuit analysis. This test is required to pass EO 002.01. To pass the test, you must apply detailed knowledge of circuit analysis in accordance with the standard.
3. References will not be provided.
4. Pass mark will be 60%.
5. You will have 500 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by academic instructor.

PO 002 SERVICE MOTOR CONTROLLERS

EO 002.02 Perform Mathematical Calculations.

1. Performance: Perform mathematical calculations.

2. Conditions:

   a. Given:

      (1) References; and
      (2) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall perform mathematical calculations, IAW refs to include:

   a. Basic mathematics;
   b. Quadratic equations;
   c. Trigonometric graphing;
   d. Trigonometric formulae and equations;
   e. Analytic geometry;
   f. Function and slope of the line;
   g. Differential calculus;
   h. Derivatives of transcendental functions; and
   i. Integral calculus.

4. Time Required.

   a. EO – 3650 minutes; and
   b. CEC – 250 minutes.

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstration – performance.

6. Training Aids.

a. TBD by Academic instructors.

7. Test Details. EO 002.02 will be tested in EC 002.02 by a written assessment administered by academic instructors IAW Annex K.

8. References. C13, and TBD by Academic instructors

9. Remarks. Nil.

5. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Basic mathematics:							
a. Review of basic mathematics: (1) Numerical fractions; (2) Algebraic expressions; (3) Equations and equation manipulation; (4) Basic trigonometry; (5) Exponents and radicals; and (6) Factoring.							
TP 2 Quadratic equations:							
a. Quadratic equations: (1) Solution by factoring; (2) Completing the square; (3) The quadratic formula; (4) Quadratic function; and (5) Equations leading to quadratic equations.							
TP 3: Trigonometric graphing							
TP 4: Trigonometric formulae and equations							
a. Solving trigonometric formula, equations and graphing: (1) Basic identities; (2) Trigonometric equations; (3) Solution of oblique triangles law of sines and cosines; and b. Applications of angles and angular movement.							
TP 5 Analytic geometry:							
a. Solving analytic geometry: (1) The conic sections; (2) The distance formula; and							

(3) The circle.					
TP 6 Functions and slope of the line:					
a. Solving functions and slope of the line: (1) Functions; and (2) Slope of the line.					
TP 7 Differential calculus:					
a. Using differential calculus to: (1) limits; (2) Derivative – Delta process; (3) Rules for differentiation; (4) Product and quotient rules; and (5) Higher order derivatives.					
TP 8 Derivatives of transcendental functions:					
a. Learning derivatives of transcendental functions: (1) Trigonometric functions; (2) Logarithmic functions; and (3) Exponential functions.					
TP 9 Integral calculus:					
a. Use integral calculus: (1) The indefinite integral; (2) Basic differential equations; (3) Power rule for integration; (4) Integration applications (electrical); (5) Integration of trigonometric, exponential and logarithmic functions; (6) Definite integral; and (7) Mean and RMS values.					

PO 002 SERVICE MOTOR CONTROLLERS

Critical Enabling Check – CEC 002.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 002.02 will be a written assessment based on content from EO 002.02
3. Pass/Fail IAW Annex K.
4. Time required: 250 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 002.02 will be a written assessment based on content from EO 002.02
2. During this written assessment, you will be required to apply a detailed knowledge of essential mathematic skills. You must pass this test to pass EO 002.02. To pass the test, you must apply a detailed knowledge of essential mathematic skills in accordance with the standard.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 250 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by academic instructor.

PO 002 SERVICE MOTOR CONTROLLERS

EO 002.03 Troubleshoot Motor Controllers

1. Performance. Troubleshoot Motor Controllers.

2. Conditions.

3. Given:

    (1) Controller;
    (2) PPE;
    (3) Testing equipment and tools;
    (4) LOTO equipment;
    (5) Access to DRMIS;
    (6) References; and
    (7) Supervision;

4. Denied: Nil;

5. Environmental:

    a. Classroom; and
    b. Workshop.

6. Standard. The Marine Technician Electrical Specialist shall troubleshoot motor controllers (AUT 4060) independently, IAW refs by:

    a. Determine safety procedures specific to an Electrical Technician in charge to include, but not limited to:

        (1) Arc Flash;
        (2) LOTO;
        (3) PPE; and
        (4) Gas-free;

    b. Interpreting electrical drawings and requirements to include:

        (1) Drawings;
        (2) Design development and installation data;
        (3) System installation;
        (4) Cable diagrams;
        (5) Wiring diagrams;
        (6) Interconnection diagrams;

(7) Schematic diagrams to include:
- (a) Diagram title;
- (b) Combine diagram;
- (c) Convention and lettering;
- (d) Graphic symbols;
- (e) Layout;
- (f) Connecting lines;
- (g) Interrupted paths;
- (h) Abbreviations;
- (i) Representation of electrical contacts; and
- (j) Identification or terminals;
- (k) Elementary diagram;
- (l) Terminal block and terminal chamber diagram;
- (m) Block diagram; and
- (n) Miscellaneous wiring diagram;

c. Interpreting special purpose drawings to include:
- (1) Book form drawing;
- (2) Cable assembly drawing;
- (3) Lead assembly drawing;
- (4) Modification drawing;
- (5) Wire ways routing drawing;
- (6) Wiring deck plan;
- (7) General arrangement;
- (8) Wire way routing;
- (9) Inspecting circuit cards (AUT0920) by:
  - (a) Inspecting control relays (AUT0930);
  - (b) Inspecting inductors (AKT0660);
  - (c) Inspecting visually, motherboards (AUT1730);
  - (d) Inspecting motherboards (ALT0650);
  - (e) Inspecting motherboards (ALT0650);
  - (f) Inspecting capacitors (AUT0890); and
  - (g) Troubleshooting printed circuit boards (ALT2110).

7. <u>Time Required.</u>

8. EO – 1900 minutes; and
9. EC – 50 minutes.

10. <u>Instructional Methods</u>.

a. Interactive lecture; and

b. Demonstration performance.

11. Training Aids.

a. Electrical Schematics;
b. Multi-fault, configurable, industrial motor controller and panel; and
c. Circuit Boards.

12. Test Details. EO 002.03 will be tested in EC 002.03 IAW Annex K.

13. References. A2, A4, A7 A36, A45 and C56.

14. Remarks. Each trainee will be provided a multi-fault, configurable, industrial motor controller and panel reflective of ship's configuration for hands-on practice. Trainees to initially train using electrical schematics, then transition into hands-on troubleshooting. Prior to first lab instructor is to review metering techniques. Gas Free TP is taught by FMF Safety.

15. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Follow Safety procedures specific to an Electrical Technician in charge by:							
1. Identifying potential Arc Flash Safety Risks 2. Assessing LOTO 3. Donning PPE in electrical 4. Identify potential gas-free safety risks. 5. Following gas-free safety protocols		300			IL	A45	
TP2 Troubleshoot motor controller, to include, but not limited to:							
1. Interpreting electrical drawings and requirements, to include, but not limited to: a) Drawings; b) Design development and installation data; c) System installation; d) Wiring diagrams; e) Interconnection diagrams f) Schematic diagrams; i) Diagram title; ii) Combine diagram; iii) Convention and lettering; iv) Graphic symbols; v) Layout; vi) Connecting line; vii) Interrupted paths; viii) Abbreviations;		150	50	1000	IL DP	A2 A4 A7 A36 C56	Trainees to initially train using electrical schematics (for 200 minutes), then transition into hands-on troubleshoot-ing. Each trainee will be provided a multi-fault,

ix) Representation of electrical contacts; and x) Identification of terminals. g) Elementary diagram; h) Terminal block and terminal chamber diagram; i) Block diagram; and j) Miscellaneous wiring diagram. 2. Interpreting Special purpose drawings to include; but not limited to: a. Book form drawing; b. Cable assembly drawing; c. Lead assembly drawing; d. Modification drawing; e. Wire ways routing drawing; f. General arrangement; g. Design 3. Visually inspecting for abnormalities, to include: a) Fault lights; b) Trouble codes; and System configuration. 4. Conducting functional test; a) Checking systematically individual components, to include, but not limited to: 1) Fuses; 2) Relays; 3) Switches; 4) Contactors; 5) Conductors; 6) Leads; 7) Test points; 8) Overloads; 9) Field devices; and 10) Circuit cards.						configurable, industrial motor controller and panel reflective of ship's configuration for hands-on practice.
TP 3 Inspect circuit cards by:						
a. Inspecting circuit cards for components, to include, but not limited to: 1) Inspecting control relays; 2) Inspecting inductors; 3) Inspecting visually, motherboards; 4) Inspecting motherboards; and 5) Inspecting capacitors. 6) Troubleshooting printed circuit boards for anomalies, to include, but not limited to: i. Burn marks;	100			IL	A2 A36 C56	Each trainee to have a circuit card with built-in defects to examine during lecture.

ii. Broken traces; iii. Utilizing test points; iv. Damaged leads; v. Damaged components; and vi. Bent pins.						
TP 4 Troubleshoot electric motors, to include, but not limited to:						
a. Troubleshooting machinery space ventilation systems, to include, but not limited to: (1) Resistance check; (2) Inspecting connection box; (3) Inspecting motor leads; (4) Meggering motor; (5) Checking motor bearings; (6) Checking mounting security; (7) Checking cable glands; (8) Conduct amp probe; (9) Checking air vents; (10) Checking air gap; (11) Checking flexible coupling; and (12) Troubleshooting various motor starters by: i. Conducting experiment in Lab Volt manual.	50	100	IL DP	A4 A36	Instructor facilitates walkthrough of electric motor maintenance. TP1.a.12 Involves phase-shift start.
TP 5 Repair electric motors, to include, but not limited to:						
a. Repairing enclosure fans (AUT2320): (1) Connecting/disconnecting motor leads; and (2) Assembling/disassembling motor. b. Testing electric motors by: (1) Testing motor rotation; and (2) Checking motor alignment.	75	75			Instructor to elaborate on motor connections, taping, and Stakons ™.

PO 002 SERVICE MOTOR CONTROLLERS

EC 002.03

The content of EO 002.03 will be assessed by EC 002.03.

EC Time: 50 mins

ASSESSMENT ACTIVITY: The trainee will have a written exercise to assess critical thinking involved in troubleshooting and repairing a fault, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- EC
- Associated EC information package

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PO 002 SERVICE MOTOR CONTROLLERS

EO 002.04 Maintain Motor Controllers

1. Performance. Maintain motor controllers.

2. Conditions.

   a. Given:

      (1) Motor controller;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) Lock-out/tag-out equipment;
      (5) References; and
      (6) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall maintain motor controllers independently, IAW refs by:

   a. Repairing motor controllers (AUT2480), to include:

      (1) Disassembling/assembling component(s);
      (2) Repairing fault and/or replace component(s);
          (a) Cable Management
      (3) Conducting bench test; and
      (4) Conducting system function test;

   b. Repairing circuit cards (AUT2130) by:

      (1) Repairing circuit card cage assembly (AUT2120); and
      (2) Testing circuit card cage assembly (AUT3120) to include:
          a. Testing circuit cards (AUT3130); and
          b. Checking for correct seating.

4. Time Required.

a. EO – 2050 minutes;
b. EC – 50 min and
c. PC – 600 min

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. Lab Volt trainer; and
b. Motor Controller.

7. Test Details. EO 002.04 will be assessed by EC 002.04 IAW Annex K.

8. References. A2, A36, C56.

9. Remarks. Multi-fault, configurable, industrial motor controllers and panels will be used to train repairing and testing motor controllers.

10. Teaching Points

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Repair motor controllers (AUT2480) by:							
a. Repairing motor controllers to include, but not limited to: 1) Disassembling/assembling component(s); 2) Repair fault and/or replace component(s), to include, but not limited to: i. Fuses; ii. Relays; iii. Switches; iv. Contactors by: A. Dressing contacts; B. Replacing coil. C. Conductors; D. Leads; E. Test points; 3) Repair Overloads by:		400	1600		IL DP	A2 A36 C56	Multi-fault, configurable, industrial motor controllers and panels will be used to train repairing and testing motor controllers. Each trainee will be provided a multi-fault, configurable, industrial motor

A. Testing parameters; B. Replacing heaters; and C. Adjusting settings. 4) Field devices by: A. Conducting bench test; and B. Conducting system function test. 5) Design and reproduce motor control schematics 6) Cable management to include; a. Cable Management Overview b. Amapacity and voltage drop c. Wiring Methods d. Cable Identification to include; i. Flexible cable e. Electromagnetic Interference					controller and panel reflective of ship's configuration for hands-on practice.
TP 2 Repair circuit cards by:					
a. Repairing circuit card cage assembly (AUT2120) by: 1) Examining for bent pins; 2) Repairing bent pins; 3) Following ESD procedures; 4) Replacing fuses; and 5) Resetting overload protection. b. Testing circuit card cage assembly (AUT3120) by: 1) Checking for correct seating; and 2) Testing circuit cards (AUT3130).	25	25		DP IL		Demonstration circuit card provided to trainees are to contain current protection devices.

PO 002 SERVICE MOTOR CONTROLLERS

EC 002.04

EC Time: 50 mins

ASSESSMENT ACTIVITY: The trainee will complete a written exercise to assess critical thinking involved in maintenance of motor controllers, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- EC
- Associated EC information package

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PC 002 - SERVICE MOTOR CONTROLLERS

PC Time: 50 mins per trainee.

ASSESSMENT ACTIVITY: The trainee will perform a practical assessment to assess the trainee's ability to troubleshoot and repair a faulted Electric Motor and Controller, IAW the standard.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- Trainer
- Schematic
- Multi-Meter
- Student Package

IMPACT OF FAILURE: TED / Remedial Training / Re-test / TRB after second unsuccessful attempt

PERFORMANCE CHECKLIST

Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely complete practical assessment checklist.

Administrative Instructions:

1. This practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from PO 002. The total time given for troubleshooting 2 x double faults is 45 minutes, this is accomplished by; giving the trainee a 5 minutes familiarization with the circuit then a total of 45 minutes, tracking time used to correct faults. For example, the trainee has finished a 5 minute famil. and is starting their first double fault, it takes them 35 minutes to find and correct the faults. For their second double fault they will have 10 minutes remaining to successfully correct the faults.

2. **Standard: The Marine Technician Electrical Specialist shall independently demonstrate the ability to Troubleshoot, Repair and Test Motor Controllers.**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

1. Ensure the trainee is given an opportunity for the maintenance routine to progress to its conclusion;

2. If the trainee experiences difficulty, provide minimal assistance to maintain the standard through PC;

3. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either;

a. Paused and corrective action taken before continuing, or halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

1. Various Components;
2. PPE (glasses, gloves, ear protection);
3. Test Tools if required
4. Supervision.
5. Instructor to pre-build the PC circuit in Lab-Volt

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Familiarization with Circuit (5 MINS) Time:			

Confirm circuit operation			Circuit must be function tested
Observe safety at all times			Failure to follow safe procedures constitutes an automatic fail
Instructor comments:			

1st DOUBLE FAULT	**Time 1st Fault:**	**Time 2nd Fault:**
Student picks faults from envelope, instructor inserts the fault, no guidance from instructor.		
Fault chosen:	Fault details (exact location of fault in circuit):	
Fault chosen:	Fault details (exact location of fault in circuit):	

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Test circuit to observe fault			
Test meter			
Use meter correctly			
Explain logic behind chosen troubleshooting technique			
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Prove faults corrected			
Instructor comments:			

2nd DOUBLE FAULT	**Time 1st Fault:**	**Time 2nd Fault:**
Student picks faults from envelope; instructor inserts the faults, no guidance from instructor.		
Fault chosen:	Fault details (exact location of fault in circuit):	

Fault chosen:	Fault details (exact location of fault in circuit):		
TEST PROCEDURES	**SAT**	**UNSAT**	**INSTRUCTOR COMMENTS**
Test circuit to observe fault			
Test meter			
Use meter correctly			
Explain logic behind chosen troubleshooting technique			
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Prove faults corrected			
Instructor comments:			

Overall Assessment	**Result** **(instructor to circle final grade)** **PASS / FAIL**
To be completed by Instructor / Invigilator	
Strengths:	Area of improvement:

Date:	**Trainee's Signature:**	**Instructor's Signature:**

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EO 003.01 Apply a Detailed Knowledge of Microprocessors

1. Performance. Apply a Detailed Knowledge of Microprocessors.

2. Conditions.

   a. Given:

      (1) Supervision;
      (2) DND Learn account;
      (3) Internet/intranet capable computer; and
      (4) References.

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Laboratory.

3. Standard: The Marine Technician Electrical Specialist shall apply a detailed knowledge of microprocessors independently, IAW refs by:

   a. Analyzing microprocessors and microcomputers to include:

      (1) Digital Architecture and Theory:
      (2) Numbering Systems;
      (3) Logic Gates; and
      (4) D-Type Flip Flops and registers.
      (5) Microcontroller Architecture and Instruction set:
      (6) Internal Registers and Instruction Set.
      (7) Additional Internal Registers and Functions.EasyPIC V7 PIC trainer:
      (8) EasyPIC v7 PIC Trainer;
      (9) Programming Sequence;
      (10) I/O Ports and Logic;
      (11) Decision Making Using Bit Testing; and
      (12) Subroutines, Multiplexing and Look-Up Tables.
      (13) Programming II- Advance Coding:
      (14) Timers;
      (15) LCD Display;
      (16) Analog to Digital Conversion; and
      (17) Interrupts.
      (18) Serial Communication:
         (a) Serial Data; and
         (b) SCADA and Networking.

4. Time required.

a. EO – 3700 minutes, and
b. CEC – 500 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstrate – performance.

6. Training Aids.

a. TBD by Academic SME

7. Test Details. EO 003.01 will be assessed by CEC 003.01 IAW Annex K TBD by Academic SME.

8. References. C17, C18, and TBD by Academic SME.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)		Method	Refs	Comments
		T	P			
TP1 Introduction to microprocessors and microcomputers to include:						
a. Digital Architecture and Theory; (1) Numbering Systems; (2) Logic Gates; and (3) D-Type Flip Flops and registers		420		IL		
b. Microcontroller Architecture (1) Internal Registers; and (2) Additional Internal Registers.		480				
c. Programming I Basic Coding: (1) EasyPIC v7 PIC Trainer; (2) Programming Sequence; (3) I/O Ports and Logic; (4) Decision Making using Bit Testing; and (5) Subroutines, Multiplexing and look-up tables.		1020				
d. Programming II- Advance Coding: (1) Timers; (2) LCD Display; (3) Analog to Digital Conversion; and						

(4) Interrupts.						
e. Serial Communication: (1) SCADA and Networking						

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS
Critical Enabling Check – CEC 003.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 003.01 will be a written assessment based on content from EO 003.01
3. Pass/Fail IAW Annex K.
4. Time required: 500 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 003.01 will be a written assessment based on content from EO 003.01
2. During this written assessment, you will be required to apply detailed knowledge of microprocessors. You must pass this test to pass EO 003.01. To pass the test, you must apply detailed knowledge of essential mathematic skills in accordance with the standard.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 500 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by academic instructor.

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EO 003.02 Apply a Detailed Knowledge of Theoretical Concepts of Control Systems

1. Performance: Apply Detailed Knowledge of Theoretical Concepts of Control Systems.

2. Conditions.

   a. Given:

      (1) Supervision;
      (2) DND Learn account;
      (3) Calculator;
      (4) Internet/intranet capable computer; and
      (5) References;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Laboratory.

3. Standard. The Marine Technician Electrical Specialist shall apply detailed knowledge of theoretical concepts of control systems independently, IAW refs by:

   a. Introducing automatic control systems by defining fundamentals of automatic control;

   b. Interacting with AC switching controllers by:

      (1) Examining bi-directional switches; and
      (2) Examining switching control of AC loads.

   c. Classifying choppers to include:

      (1) Examining voltage step-down choppers;
      (2) Examining step-up choppers;
      (3) Examining two-quadrant choppers;
      (4) Examining four-quadrant choppers; and
      (5) Examining switching control circuits for chopper converters.

   d. Classifying inverters by:

      (1) Examining functions and features of inverters;
      (2) Examining Inverter applications;
      (3) Examining types of inverters;
      (4) Examining half-bridge rectifiers;
      (5) Examining full-bridge configurations; and

(6) Examining three phase inverters.

e. Inspecting adjustable speed DC motor drives to include:

(1) Examining speed control methods of separately excited DC motors;
(2) Examining chopper-controlled DC motor devices;
(3) Examining DC motor drives using phase-controlled thyristor converters; and
(4) Demonstrate control of series motors.

f. Controlling solid-state AC motor control to include:

(1) Examining voltage source inverters;
(2) Examining current source inverters;
(3) Examining phase controlled cycloconverters;
(4) Examining load commutated inverters; and
(5) Examining adjustable speed drives using squirrel cage induction motors.

g. Utilizing power supply systems to include examining power supply components;

h. Analyzing synchro and servo motors to include:

(1) Examining synchro transmitters' construction and basic operation;
(2) Examining synchro receivers;
(3) Examining control systems;
(4) Examining the construction and basic operation of differential transmitters;
(5) Examining construction and basic operation of differential receivers;
(6) Examining construction and basic operation of servo amplifiers; and
(7) Examining AC and DC servo motors.

i. Troubleshooting programmable logic controllers (PLCs) to include but are not limited to:

(1) Troubleshooting plastic compress melt unit (AYT0420);
(2) Troubleshooting reverse osmosis desalination (ROD) unit (AOT1960);
(3) Troubleshooting auxiliary steam generator (AIT0910);
(4) Troubleshooting chiller units (AWT0220); and
(5) Troubleshooting black & grey water treatment system (AOT1830) by:
(a) Troubleshooting black water system (AOT1840); and
(b) Inspecting black water treatment system (AOT0450).

4. Time required.

a. EO – 12300 (7850 Controls + 4450 PLC minutes); and
b. CEC – 100 minutes.

5. Instructional Methods.

a. Interactive lecture; and

b. Demonstrate – performance.

6. Training Aids.

a. TBD by Academic instructors.

7. Test Details. EO 003.02 will be assessed by CEC 003.02 administered by academic instructors IAW Annex K.

8. References. A29, A30, A36, A57, C1, C2, C4, C28, C29, C30, C31, C37, C38, C54, C55,C56, C57 and B1.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Introducing automatic control systems by: 1.1 Defining Fundamentals of Automatic control to include: a. Define and state purpose; b. Describe variety of electronic controllers (PID, PLC, Computer etc) c. Describe how automatic controls are utilized in the ships propulsion machinery (ex. temperature, level control, and include control parameters such as time lag, time constant, dead time, 1st and 2nd order lag, disturbance, and offset. d. On-Off control: (1) What it means; (2) Characteristics; (3) How it is used; and (4) Use applications as examples. e. Sequential Control: (5) What it means; (6) Characteristics; (7) How it is used; and (8) Use applications as examples. f. Proportional-Integral-Derivative (PID) Control Introduction: (9) Explain principle/ theory; (10) Characteristics of PID Control (explain the P, I, D, PD, and PID actions respectively using step or ramp input; and (11) Describe components comprising a PID control system, including (in block format) sensing unit, Transducer, manipulator and controller.							
TP2: Interact with AC switching controllers by:							
a. Examining bi-directional switches, to include, but not limited to:		50			IL	C28 C29	Laboratory experiments

(1) AC switches using thyristors; and (2) AC switches using other devices.					C30 C31 C37 C38	to support knowledge.
b. Examining switching control of AC loads to include by not limited to: (1) Single phase resistance loads – integral half cycle control; (2) Phase control – resistive loads; (3) Single phase R-L loads; (4) Thyristor controlled inductors; (5) Static VAR compensation circuits; and (6) Three phase applications of switching control.	50		150	IL		
TP3: Classify choppers by:						
a. Examining voltage step-down choppers, to include, but not limited to: (1) Power circuit configuration and working principles; (2) Voltage relationships; (3) Duty cycle range of practical choppers; (4) Chopper operation when load presents a “back EMF”; (5) Use of filter on input side; and (6) Use of filter on the output side.	25			IL	C28 C29 C30 C31 C37 C38	Laboratory experiments to support knowledge.
b. Examining step-up choppers including but not limited to: (1) Power circuit configuration and working principles; (2) Voltage relationships; (3) Duty cycle range of practical choppers; (4) Chopper operation when load presents a “back EMF”; (5) Use of filter on input side; and (6) Use of filter on the output side.	15			IL		
c. Examining two quadrant choppers, to include, but not limited to: (1) Waveforms; and (2) Switching circuitry.	10			IL		
d. Examining four quadrant choppers, including but not limited to: (1) Waveforms; and (2) Switching circuitry.	25			IL		
e. Examining switching control circuits for chopper converters, including but not limited to:	15		150	IL		Laboratory experiments to support knowledge.

(1) Functional requirements of the switching control circuits.						
TP4: Classify inverters by:						
a. Examining functions and features of inverters.	20			IL	C28 C29 C30 C31 C37 C38	
b. Examining inverter applications including but not limited to: (1) DC link inverters; (2) Adjustable speed AC drives; and (3) Uninterruptable power supplies.	15			IL		
c. Examining types of inverters, to include, but not limited to: (1) Voltage source inverters; (2) Current source inverters; and (3) Current regulated inverters.	50			IL		
d. Examining half-bridge rectifiers, to include, but not limited to: (1) Circuit configuration switching; (2) Analysis of half-bridge inverter with inductive load; (3) Adjustment of AC frequency and AC voltage; and (4) Output waveform considerations; and sinusoidal pulse width modulation using half-bridge rectifiers.	50			IL		
e. Examining full-bridge configurations, to include, but not limited to: (1) Single phase full wave bridge topology; (2) Control of the AC output voltage – pulse width modulation; and (3) Shaping of the output waveform – sinusoidal pulse width modulation.	35			IL		
f. Examining three phase inverters, to include, but not limited to: (1) Circuit configuration and switching sequence; (2) Three phase inverters feeding a delta connected load; (3) Three phase inverter feeding wye connected load; (4) Voltage control and waveform shaping in three phase converters; (5) Filtering requirements; and (6) Inverter operation with reverse power flow.	95		150	IL		Laboratory experiments to support knowledge.
TP5: Examine adjustable speed DC motor drives by:						

a. Examining speed control methods of separately excited DC motors, to include, but not limited to: (1) Converter restrictions; (2) Speed control ranges with separate excitation; and (3) Regenerative braking and reversing.	55			IL	C28 C29 C30 C31 C37 C38	Instructor to include quick review of DC motor basics.
b. Examining chopper-controlled DC motor devices, to include, but not limited to: (1) Driving and braking considerations; (2) Rotation reversal; and (3) Static four quadrant chopper.	30			IL		
c. Examining DC motor drives using phase-controlled thyristor converters, to include, but not limited to: (1) Dual bridge configuration; (2) Speed control scheme for a DC motor drive using a dual converter with circulating (3) current; and (4) Sequence of operations during a rotation reversal.	25			IL		
d. Demonstrate control of series motors.		250		DP		Laboratory experiments to support knowledge.
TP6: Control solid state AC motors control by:						
a. Examining voltage source inverters;	40			IL	C28 C29 C30 C31 C37 C38	
b. Examining current source inverters;	25			IL		
c. Examining phase controlled cycloconverters;	50			IL		
d. Examining load commutated inverters; and	50			IL		
e. Examining adjustable speed drives using squirrel cage induction motors, to include, but not limited to: (1) Induction motor basics; (2) Open loop speed control of an induction motor fed from a voltage source inverter or cycloconverter; and (3) Closed loop control using a voltage source inverter.	70	250		DP		Laboratory experiments to support knowledge.
TP7: Utilize power supply systems by:						
a. Examining power supply components to include but not limited to: (1) Linear regulators;	150			IL	C28 C29 C30	

(2) Pre-regulators; (3) Off-line switchers; (4) SMPS converter circuit topologies; (5) Controllers for SPMS; and (6) UPS construction.					C31 C37 C38	
TP8: Analyzing synchro and servo motors by:						
a. Examining synchro transmitters' construction and basic operation;	75			IL	C28 C29 C30 C31 C37 C38	
b. Examining synchro receivers to include but not limited to: (1) Construction and basic operation; and (2) Differences between transmitters and receivers.	10			IL		
c. Examining control systems including but not limited to: (1) Closed loop system (TX/RX); (2) Analysis of systems – controlled variable response to set-point changes.	50			IL		
d. Examining construction and basic operation of differential transmitters;	20			IL		
e. Examining construction and basic operation of differential receivers;	80			IL		
f. Examining construction and basic operation of servo amplifiers; and	50		150	IL		
g. Examining AC and DC servo motors				DP		Laboratory experiments to support knowledge.
TP9: Troubleshooting programmable logic controllers (PLCs) by:						
a. Applying a detailed knowledge of PLC controllers, to include, but not limited to: (1) Basics of programming to include: i. Programming flow charts; ii. Programming symbology; and iii. Ladder logic. (2) Functionality; (3) PLC machinery interface; (4) Process of a scan cycle; and (5) User interface as it relates to: i. Troubleshooting plastic compress melt unit; ii. Troubleshooting reverse osmosis	400	5150		IL DP	A29 A30 A36 A57 C1 C2 C4 C28 C54 C55 C56 C57	PLCs to be provided to trainees for practical programming and examination.

desalination (ROD) unit; iii. Troubleshooting auxiliary steam generator; iv. Troubleshooting chiller units; and v. Troubleshooting black & grey water treatment system by: 1. Troubleshooting black water system; and 2. Inspecting black water treatment system.					

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS
Critical Enabling Check – CEC 003.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 003.02 will be a written assessment based on content from EO 003.02
3. Pass/Fail IAW Annex K.
4. Time required: 100 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 003.02 will be a written assessment based on content from EO 003.02
2. During this written assessment, you will be required to apply detailed knowledge of essential mathematic skills. You must pass this test to pass EO 003.02. To pass the test, you must apply detailed knowledge of essential mathematic skills in accordance with the standard.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 100 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by the academic instructor.

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EO 003.03 Troubleshoot Control Systems

1. Performance. Troubleshoot Control Systems.

2. Conditions.

   a. Given:

      (1) Applicable equipment and/or control system;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard.The Marine Technician Electrical Specialist shall troubleshoot control systems independently, IAW refs by:

   a. Troubleshooting human/machine interface (HMI) (ALT1970) to include
      (1) but not limited to:

      (2) Troubleshooting integrated platform control system (IPCS) man machinery interface (ALT1980);

      (3) Troubleshooting the integrated platform management system (IPMS) monitors
      (4) Troubleshooting diesel generator local control panel; and

      (5) Troubleshooting local operating panel (ALT2010) by inspecting local operating panel;

   b. Investigating access modules to include but not limited to:

      (1) Investigating internal integrated platform management system faults by:
         (a) Analyzing digital data;
         (b) Collecting equipment health monitoring data (AFT0970);

(2) Investigating damage control systems by;
(3) Analyzing components of fire alarm systems;
(4) Examine loops and system interactions; and
(5) Analyzing zoning, including addressable and non-addressable systems.

(6) Investigating remote terminal unit (RTU) by conducting built-in test equipment (BITE) test.

(7) Inspecting digital input/output processor (WAGO)

(8) Investigating bus access modules (BAMS) by inspecting bus access modules (BAMS) (ALT0480); and

(9) Investigating analog-digital acquisition modules (ADAM) by inspecting RTU accessory sub components.

4. Time Required.

a. EO – 600 minutes; and
b. EC – 50 min.

5. Instructional Methods.

a. Interactive lecture;
b. Field Trip; and
c. Demonstration – performance.

6. Training Aids.

a. Maintenance Trainer

7. Test Details. EO 003.01 will be assessed by EC 003.01 IAW Annex K.

8. References. A4, A8, A9, A10, A12, A29, A30,A36, A57, C1, C2, C4, C16, C24, C25, C26, C27, C28, C29, C30, C32, C54 and C55

9. Remarks. Nil.

10. Teaching Points.

	Sub Teaching Points	Instructional	Method		Comments

Teaching Point		Technique (Time in min)				Refs	
		T	D	P			
TP 1: Troubleshoot human/machine interface (HMI) by:							
a. Troubleshooting integrated platform management system (IPMS) man machinery interface by: (1) Explaining common HMI components to include: i. Purpose; ii. Components: 1. Monitors; a) 20-inch high-resolution (1280 x 1024 pixel) rugged color liquid crystal display (LCD) b) Rugged QWERTY keyboard c) Panel mount trackball d) Computer system e) Speaker f) Fan assembly g) Terminal block panel assembly h) EMI power filter and diode i) Throttle button panel assembly j) Pushbuttons k) Ethernet switch l) 120V AC power receptacle m) Power supply. n) Relays 2. Power supplies; and 3. Processors. iii. Network connections. (2) Rectifying HMI faults, to include, but not limited to: i. Function tests; and ii. Common faults.		150	50		IL DP	A4 A8 A9 A10 A12 A29 A30 A36 A57 C1 C2 C4 C16 C24 C25 C26 C27 C28 C29 C30 C32 C54 C55 C56 C57	
b. Troubleshooting integrated platform management system (IPMS) monitors by: (1) Recognizing monitor types;		50			IL DP		

(2) Identifying video connection points; (3) Identifying power supplies; and (4) Configuring screen resolution.					
c. Troubleshooting generator local control panel by recognizing components, to include, but not limited to: (1) Recognize common HMI components; (2) Circuit Breakers; (3) Fuse blocks; (4) Power supplies; (5) WAGO device.	50		IL DP		Trainees to investigate various faults introduced on HMI systems.
TP 2: Troubleshooting access modules by:					
a. Troubleshooting internal integrated platform management system (IPMS) faults by: (1) Analyzing distributed control systems; and (2) Analyzing supervisory control and data acquisition (SCADA) systems.	50	50	IL DP	A29 A30 A36 A57 C1 C2 C4 C28 C54 C55 C56 C57	
b. Troubleshooting damage control sub-systems by recognizing components, to include but not limited to: (1) BAMs; (2) PDDBs i. Filters; ii. Circuit breakers; iii. Fuse blocks; iv. Power supplies; and v. Terminal blocks (3) Analyzing components of fire detection systems (4) Examine loops and system interactions; and (5) Analyzing zoning, including addressable and non-addressable systems.	50				
c. Troubleshooting remote terminal unit (RTU) by recognizing components to include, but not limited to: (1) EMI Filters; (2) Circuit breakers; (3) Fuse blocks; (4) Terminal blocks (5) Power supplies (6) VME Chassis (7) The various RTU circuit cards	50				Trainees to perform BITE test on an RTU.

d. Inspecting digital input/output processor (WAGO) by: (1) Explaining WAGO modules, to include, but not limited to: i. Purpose; ii. Function; iii. Field bus coupler; iv. Bus configuration; and	25				
e. Troubleshooting bus access modules (BAMs) by: (1) Inspecting bus access modules (BAMs), to include, but not limited to: i. Function of BAM types: 1. Type 1; 2. Type 2; and 3. Type 3. ii. Configuration of HRC and LCUs; iii. HFCP connection; iv. TAU connection; v. AFFF proportioners; vi. Galley fire extinguishing agent release connection; vii. Hangar gas monitoring; viii. Smoke clearing fans; and ix. Citadel pressure monitoring.	50		IL DP		
f. Troubleshooting analog-digital acquisition modules (ADAM) by: (1) Explaining: i. Purpose; and ii. Function.	25				Written quiz to support understanding of training knowledge of TP

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EC 003.03

EC Time: 50 minutes.

ASSESSMENT ACTIVITY: The trainee will have a written assessment on the content of EO 003.03 troubleshoot control systems, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- EC

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EO 003.04 Repair Control Systems

1. Performance. Repair Control Systems.

2. Conditions.

   a. Given:

      (1) Applicable equipment and/or control system;
      (2) PPE;
      (3) Test equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall repair control systems independently, IAW refs by:

   a. Maintaining Human/Machine interfaces to include, but not limited to:

      (1) Repairing DG local control panel (ALT1060);
      (2) Testing DG local control panels (ALT1540);
      (3) Repairing QM local operating panel (ALT1340).
      (4) Testing QM local operating panels (ALT1750);
      (5) Testing portable operating units (POU) (ALT1670); and
      (6) Testing onboard training systems (OBTS) (ALT1650)

   b. Maintaining Bus Access modules (BAMS) (ALT0950) to include, but not limited to:

      (1) Connecting/disconnecting BAMS and associated devices (ALT0380);
      (2) Calibrating power distribution control systems (AUT0560);
      (3) Testing bus access modules (BAMs) (ALT1470); and
      (4) Testing remote terminal units (RTU) (ALT1760) by:
         (a) Testing components with built-in test equipment (BITE) (ALT1520);

(b) Performing analog/digital ground checks (ALT0910);
(c) Testing printed circuit boards (ALT1710);
(d) Testing network communication cables (ALT1640);
(e) Testing power data distribution boxes (PDDB) (ALT1680);
(f) Testing WAGO modules (ALT1790);
(g) Testing power level angle actuators (ALT1690); and
(h) Operating virtual remote terminal units (VRTUs) (HT0170).

4. Time Required.

a. EO – 1450 minutes; and
b. EC – 50 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. Working IPMS system with local operating panel (QM Panel) and BAM; and
b. Working control system that can have a fault introduced into it.

7. Test Details. EO 003.02 will be assessed by EC 003.02 IAW Annex K.

8. References. A13, A14, A15, C33, C34, C35, C36, D1, D2, D3, D4, D5, D6, D7, D8, D9, and D10.

9. Remarks. Fundamental instructional knowledge for the comprehension of circuits and electronic components shall be provided at the beginning of this EO.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
TP1: Maintaining Human/Machine interfaces to include, but not limited to:							
a. Testing portable operating units (POU) by: (1) Identifying hardware and software; (2) Operating software diagnostics; and		50	50	100	IL DP	A13 A14 A15 C33 C34	

(3) Repair/replacement procedures.					C35 C36	
b. Testing onboard training systems (OBTS) by: (1) Describing the purpose of the OBTS; (2) Performing a flash up of the system; and (3) And loading training scenarios.	50	25	100	IL DP		Trainees should be able to boot up and load training program-ming to the OBTS.
TP2: Maintain Bus Access modules (BAMS), to include, but not limited to:						
a. Familiarize trainee with BAMS by: (1) Connecting/disconnecting BAMS and associated devices; and (2) Testing bus access modules (BAMs).	50	25	50	IL	D1 D2 D3 D4 D5 D6 D7 D8 D9 D10	Instructor to demonstrate on functional BAM.
b. Testing remote terminal units (RTU) by: (1) Testing components with built-in test equipment (BITE); (2) Performing analog/digital ground checks; (3) Testing printed circuit boards; (4) Testing network communication cables; (5) Testing power data distribution boxes (PDDB); (6) Testing WAGO modules; and (7) Operating virtual remote terminal units.	200	700		IL DP		

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EC 003.04

EC Time: 50 minutes.

ASSESSMENT ACTIVITY: The trainee will have a written assessment on the content of EO 003.04 repair control systems, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- EC

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EO 003.05 Repair Engine Control Systems

1. Performance. Repair Engine Control Systems.

2. Conditions.

   a. Given:

      (1) Applicable equipment and/or control system;
      (2) PPE;
      (3) Test equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall repair engine control systems independently, IAW refs by:

4. Repairing Governor circuits by:

   a. Repairing main diesel engine electronic governor (AVT1140); and
   b. Testing propulsion diesel engine controls (ALT1730) by testing diesel engine electronic governor (AVT1230);
   c. Repair Engine Control systems by:
      (1) Installing propulsion control system components (AUT1850);
      (2) Assembling/disassembling diesel engine electric starting system components (AVT0300); and
      (3) Testing momentum modules (ALT1630).

5. Time Required.

   a. EO – 1450 minutes; and
   b. EC – 300 minutes.

6. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

7. Training Aids.

a. Running diesel engine with an electric governor and working controls.

8. Test Details. EO 003.03 will be assessed by EC 003.03 IAW Annex K.

9. References. A10, A16, A17, C22 and C25.

10. Remarks. Nil.

11. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1: Repairing Governor circuits by:							
a. Repairing main diesel engine electronic governor by: (1) Describing the governor box; (2) Identifies governor components; (3) Describe governor functions, including: i. Speed measuring system control; ii. Fuel limited system; iii. Speed droop system; iv. Speed sensor; and v. Frequency divider.		100			IL	A10 A16 A17 C22 C25	
b. Testing propulsion diesel engine controls by: (1) Describing principles of speed measuring control to include, but not limited to: i. Electronic mode; ii. Module 721 (digital control); iii. PLC; iv. PGA-EG; and v. Pneumatic mode. (2) Testing diesel engine electronic governor by: i. Identifying and repairing: 1. Types of faults; 2. Remote control mode; 3. Local control mode; and		150					Written quiz to support training of TP1.

4. Mechanical mode.						
TP2: Repair Engine Control systems by:						
a. Installing propulsion control system components;	25	200		IL DP	A10 A16 A17 C22 C25	
b. Assembling/disassembling diesel engine electric starting system components, to include, but not limited to: (1) Components of an electrical starter; (2) Operation of the starter-mounted solenoids; and (3) Operation of the overrunning clutch and drive pinion.	50	200		IL DP		Trainees to complete an overhaul of an engine starter, hook up to an engine and trial.
c. Calibrating electrical propulsion control system components by: (1) Testing power lever angle actuators.	25			IL		
d. Testing momentum modules.	25			IL		
e. Troubleshooting engine control systems by: (1) Interpreting engine start permissives; (2) Examining field device signal processing to control circuits; (3) Analyzing pyrometer data and fire detection systems in engine enclosures; (4) Examining engine mechanical governor; (5) Examining engine electrical/electronic governor; and (6) Analyzing emergency stop conditions.	200	75	400	IL FT	A10 A16 A17 C22 C25	TEL or universal engine starting/control system available on computer for trainees to manipulate. Trainees to be given a tour of different classes of ships to examine different types of control systems. Written quiz to support understanding of training knowledge of TP

EC 003.05

EC Time: 25 mins per trainee

ASSESSMENT ACTIVITY: The trainee will troubleshoot Engine Control utilizing Lab-Volt and a Governor circuit to find a single fault, IAW refs.

At the time of testing, the trainee will use a random selection method (such as drawing a scenario from a lot). The trainee will be role-playing as the on-watch technician. The instructor will induce the selected fault within the control system. The trainee will proceed to use relevant schematics and technical manuals to troubleshoot and identify the potential fault. The trainee will signal to the instructor that they have identified the fault, after which the instructor will confirm that the fault has been identified. The trainee will then proceed to repair the fault(s) and confirm the proper functionality of the system with the instructor present. An example of an EC would be as follows: The trainee receives a notification that the Governor Cct has a fault. Utilizing drawings and technical manuals and their understanding of troubleshooting techniques, the trainee will troubleshoot the system and identify the fault to the instructor. The trainee will rectify the fault. The instructor will have the opportunity to ask probing questions and fully develop the trainee's solution.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- Trainer
- Schematic / System Block Diagram
- Multi-Meter

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

ENABLING CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely complete practical assessment checklist.

Administrative Instructions:

1. This practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from PO 003.

2. **Standard: The Marine Technician Electrical Specialist shall independently demonstrate the ability to Troubleshoot, Repair and Test Electronic Control systems.**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

1. Ensure the trainee is given an opportunity for the maintenance routine to progress to its conclusion;
2. If the trainee experiences difficulty, provide minimal assistance to maintain the standard through EC;
3. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either;
   a. Paused and corrective action taken before continuing, or halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

1. Various Components;
2. PPE (glasses, gloves, ear protection);
3. Test Tools if required
4. Supervision.
5. Instructor to pre-build the EC circuit in Lab-Volt

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Familiarization with Circuit (5 MINS) Time:			
Confirm circuit operation			Circuit must be function tested
Observe safety at all times			Failure to follow safe procedures constitutes an automatic fail
Instructor comments:			
1st FAULT (20 MINS) Time 1st Fault:			
Student picks faults from envelope, instructor inserts the fault, no guidance from instructor.			
Fault chosen:	Fault details (exact location of fault in circuit):		

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Test circuit to observe fault			
Test meter			
Use meter correctly			
Explain logic behind chosen troubleshooting technique			
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Prove faults corrected			
Instructor comments:			

Overall Assessment	**Result** **(instructor to circle final grade)** **PASS / FAIL**

To be completed by Instructor / Invigilator		
Strengths:		Area of improvement:
Date:	**Trainee's Signature:**	**Instructor's Signature:**

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS

EO 003.06 Apply a Detailed Knowledge of Theoretical Concepts of Digital Systems

1. Performance. Apply a Detailed Knowledge of Theoretical Concepts of Digital Systems.

2. Conditions.

   a. Given:

      (1) Supervision;
      (2) DND Learn account;
      (3) Internet/intranet capable computer; and
      (4) References;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Laboratory.

3. Standard. The Marine Technician Electrical Specialist shall apply detailed knowledge of theoretical concepts of digital systems independently, IAW refs by:

   a. Explaining concepts of digital logic;

   b. Comparing number system and codes;

   c. Analyzing logic gate sand Boolean algebra;

   d. Comparing combinational logic circuits;

   e. Explaining flip flops, counters, registers;

   f. Conducting digital arithmetic operations and circuits;

   g. Analyzing counters and registers; and

   h. Analyzing MSI logic circuits.

4. Time required.

   a. EO - 4450 mins, and
   b. CEC - 300 minutes

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstrate – performance.

6. Training Aids.

a. TBD by Academic Instructors.

7. Test Details. EO 003.06 will be assessed by CEC 003.06 IAW Annex K.

8. References. C38, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24, D25, D26, D27, D28, D29, D30, D31, D32, D33, D34, D35, D36, D37, D38, D39, D40, D41, D42, D43, D44, D45, D46, D47, D48, D49, D50

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Explain concepts of digital logic							
a. Numerical representation; b. Digital and analog systems; c. Digital number systems; d. Representing binary quantities; e. Digital circuits; f. Parallel and serial transmission; g. Memory; h. Digital computers; and i. Switch representation of logic functions.							
TP2 Comparing number systems and codes							
a. Binary to decimal conversions; b. Decimal to binary conversions; c. Octal number system; d. Hexadecimal number system; e. BCD code; f. Excess-3 code; g. Grey code; h. Alphanumeric code; and i. Parity method of error detection.							
TP3 Analyze logic gate sand Boolean algebra							
a. Boolean constants and variables; b. Truth tables; c. OR operation; d. AND operation; e. NOT operation; f. Describing logic circuits algebraically;							Laboratory experiment to support knowledge

g. Evaluating logic circuits outputs; h. Implementing circuits from Boolean expressions; i. NOR gates and NAND gates; j. Boolean and DeMorgan theorems; k. Boolean identities and proofs; l. Universality of NAND gates and NOR gates; m. Alternate logic gate representations; and n. IEEE/ANSI stand logic symbols.						
TP4 Compare combinational logic circuits						
a. Sum of products form b. Simplifying logic circuits c. Algebraic simplifications d. Proof by perfect induction e. Combinational logic, min-terms and max-terms f. Designing combinational logic circuits g. Karnaugh map method h. Exclusive OR and exclusive NOR circuits i. Parity generator and checker j. Inhibit circuits k. Basic characteristics of digital ICs l. Troubleshooting digital systems m. Internal digital IC faults n. External faults o. Troubleshooting case study						Laboratory experiment to support knowledge
TP5 Explain flip flops						
a. Introduction to flip flops b. NAND gate latch c. NOR gate latch d. Troubleshooting case study e. Clock signals and clocked flip flops f. Clocked SET CLEAR flip flops g. Clocked J-K flip flop h. Clocked D flip flops i. D latch j. Asynchronous inputs k. IEEE/ANSI symbols l. Flip flop timing considerations m. Detecting an input sequence n. Data storage and transfer o. Serial data transfer, shift registers p. Frequency division and counting q. Schmitt trigger devices r. One shot (monostable Multivibrator) s. Analysing sequential circuits						Laboratory experiments to support knowledge

t. Astable multivibrators u. Troubleshooting flip flops circuits						
TP6 Conduct digital arithmetic operations and circuits						
a. Binary addition b. Repressing signed numbers c. Addition in the 2s complement d. Subtraction in the 2s complement e. Multiplication of binary numbers f. Binary division g. BCD additions h. Hexadecimal arithmetic i. Arithmetic circuits j. Parallel binary adder k. Design of a full adder l. Complete parallel adders with registers m. Carry propagation n. Integrated circuits parallel adder o. 2s complement system p. BDR adder q. Binary multipliers r. Complex arithmetic ICs s. Comparators circuit t. Parity circuits u. IEEE/ANSI symbols v. Troubleshooting case study						Laboratory experiment to support knowledge
TP7 Analyze counters and registers						
a. Asynchronous (ripple) counters b. Counters with MOD numbers greater than 2n c. IC asynchronous counters d. Asynchronous down counter e. Propagation delay in ripple counters f. Synchronous (parallel) counters g. Synchronous down and up/down counters h. Pre-settable counters i. The 74193 (LS193/HC 192) counter j. IEEE/ANSI dependency notation k. Decoding a counter l. Decoding glitches m. Cascading BCD counters n. Synchronous counter design o. Shift-register counters p. Counter applications, frequency counter and digital clock q. Integrated circuit registers r. Parallel in/parallel out registers s. Serial in/serial out registers t. Parallel in/serial out registers						Laboratory experiments to support knowledge

u. Serial in/parallel out registers v. IEEE/ANSI register symbols w. Troubleshooting counters and registers						
TP8 Analyze MSI logic						
a. Encoders and decoders, terminology b. Decoders c. BCD to 7 segment decoders/drivers d. Liquid crystal displays e. Encoders f. Decoder and encoder IEEE/ANSI symbols g. Troubleshooting decoders and encoders h. Multiplexers i. Multiplexer application j. Multiplexers to replace random logic k. De-multiplexers l. Multiplexer/ de-multiplexer IEEE/ANSI symbols m. Troubleshooting multiplexer/de-multiplexers circuits n. Magnitude comparator o. Code converters p. Digital IC terminology q. Standard TTL series characteristics r. TTL open collector outputs s. Tristate (3-stage) TTL t. Tristate registers u. Data bussing						Laboratory experiments to support knowledge

PO 003 SERVICE ELECTRONIC CONTROL SYSTEMS
Critical Enabling Check – CEC 003.06

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 003.06 will be a written assessment based on content from EO 003.06
3. Pass/Fail IAW Annex K.
4. Time required: 300 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 003.06 will be a written assessment based on content from EO 003.06
2. During this written assessment, you will be required to apply detailed knowledge of theoretical concepts of Digital systems. You must pass this test to pass EO 003.06. To pass the test, you must apply detailed knowledge of theoretical concepts of Digital system skills in accordance with the standard.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 300 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by the academic instructor.

PO 004 MAINTAIN FIELD DEVICES

EO 004.01 Apply detailed knowledge of electronic circuit fundamentals

1. Performance. Apply detailed knowledge of electronic circuit fundamentals.

2. Conditions.

   a. Given:

      (1) Supervision;
      (2) DND Learn account;
      (3) Internet/intranet capable computer; and
      (4) References;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Laboratory.

3. Standard. The Marine Technician Electrical Specialist shall apply detailed knowledge of electronic circuit fundamentals, IAW refs by:

   a. Defining semiconductors;

   b. Applying diode theory;

   c. Recognizing diode circuits;

   d. Identifying special purpose diodes;

   e. Identifying bipolar transistors;

   f. Applying transistor fundamentals;

   g. Recognizing transistor biasing;

   h. Utilizing AC models;

   i. Utilizing voltage amplifiers; and

   j. Utilizing power amplifiers.

4. Time Required.

   a. EO – 3300 minutes; and
   b. CEC – 300 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. TBD by Academic Instructors.

7. Test Details. EO 004.01 will be assessed by EC 004.01 IAW Annex K.

8. References. C39, C40, and TBD by Academic Instructors.

9. Remarks. Laboratory experiments to confirm knowledge.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Introduction to semiconductors							
a. Semiconductor theory b. Doping a semiconductor c. Unbiased diode d. Forward bias e. Reverse bias f. Breakdown g. Energy levels and the energy hill h. Barrier potential and temperature i. The reversed biased diode					IL	C39 C40	Laboratory experiment to confirm knowledge
TP 2 Apply diode theory							
a. Schematic symbol: b. Diode curve c. Forward and reverse regions d. Ideal diode e. Second and third approximation f. Approximation g. Data sheet h. Bulk/diode resistance i. Load line					DP	C39 C40	Laboratory experiment to confirm knowledge
TP 3 Recognize diode circuits							
a. Input transformer b. Half wave rectifier c. Full wave rectifier d. Bridge rectifier e. Capacitor input filter f. Calculating other quantities g. Troubleshoot					IL / DP	C39 C40	Laboratory experiment to confirm knowledge

h. Read data sheet to include (1) fuses (2) real transformers (3) design guidelines; and (4) surge current. i. RC and LC filters j. Voltage multipliers k. Clippers and clampers l. Peak to peak detector and m. DC return					
TP4 Identify special purpose diodes					
a. Zener diode b. Loaded zener regulator c. Optoelectronic devices d. Schottky diode e. Varactor f. Varistors g. Data sheet h. Troubleshooting i. Load lines j. Second approximation k. Ripple zener drop out point l. LED design guidelines m. reverse recovery time n. varactor characteristics o. other diodes				C39 C40	Laboratory experiment to confirm knowledge
TP5 Identify bipolar transistors					
a. unbiased transistor b. biased transistor c. CE connection d. Transistor characteristics e. Transistor approximations f. Data sheets g. Energy viewpoint h. Base curves, cut off and breakdown i. Third approximation j. Base spreading resistance k. Ebers-Moll model l. CB connection				C39 C40	Laboratory experiment to confirm knowledge
TP6 Apply transistor fundamentals					
a. Variations in current gain b. Load line c. Operating point d. Proof of the load line e. Saturation f. Transistor switch g. Transistor current source h. Optoelectronic devices				C39 C40	
TP7 Recognize transistor biasing					

a. Voltage-divider bias b. Voltage-divider analysis c. Voltage-divider load line and Q-point d. Two supply emitter bias e. PNP transistor f. Other types of bias i. Emitter feedback bias ii. Collector feedback bias iii. Collector and emitter feedback bias iv. Voltage divider bias g. Troubleshooting					Laboratory experiment to confirm knowledge
TP8 Utilize AC models					
a. Coupling capacitor b. Bypass capacitor c. Superposition in amplifiers d. Small signal operation e. AC resistance of the emitter diode f. AC beta g. Common emitter amplifier h. AC model of a CE amplifier i. AC quantities on the data sheet j. H parameters					
TP9 Utilize voltage amplifiers					
a. Highlights of a CE amplifier b. Voltage gain c. Predicting voltage gain d. Simplified analysis e. Swamped amplifier f. Cascaded stages g. Output impedance h. Negative feedback i. Cascade stages j. H parameters k. Common base amplifiers l. Common collector amplifier m. Maximum unclipped output n. Cascading CE and CC					Laboratory experiment to confirm knowledge
TP10 Utilize power amplifiers					
a. AC load line b. Limits on signal swing c. Class A operation d. Transistor power rating e. AC saturation and cut off f. AC output compliance g. Thermal resistance h. Class B operation i. Class B power formulae j. Biasing class B amplifiers					Laboratory experiment to support knowledge

k. Class B driver					

PO 004 MAINTAIN FIELD DEVICES
Critical Enabling Check – CEC 004.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 004.01 will be a written assessment based on content from EO 004.01
3. Pass/Fail IAW Annex K.
4. Time required: 300 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 004.01 will be a written assessment based on content from EO 004.01
2. During this written assessment, you will be required to Apply detailed knowledge of electronic circuit fundamentals. You must pass this test to pass EO 004.01. To pass the test, you must Apply detailed knowledge of electronic circuit fundamentals skills in accordance with the standard.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 300 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by the academic instructor.

PO 004 MAINTAIN FIELD DEVICES

EO 004.02 Apply detailed knowledge of electrical circuit fundamentals

1. Performance. Apply detailed knowledge of electrical circuit fundamentals.

2. Conditions.

   a. Given:

      (1) Supervision;
      (2) DND Learn account;
      (3) Internet/intranet capable computer; and
      (4) References.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and
      (2) Laboratory.

3. Standard. The Marine Technician Electrical Specialist shall apply detailed knowledge of electrical circuit fundamentals, IAW refs by:

   a. Identifying field effect transistors (FET);
   b. Utilizing FET circuits;
   c. Recognizing frequency effects;
   d. Applying op-amp theory;
   e. Recognizing negative feedback;
   f. Utilizing linear op-amp circuits;
   g. Utilizing non-linear op-amp circuits;
   h. Designing regulated power supplies; and
   i. Applying oscillatory theory.

4. Time Required.

   a. EO – 3300 minutes; and
   b. CEC – 300 minutes.

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstration – performance.

6. Training Aids.

   a. TBD by Academic Instructors.

7. Test Details. EO 004.02 will be assessed by CEC 004.02 IAW Annex K.

8. References. C41, C42, and TBD by Academic Instructors.

9. Remarks. Laboratory experiments to support knowledge.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Identify field effect transistors							
a. JFET b. Biased JFET c. Drain curves d. Transconductance curve e. JFET approximation f. MOSFET depletion mode g. MOSFET enhancement mode h. Data sheets i. Saturation j. Mathematical derivation					IL	C41 C42	
TP 2 Utilizing FET circuits							
a. Self-bias JFETs b. Graphical solution of self-bias c. Solution with universal JFET curve d. Transconductance e. JFET amplifiers f. JFET analog switch g. Depletion mode MOSFET amplifiers h. Enhancement mode MOSFET amplifiers i. Other JFET biasing 1) Voltage divider biasing 2) Source bias 3) Current source bias j. Output impedance of the source follower k. Other JFET applications					DP	C41 C42	Laboratory experiments to support knowledge
TP 3 Recognizing frequency effects							
a. Frequency response of an amplifier 1) Input coupling capacitor 2) Output coupling capacitor b. Emitter bypass capacitor c. Collector bypass circuit d. Miller's theorem e. High frequency bipolar analysis f. Total frequency response					IL / DP	C41 C42	Laboratory experiment to support knowledge

g.Decibels 1)Decibels voltage gain 2)Voltage gain outside the midband h. Proof of Miller's theorem i. High frequency FET analysis j. dBM k. Power and voltage gain l. Bode plots m. Risetime bandwidth relation n. Stray effects o. Several critical frequencies					
TP 4 Applying op-amp theory					
a. Integrated circuits (ICs) b. Differential amplifier c. Two input characteristics d. AC analysis of a differential amplifier e. Output offset voltage f. Common mode gain g. Current mirror h. Differential amplifier with current mirrors i. Small signal frequency response j. Large signal frequency response k. Power bandwidth l. Operational amplifier (op amp) m. Op amp characteristics n. Popular op amps o. Other linear ICs				C41 C42	Laboratory experiment to support knowledge
TP 5 Recognizing negative feedback					
a. Non-inverting voltage feedback b. Open loop and closed loop voltage gains c. Input and output impedance d. Other benefits of negative feedback e. Inverting voltage feedback f. Bandwidth g. Exact value of feedback fraction h. More on inverting voltage feedback i. Non-inverting current feedback j. Inverting current feedback k. More on closed loop bandwidth				C41 C42	Laboratory experiment to support knowledge
TP 6 Utilizing linear op-amp circuits					
a. Voltage amplifiers 1)Non-inverting amplifiers 2)Inverting amplifiers b. Op amp inverting circuits c. The summing amplifier d. Current boosters e. Voltage controlled current sources				C41 C42	Laboratory experiments to support knowledge

f. Differential and instrument amplifier g. Active filters 1) Automatic gain control 2) Howland current course 3) Other active filters					
TP 7 Utilizing non-linear op-amp circuits					
a. Active diode circuits i. Peak detector ii. Active clipper and clamper b. Comparator c. Schmitt trigger i. Moving the trip points d. The integrator e. The differentiator f. Relaxation oscillator g. Window comparator				C41 C42	Laboratory experiment to support knowledge
TP 8 Designing regulated power supplies					
a. Voltage feedback operation b. Current limiting c. Power supply characteristics d. Three terminal IC regulators e. Current boosters f. DC to DC converter g. Switching regulators i. Foldback limiting				C41 C42	Laboratory experiments to support knowledge
TP 9 Applying oscillatory theory					
a. Theory of sinusoidal oscillation b. The Wien bridge oscillator c. Other RC oscillators d. The Colpitts oscillator e. Other LC oscillators i. Crystal oscillators f. Unwanted oscillators g. Other ways to reduce loopgain $A\beta$ to unity				C41 C42	Laboratory experiments to support knowledge

PO 004 MAINTAIN FIELD DEVICES
Critical Enabling Check – CEC 004.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 004.02 will be a written assessment based on content from EO 004.02
3. Pass/Fail IAW Annex K.
4. Time required: 300 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 004.02 will be a written assessment based on content from EO 004.02
2. During this written assessment, you will be required to Apply detailed knowledge of electrical circuit fundamentals. You must pass this test to pass EO 004.02. To pass the test, you must Apply detailed knowledge of electrical circuit fundamentals skills in accordance with the standard.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 300 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by the academic instructor.

PO 004 MAINTAIN FIELD DEVICES

EO 004.03 Troubleshoot Field Devices

1. Performance. Maintain Field Devices.

2. Conditions.

   a. Given:

      (1) Field devices;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall troubleshoot field devices independently, IAW refs by:

   a. Troubleshoot control sensors to include:

      (1) Temperature sensors;
      (2) Pressure sensors;
      (3) Level sensors;
      (4) Reed switches;
      (5) Limit switches;
      (6) Thyristors;
      (7) Tension transducers;
      (8) Synchro-servos;
      (9) Thermistor;
      (10) Explosive gas detectors;
      (11) Optical sensors;
      (12) Photo sensors;
      (13) Speed sensors;
      (14) Carbon monoxide sensor;
      (15) Vacuum sensors;
      (16) Pyrometer;
      (17) Resistive temperature devices;
      (18) Smoke detector;
      (19) Heat detectors;
      (20) Thermocouples;
      (21) Flow sensor; and
      (22) Leak detector;

(23) Inspecting field devices, to include:
(24) Explosive gas detector;
(25) Speed sensors;
(26) Tachogenerator;
(27) Optical sensors;
(28) Photo sensors;
(29) Propulsion diesel engine enclosure carbon monoxide sensor;
(30) Thermistors;
(31) Tension transducer;
(32) Thermistors;
(33) Rudder angle indicator;
(34) Feedback transmitter; and
(35) Shaft knot indicators;
(36) Troubleshooting control electronics to include, but not limited to:
(37) Inspecting power electronics (AUT1360) by;
i. Examining rectifier firing boards;
ii. Examining voltage phase monitoring cards;
iii. Examining EMI filters and protection; and
iv. Inspecting low voltage power supplies.

4. Time Required.

a. EO – 650 minutes; and
b. EC – assessed during EC 004.04.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. Temperature sensors;
b. Pressure sensors;
c. Level sensors;
d. Reed switches;
e. Limit switches;
f. Thyristors;
g. Tension transducers;
h. Synchro-servos;
i. Thermistor;
j. Explosive gas detectors;
k. Optical sensors;
l. Photo sensors;
m. Speed sensors;
n. Carbon monoxide sensor;
o. Vacuum sensors;
p. Pyrometer;
q. Resistive temperature devices;
r. Smoke detector;
s. Heat detectors;

t. Thermocouples;
u. Flow sensor; and
v. Leak detector.

7. Test Details. EO 004.03 will be assessed by EC 004.04.

8. References. A17, A18, A19, A20, A21, A27, C25, C26, C27, and C45.

9. Remarks. Instructor to have demonstration pieces for each type of field device that can be assembled/disassembled. Troubleshoot Field Devices (ALT1870) replaced with Maintain Field Devices.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Maintain control sensors, to include, but not limited to:							
a.	Describing operation of field devices, to include, but not limited to: (1) Temperature sensors; (2) Pressure sensors; (3) Level sensors; (4) Reed switches; (5) Limit switches; (6) Thyristors; (7) Tension transducers; (8) Synchro-servos; (9) Thermistor; (10) Explosive gas detectors; (11) Optical sensors; (12) Photo sensors; (13) Speed sensors; (14) Carbon monoxide sensor; (15) Vacuum sensors; (16) Pyrometer; (17) Resistive temperature devices; (18) Smoke detector; (19) Heat detectors; (20) Thermocouples; (21) Flow sensor; and (22) Leak detector.	400			IL DP	A17 A18 A19 A20 A21 A27 C25 C26 C27 C45	Instructor to have demonstration model of sensors/detectors available for trainees.
b.	Inspecting field devices, to include, but not limited to: (1) Inspecting rudder angle indicator; (2) Inspecting feedback transmitter; and (3) Inspecting shaft knot indicators.	50			IL DP	A17 A18 A19 A20 A21 A27 C25 C26	Instructor to have demonstration pieces for each type of field device that can be assembled/disassembled.

					C27 C45	
b. Inspecting power electronics by: (1) Examining rectifier firing boards; (2) Examining voltage phase monitoring cards; (3) Examining EMI filters and protection; and (4) Inspecting low voltage power supplies.	200			IL	A17 A18 A19 A20 A21 A27 C25 C26 C27 C45	Instructor to provide demonstration model of rectifier firing board, voltage phase monitoring cards, EMI filters and protection and low voltage power supplies.

PO 004 MAINTAIN FIELD DEVICES

EC 004.03

The content of EO 004.03 will be tested in EC 004.04

PO 004 MAINTAIN FIELD DEVICES

EO 004.04 Repair Control Sensors

1. Performance. Repair Control Sensors.

2. Conditions.

   a. Given:

      (1) Field devices;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall repair control sensors, IAW refs by:

   a. Maintaining rudder angle indicator (BAT0270) by:

      (1) Repairing feedback transmitter (BAT0210); and
      (2) Repairing steering feedback electrical control unit (BAT0290).

4. Time Required.

   a. EO – 375 minutes; and
   b. PC – 50 minutes.

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstration – performance.

6. Training Aids. Visual representation of a Rudder Angle Indicator.

7. Test Details. EO 004.04 shall be assessed in PC 004.

8. References. A38, A40

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Repair control sensors, to include: Maintain rudder angles Indicator by;							
a.	Repairing feedback transmitter; and	150	50		IL DP	A38 A40	
b.	Repairing steering feedback electrical control unit.	100					
c.	Inspecting shaft knot indicators.	75					

PC 004 – Maintain Field Devices

PC Time: 50 mins

ASSESSMENT ACTIVITY: The trainee will have a written assessment on the contents of 4.03 and 4.04, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- PC

IMPACT OF FAILURE: TED / Remedial Training / Re-test / TRB after second unsuccessful attempt

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

EO 005.01 Maintain Generators

1. Performance: Maintain Generators.

2. Conditions:

   a. Given:
      (1) Power generation system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:
      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall maintain generators independently (AUT 4030), IAW refs by:

      (1) Inspecting automatic voltage regulators (AUT0800);
      (2) Inspecting enclosure fans;
      (3) Inspecting enclosure heaters;
      (4) Inspecting terminal block components;
      (5) Inspecting leak detector; and
      (6) Inspecting DECS.

4. Time Required:

   a. EO – 900 minutes; and
   b. EC – 0 minutes

5. Instructional Methods:

   a. Interactive lecture;
   b. Simulation; and
   c. Demonstration – performance.

6. Training Aids: Lab Volt.

7. Test Details: EO 005.01 will be assessed by an in-class activity with feedback provided by instructors.

8. References: A28 and A34.

9. Remarks: Task Troubleshooting primary power generation (CT0440) is covered within task Troubleshoot generators (AUT 4030). Moved 600 minutes from PO 005 PC into Instruction of 5.01 to facilitate lab activities. PC will remain at 300 minutes total.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Troubleshoot Power Generation by:							
a.	Inspecting automatic voltage regulators, to include, but not limited to: (1) Rotating rectifier; (2) Main exciter; (3) Pilot exciter; (4) PCB regulator to include: i. Power supply; ii. Under-speed protection; and iii. Regulator circuit. (5) PCB firing unit; (6) Fault logic system; (7) Active and reactive load sharing; (8) No load and short circuit conditions; and (9) Limitation of output voltage.	400			IL	A28 A34	
b.	Troubleshooting generators, to include, but not limited to: (1) Generator excitation; (2) Generator under load; (3) Regulation of output voltage; (4) De-excitation; and (5) Function of the control loop.	50	150	300	SIM DP IL	A28 A34	Voltage regulation through Lab Volt experiments

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

EC 005.01

The content of EO 005.01 will be assessed by an in-class activity with feedback provided by instructors.

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

EO 005.02 Maintain Motor Alternator

1. Performance. Repair Motor Alternators

2. Conditions.

   a. Given:

      (1) Power generation system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall repair Motor Alternators independently, IAW refs by:

   a. Troubleshooting motor alternator sets (AUT4050).

4. Time Required:

   a. EO – 100 minutes; and
   b. EC – 0 minutes.

5. Instructional Methods:

   a. Simulation; and
   b. Demonstration – performance.

6. Training Aids: Nil.

7. Test Details: EO 005.02 will be assessed by an in-class activity with feedback provided by the instructor.

8. References: A28, and A34.

9. Remarks: Instructor to provide the trainee with fault, trainee to solve using troubleshooting guide. Assessment will be done as an in-class activity with feedback given by the instructor. Removed - Repairing automatic voltage regulators (AUT2000) by:

a. Applying voltage regulation theory;
b. Repairing automatic voltage regulators;
c. Adjusting automatic voltage regulators (AUT0100); and
d. Assembling/disassembling automatic voltage regulators (AUT0320).
e. Covered in Maintain in EO 5.01

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Maintain Motor Alternators by:							
a. Troubleshooting motor alternator sets, to include: (1) Review motor theory; (2) Review alternator theory; and b. Purpose of MA.		100			IL	A28 A34	Review to consist of high level overview motor alternator theory.

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

EC 005.02

The content of EO 005.02 will be assessed by an in-class activity with feedback provided by instructors.

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

EO 005.03 Test Primary Power Generation Systems

1. Performance. Test Primary Power Generation Systems.

2. Conditions.

   a. Given:

      (1) Power generation system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall test primary power generation systems independently, IAW refs by:

   a. Testing generators to include:

      (1) Measuring electrical parameters on generator windings (AUT1900); and
      (2) Testing generator main field windings (AUT3380);

   b. Testing voltage regulation (AUT3670) to include:
      (1) Test voltage regulation by building and troubleshooting a Power generation and Distribution Circuit – Lab Volt.

4. Time Required:

   a. EO – 1050 minutes; and
   b. PC – 300 minutes.

5. Instructional Methods. Interactive Lecture.

6. Training Aids. Nil.

7. Test Details. EO 005.03 will be assessed by EC 005.03 IAW Annex K.

8. References. A28, and A34.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Test generator, to include, but not limited to:							
a.	Measuring electrical parameters on generator windings by: (1) Testing generator field windings to include: i. Testing continuity; ii. Testing resistance; and iii. Testing thermistors.	25			IL	A28 A34	
TP 2 Test voltage regulation by:							
a.	Build and troubleshoot a Power generation and Distribution Circuit – Lab Volt	25		1000	IL	A28 A34	

PO 005 MAINTAIN PRIMARY POWER GENERATION SYSTEMS

PC 005

PC Time: 25 mins per trainee

ASSESSMENT ACTIVITY: The trainee will have a practical assessment it will be performed on the switchboard circuit to be built in Lab-Volt. The trainee will troubleshoot a single fault in the circuit. The trainee will be role-playing as the on-watch technician. The trainee will use relevant schematics and technical manuals to troubleshoot and identify the potential fault(s) that are occurring and discuss his/her trouble-shooting methodology and thought process with the instructor. An example of an PC would be as follows: The trainee receives a notification that the generator cct. has a fault. Utilizing drawings and technical manuals and their understanding of troubleshooting techniques, the trainee will troubleshoot the system and identify a list of potential faults to the instructor. The trainee will then identify means of eliminating the potential sources systematically to arrive at the correct diagnosis. Employing critical thinking skills, the trainee will develop a plan to repair and test the system. The instructor will have the opportunity to ask probing questions and fully develop the trainee's solution.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- Trainer
- Schematic / System Block Diagram
- Multi-Meter

IMPACT OF FAILURE: TED / Remedial Training / Re-test / TRB after a second unsuccessful attempt

PERFORMANCE CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely complete practical assessment checklist.

Administrative Instructions:

1. This practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from PO 005.

2. **Standard: The Marine Technician Electrical Specialist shall independently demonstrate the ability to Maintain Primary Power Generation Systems.**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

1. Ensure the trainee is given an opportunity for the maintenance routine to progress to its conclusion;

2. If the trainee experiences difficulty, provide minimal assistance to maintain the standard through PC;

3. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either;

a. Paused and corrective action taken before continuing, or halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

1. Various Components;
2. PPE (glasses, gloves, ear protection);
3. Test Tools if required
4. Supervision.
5. Instructor to pre-build the PC circuit in Lab-Volt

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Familiarization with Circuit (5 MINS) Time:			
Confirm circuit operation			Circuit must be function tested
Observe safety at all times			Failure to follow safe procedures constitutes an automatic fail
Instructor comments:			
1st FAULT (20 MINS) Time 1st Fault:			
Student picks faults from envelope, instructor inserts the fault, no guidance from instructor.			
Fault chosen:	Fault details (exact location of fault in circuit):		

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Test circuit to observe fault			
Test meter			
Use meter correctly			
Explain logic behind chosen troubleshooting technique			
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Prove faults corrected			
Instructor comments:			

Overall Assessment	**Result** **(instructor to circle final grade)** **PASS / FAIL**

To be completed by Instructor / Invigilator		
Strengths:		Area of improvement:
Date:	**Trainee's Signature:**	**Instructor's Signature:**

PO 006 MAINTAIN ELECTRICAL DISTRIBUTION SYSTEM

EO 006.01 Troubleshoot Electrical Distribution Systems

1. Performance. Troubleshoot Electrical Distribution Systems.

2. Conditions.

   a. Given:

      (1) Electrical distribution system;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall troubleshoot electrical distribution systems independently, IAW refs by:

   a. Troubleshooting power generation and distribution system (CT0440) to include:

      (1) Troubleshooting electrical switchboard control (ALT1910);
      (2) Inspecting synchroscope (AUT1510);
      (3) Inspecting electrical switchboard transducers (AUT1080);
      (4) Inspecting switchboard Terminal box (AUT1490);
      (5) Inspecting electrical switchboard breakers (AUT1060) to include, but not limited to:
         (a) Inspecting 440 VAC distribution breakers (AUT0750).

      (6) Examining switchgear components to include, but not limited to:
         (a) Inspecting 440 VAC distribution panels (AUT0760);
         (b) Inspecting backplane (AUT0810);
         (c) Inspecting bus bars (AUT0850); and
         (d) Inspecting electrical disconnect links (AUT1020).

   b. Troubleshooting bus transfer switches by inspecting switching unit auto (SABT) (AUT1500).

(1) Time Required.

c. EO – 600 minutes; and
d. EC – 0 minutes.

4. Instructional Methods.

a. Interactive lecture;
b. Field trip; and
c. Demonstration – performance.

5. Training Aids.

a. Power transducer;
b. Linear transducer;
c. Rotary transducer;
d. MCOS;
e. ACOS; and
f. SABT.

6. Test Details. EO 006.01 will be assessed by an in-class activity with feedback provided by instructors.

7. References. A29, A30, A31, A32, A35, A36 and C48.

8. Remarks. Students will complete an in-class quiz during EO 006.01.

9. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Troubleshoot power generation and distribution system, to include:							
a. Troubleshooting electric switchboard control (ECS) by: (1) Examining switchboard control components, to include, but not limited to: i. Associated I/O cards ii. Surge protectors iii. Differential Current protection		30	50		IL DP	A29 A30 A31 A32 A35 A36 C48	SIM – virtual environment consisting of different switchboards from different platforms.
b. Inspecting synchroscope by:		10	25		IL DP		

(1) Examining synchroscope components and operation; and (2) Utilizing synchroscopes to troubleshoot paralleling abnormalities.						
c. Inspecting electrical switchboard transducers by: (1) Examining power transducers; and (2) Examining linear transducers; (3) Examining rotary transducers; and (4) Analyzing application of switchgear protection.	25	75		IL DP FT		
d. Inspecting switchboard tangle box by: (1) Examining connections between generators and switchboards.	25			IL		SIM – virtual environment consisting of different switchboards from different platforms.
e. Inspecting electrical switchboard breakers by: (1) Disassemble moulded circuit breakers; and (2) Disassemble air circuit breakers, to include: i. Inspecting 440 VAC distribution breakers.	20	40	120	IL DP	A36	Assembly of CCT breakers to be included in EO 006.02. To include high voltage (1000V+).
f. Examining switchgear components to include but not limited to: (1) Inspecting bus-bars; and (2) Inspecting electrical disconnect links.		100		FT		Field trip to various classes of ship, to show differences in switchgear components and construction.
TP2 Troubleshoot bus transfer switches by:						
a. Disassembling components of bus transfer switches.	20	100		IL DP	A35 A36	To include, but not limited to, ACOS, MCOS & SABT.
b. Inspecting switching unit auto (SABT) by: (1) Reviewing SABT technical safety bulletin.	10			IL	A32	

PO 006 MAINTAIN ELECTRICAL DISTRIBUTION SYSTEM

EC 006.01

The content of EO 006.01 will be assessed by an in-class activity with feedback provided by instructors.

PO 006 MAINTAIN ELECTRICAL DISTRIBUTION SYSTEM

EO 006.02 Maintain Electrical Distribution Systems

1. Performance. Maintain Electrical Distribution Systems.

2. Conditions.

   a. Given:

      (1) Electrical Distribution System;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall repair electrical distribution systems independently, IAW refs by:

   a. Repairing power generation and distribution systems to include:

      (1) Calibrating power generation control system (AUT0580); and
      (2) Calibrating electrical switchboard parameters (AUT0520);
      (3) Testing bus bars (AUT3060) by:
         (a) testing electrical disconnect links (AUT3260).

   b. Repairing breakers (AUT2040) by:

      (1) Adjusting circuit (CCT) breakers (AUT0130); and
      (2) Repairing electrical switchboard breakers (AUT2270);
      (3) Testing electrical switchboard breakers (AUT3300);
      (4) Testing breakers (AUT3050); and
      (5) Testing 440 VAC distribution panels (AUT2990).

   c. Repairing bus transfer switches by:

      (1) Adjusting the automatic change-over switching unit (AUT0080); by

(a) Calibrating automatic change-over switching unit (AUT0430).

(2) Adjusting switching unit auto (SABT) (AUT0260);

(3) Adjusting manual change-over switching unit (AUT0170); and

(a) Calibrating manual change-over switching unit (AUT0550).

(4) Assembling transfer switches by:

(a) Repairing switching unit auto (AUT2710);

(b) Repairing switching unit auto (SABT) (AUT2720);

(c) Repairing switching unit manual (AUT2730); and

(d) Testing switching unit auto (AUT3750);

(e) Testing switching unit auto (SABT) (AUT 3760); and

(f) Testing switching unit manual (AUT3770).

4. Time Required.

a. EO – 600 minutes;
b. PC – 50 minutes.

5. Instructional Methods:

a. Interactive lecture;
b. Demonstration – performance.

6. Training Aids:

a. Power transducer;
b. Linear transducer;
c. Rotary transducer;
d. Moulded circuit breaker;
e. Air circuit breaker;
f. MCOS;
g. ACOS and
h. SABT.

7. Test Details: EO 006.01 and 006.02 will be assessed by a written PC.

8. References: A29, A30, A31, A32, A35, A36 and C48.

9. Remarks: Students will complete an in-class quiz during EO 006.02.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)	Method	Refs	Comments

	T	D	P			
TP1 Repairing power generation and distribution systems by:						
a. Calibrating electrical switchboard parameters to include but not limited to: (1) Adjusting set points IAW REFs including but not limited to: i. Load/no load testing; ii. Paralleling conditions; and iii. Setting up droop.	50			IL	A29 A30 A31 A32 A35 A36 C48	RF guest speaker to explain to trainees the concepts of no-load, full-load and droop, and how to adjust them. (Shadowing RF)
TP 2 Repair Circuit Breakers by:						
a. Repairing breakers, to include: (1) Disassembly and reassembly, calibration, and testing, to include, but not limited to: i. Adjusting circuit (cct) breakers; ii. Repairing electrical switchboard breakers;	150	200		IL DP		Requires breakers identified in EO 006.01.
TP 3 Repair bus transfer units by:						
a. Adjusting automatic change-over switching unit by: (1) Examining component parts of ACOS and how they interact.	25			IL DP	A29 A30 A31 A32 A35 A36 C48	Instructor to discuss the different faults which may arise.
b. Adjusting switching unit auto (SABT) by: (1) Examining component parts of SABT and how they interact.	25			IL DP		
c. Adjusting manual change-over switching unit by: (1) Examining component parts of MCOS and how they interact.	25			IL DP		
d. Assembling transfer switches by: (1) Conducing corrective maintenance by: i. Repairing switching unit auto; ii. Repairing switching unit auto (SABT); iii. Repairing switching unit manual		200		DP		Instructor to introduce random faults into units provided to trainees.

TP 4 Test power generation and distribution systems by:					
a. Testing bus bars by: i. Testing electrical disconnect: and ii. Assisting with load banking.	50		IL	A36	RF guest speaker to explain to trainees the concepts of no-load, full-load and droop, and how to adjust them. (Shadowing RF)
TP 5 Test Circuit Breakers by:					
a. Testing breakers, to include: i. Testing electrical switchboard breakers; and ii. Testing 440 VAC distribution panel.	50		IL DP	A29 A30 A31 A32 A35 A36 C48	Requires breakers identified in EO 006.01.
TP 6 Repair bus transfer units by:					
a. Testing transfer switches by: i. Testing switching unit auto; ii. Testing switching unit auto (SABT); iii. Testing switching unit manual.	50	50	DP	A37 A38 A32	Instructor to describe common faults into units provided to trainees.

PO 006 MAINTAIN ELECTRICAL DISTRIBUTION SYSTEM

EC 006.02

The content of EO 006.02 will be assessed by an in-class activity with feedback provided by instructors.

PO 006 MAINTAIN ELECTRICAL DISTRIBUTION SYSTEM
PC 006

PC Time: 50 minutes

ASSESSMENT ACTIVITY: To assess the trainees' ability to maintain electrical distribution systems they will perform a written assessment on the contents of PO 006, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- PC
- PC Student Package

IMPACT OF FAILURE: TED / Remedial Training / Re-test / TRB after second unsuccessful attempt

PO 007 SERVICE FLUID POWER SYSTEMS

EO 007.01 Apply fluid power theory

1. Performance. Apply fluid power theory.

2. Conditions.

   a. Given:

      (1) References;
      (2) Internet capable computer; and
      (3) Academic Instructor;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall apply fluid power theory independently, IAW refs by:

   a. Defining units;
   b. Listing basic fluid properties;
   c. Differentiating viscosity of fluid;
   d. Computing pressure measurements;
   e. Relating the relationship between pressure and elevation;
   f. Comparing flow of fluids;
   g. Analysing conservation of energy;
   h. Appling general energy equations;
   i. Classifying pumps and motors;
   j. Explaining Reynolds number, laminar flow and turbulent flow;
   k. Introducing fluid power;
   l. Describing hydraulic fluids;
   m. Utilizing directional control valves;
   n. Inspecting flow control;
   o. Inspecting pressure control;
   p. Describing the function of accumulators;
   q. Describing the function of actuators;
   r. Indicating hydraulic colouring;
   s. Interpreting hydraulic circuits; and
   t. Troubleshooting hydraulics.

4. Time Required.

   a. EO – 4600 minutes; and
   b. CEC – 600 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. TBD by Academic Instructors.

7. Test Details. EO 007.01 will be tested in CEC 007.01.

8. References. C50, C51, and TBD by Academic Instructors.

9. Remarks. Nil

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Define units to include:							
a. SI units b. English gravitational system c. Metric prefixes d. Consistency in units e. Force f. Acceleration due to gravity g. Weight h. Cancellation procedures						C50 C51	
TP 2 List basic fluid properties to include:							
a. Compressibility b. Mass, weight and density c. Specific weight d. Class examples e. Viscosity of fluids.						C50 C51	
TP 3 Compute pressure measurements							
a. Definition of pressure b. Pascal's law c. How pressure is created d. Hydraulic pwer transmission e. Absolute and guage pressure In SI and English						C50 C51	
TP 4 Relate the relationship between pressure and elevation							
a. Fluid flow rate (1) Q=Av (2) W=Y(gamma) Q (3) M= (rho) Q						C50 C51	

iiii. Continuity equation					
TP 5 Analyze conservation of energy					
a. Forms of energy in a field b. Bernoulli's equation c. Restrictions on Bernoulli's equations d. Application of Bernoulli's equation e. Torricelli's theorem				C50 C51	
TP 6 Apply general energy equations					
a. Nomenclature of energy losses and additions b. General energy equation. c. Forms of energy in a field.				C50 C51	
TP 7 Classify pumps and motors					
a. Mechanical efficiency of pumps b. Power delivered to motors c. Mechanical efficiency of				C50 C51	
TP 8 Introduce fluid power					
a. Hydraulic Safety b. Basic hydraulic theory c. Basic hydraulic circuits d. ISO and ANSI symbols e. Series and parallel circuits f. Forces/pressure relationship g. Speed flow relationship h. Advantages and disadvantages of fluid power				C50 C51	Laboratory experiments to support knowledge
TP 9 Describe hydraulic fluids					
a. Fluid types b. Fluid properties c. Dynamic and kinematics viscosity d. Viscosity index e. Viscosity measurements methods f. Rotating drum viscometer g. Capillary tube h. Saybolt universal viscometer i. SAE Viscosity measurement/ Viscosity tables j. Fluid conductors (1) Types (2) Advantages (3) Fittings				C50 C51	

k. Pressure and flow measurement l. Contamination detection and control m. Operation of pressure gauge and flow meter n. Seals and leakage o. Fluid conditioners (1) Filters (2) Heaters (3) Coolers (4) Reservoirs.					
TP 10 Hydraulic pumps					
a. Positive and non-positive displacement b. Construction and accessories c. Vacuum d. Gear, vane and piston pumps e. Pressure compensation and load sensing f. High/low volume circuit and closed loop circuits				C50 C51	
TP 11 Utilize directional control valves					
a. check valves b. pilot pressure and pilot operated check valves c. shuttle valves d. DCV centres and control methods e. Rotary and spool type DCVs f. Solenoid controlled, pilot operated DCV g. Proportional\infinite position DCVs.				C50 C51	Laboratory experiments to support knowledge
TP 12 Understand flow control					
a. Relationship between (1) Flow (2) Speed (3) Pressure drop. b. Basic restrictor type of flow control valve(FCV) c. Meter in and meter out, bleed off d. Pressure compensated FCV e. Deceleration valves				C50 C51	

TP 13 Understand pressure control					
a. Introduction to the six main pressure control valves b. Relief valves (1) Pressure override (2) Compound relief valve (3) Venting c. Unloading valve d. Sequence valve e. Counterbalance valve f. Reducing valve g. Brake valve				C50 C51	Laboratory experiments to support knowledge
TP 14 Describe the function of accumulators					
a. Accumulator types (1) Weight (2) Spring (3) Gas b. Gas laws c. Accumulator circuits (1) Leakage compensation (2) Emergency supply Shock absorber.				C50 C51	Laboratory experiments to support knowledge
TP 15 Describe the function of actuators					
a. Linear actuators b. Rotary actuators including motors c. Actuator accessories (1) Cushioning (2) Mounts d. Intensifier e. Limited rotary actuator (torque generator) f. Pressure compensation.				C50 C51	Laboratory experiments to support knowledge
TP 16 Interpret hydraulic circuits					
a. Introduction to Theory of Hydraulic troubleshooting; and b. Troubleshoot various Hydraulic Circuits using drawings.				C50 C51	Laboratory experiments to support knowledge
TP 17 Troubleshoot hydraulics					
a. Introduction to practical hydraulic troubleshooting procedure				C50 C51	

PO 007 APPLY FLUID POWER THEORY

Critical Enabling Check – EO 007.01

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 007.01 will be a written assessment based on content from EO 007.01
3. Pass/Fail IAW Annex K.
4. Time required: 600 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 007.01 will be a written assessment based on content from EO 007.01
2. During this written assessment, you will be required to apply fluid power theory. You must pass this test to pass EO 007.01.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 600 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by the academic instructor.

PO 007 SERVICE FLUID POWER SYSTEMS

EO 007.02 Apply detailed knowledge of pneumatic systems and electrically operated and controlled fluid power

1. Performance. Apply detailed knowledge of pneumatic systems and electrically operated and controlled fluid power.

2. Conditions.

   a. Given:

      (1) References;
      (2) Internet capable computer; and
      (3) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall apply a detailed knowledge of pneumatic systems and electrically operated and controlled fluid power independently, IAW refs by:

   a. Describing pneumatics;
   b. Advanced control of hydraulic pumps;
   c. Electro-hydraulic control of fluid power systems;
   d. Electro-hydraulic servo systems.
   e. Proportional Valves;
   f. Electro-hydraulic servo valves;
   g. Hydrostatic transmissions;
   h. Examples of the types of electrical control used on CPF fluid power systems;
   i. Electrical/ Electronic monitoring and control of fluid power systems;
   j. Electro-hydraulic troubleshooting procedures; and
   k. Practical Identification of electro-hydraulic system problems.

4. Time Required.

   a. EO – 4500 minutes; and
   b. CEC – 1200 minutes.

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstration – performance.

6. Training Aids.

a. TBD by Academic Instructors.

7. Test Details. EO 007.02 will be assessed by CEC 007.02 IAW Annex K.

8. References. C51, C52, C53, and TBD by Academic Instructors.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Describe pneumatics						
a. Differences/ advantages/ disadvantages of pneumatic systems compared to hydraulic systems b. Air compression c. Gas laws d. Compressor types e. Moisture in air f. Cooling methods g. Filters, regulators and lubricators h. Quick air exhaust circuit					C51 C52 C53	Laboratory experiment to support knowledge
TP 2 Advanced control of hydraulic pumps						
a. Meter in speed control b. Bleed off speed control c. Pressure compensated flow control d. Open centre systems e. Pressure compensated pump operation f. Load sensing pump operation g. Analysis of hydraulic system containing a load sensing pump and infinite positioning pressure compensated directional control valves.					C51 C52 C53	
TP 3 Define electro-hydraulic systems						
a. Ladder diagrams b. Relays, limit switches c. Fluid power circuits controlled by ladder logic/ diagrams					C51 C52 C53	Lab experiment to support knowledge.
TP 4 Proportional valves						
a. Hydraulic Joysticks b. Safety Issues c. Proportional solenoid construction d. Proportional DCV examples					C51 C52 C53	

e. Spool feedback – LVDT (PP11) f. Hydrostatic Module (Example on C-rast system) g. Proportional relief valve h. Proportional flow control i. Proportional reducing valve					
TP 5 Electro-hydraulic servo systems					
a. Positional System b. Velocity System c. Force/ Pressure System d. Flapper Design Servo Valve e. Symbols f. Operation g. Amplifier Adjustments				C51 C52 C53	
TP 6 Hydrostatic Transmissions					
a. Component Functions b. Displacement Control Options (1) Manual Control (2) Remote Control (3) Electro-Hydraulic control				C51 C52 C53	Laboratory experiments to support knowledge
TP 7 Introduction to electro-hydraulic control of a CPF System					
a. C-RAST overview (1) System Schematic overview (2) Indicate the location of hydrostatic transmissions (3) Indicate position for proportional DCV (4) Indicate the position of Servo Valves b. Discuss the function of components (1) Bi-directional Pumps (2) Servo Valves (3) Proportional DCV (4) Accumulator (5) Transmission Circuits (6) Brake Valve				C51 C52 C53	
TP 8 Introduction to system control and monitoring					
a. Describe the monitoring of an electrically controlled hydraulic system using a programmable logic controller.				C51 C52 C53	Lab experiment to support knowledge

TP 9 Troubleshoot electro-hydraulic systems					
a. Introduction of the correct method of troubleshooting electro-hydraulic circuits using schematics and simulated faults				C51 C52 C53	Laboratory experiments to support knowledge

PO 007 APPLY FLUID POWER THEORY
Critical Enabling Check – EO 007.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 007.02 will be a written assessment based on content from EO 007.02
3. Pass/Fail IAW Annex K.
4. Time required: 1200 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 007.02 will be a written assessment based on content from EO 007.02
2. During this written assessment, you will be required to apply fluid power theory. You must pass this test to pass EO 007.02.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 1200 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by the academic instructor.

PO 007 SERVICE FLUID POWER SYSTEMS

EO 007.03 Troubleshoot Electro-Hydraulics

1. Performance: Troubleshoot Electro-Hydraulics.

2. Conditions:

   a. Given:

      (1) Electro-Hydraulic system;
      (2) PPE;
      (3) Testing equipment and applicable tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall troubleshoot hydraulics independently, IAW refs by:

   a. Troubleshooting steering control system (BAT0390) by:

      (1) Inspecting electronic steering system (BAT0100);
      (2) Inspecting helm unit (BAT0130);
      (3) Troubleshooting autopilot system (BAT0380); and
      (4) Inspecting auto-steering unit (BAT0060).

   b. Repairing Steering control systems:

      (1) Repairing auto-steering unit (BAT0180); and
      (2) Repairing helm units (BAT0220).

   c. Maintaining Control Electronics to include, but not limited to:

      (1) Repairing power supplies (AUT2590):
      (2) Repairing power electronics (AUT2570);
      (3) Calibrating power electronics (AUT0570);

(4) Calibrating automatic control systems (AUT0440); and
(5) Testing power electronics (AUT3580) by;
(6) Testing amplifiers (AUT3010);
(7) Testing diodes (AUT3220);
(8) Troubleshooting engine control systems (ALT1920).
(9) Inspecting hydraulic controls (BAT0140);
(10) Inspecting hydraulic directional control valves (AQT0030);
(11) Repairing hydraulic steering controls (BAT 0240);
(12) Repairing hydraulic power unit (AKT1840); and
(13) Testing hydraulic steering control (BAT0350).

4. Time Required.

a. EO – 850 minutes;
b. EC - 400 minutes; and
c. PC – 300 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. Hydraulics trainer.

7. Test Details. EO 007.03 will be assessed by PC 007 Part 1.

8. References. A8, A13, A29, A30, C37, C54, C56, and C57.

9. Remarks. Requires additional hydraulic upgrade (4x4 vv, servo vv) to include a closed loop feedback to replicate F.U. steering control.

10. Teaching Points.

Teaching Point	Instructional Technique (Time in min)			Method	Refs	Comments
	T	D	P			
TP1 Troubleshoot control electronics by:						
a. Troubleshooting steering control system by:	600				A8 A13 A29 A30	

(1) Inspecting electronic steering system to include but limited to: (a) Theory of Operation; (b) Helm/rudder interface; and (c) Operation of azimuth drives. (2) Inspecting helm unit to include but not limited to: (a) Signaling of requested Rudder; (b) Signaling of feedback to rudder/ Z-drive angle indicators; and (c) Plug and Play components. (3) Troubleshooting auto pilot system (Halifax Class): (a) Rudder feedback signaling (synchro servers). (4) Inspecting Auto steering unit (MCDV): (a) Z-Drive Position signaling				A36 A57 C37 C54 C56 C57	
TP2 Repairing Steering Control System					
a. Repairing Steering Control Systems (1) Repairing auto steering unit by: (a) Examining INS interface Circuits; (b) Inspecting backplane; (c) Circuit component cards to include but not limited to: i. Microprocessor cards; ii. 4 Channel D/A cards; iii. Paramour cards; iv. Input interface cards; v. Relay cards; vi. Watchdog cards; and vii. 2 Channel S/D cards.					
TP 3 Repairing the Helm Unit					

a. Repairing Helm Units By; (1) Replacing the Helm Unit b. Repairing power supplies: (1) Repairing power electronics; (2) Calibrating power electronics; (3) Calibrating automatic control systems; and (4) Testing power electronics by; (a) Testing amplifiers; and (b) Testing diodes.					
TP 4 Gather Resources					
a. Adhering to proper lock-out/tag-out procedures; b. Mustering tools required; c. Mustering appropriate PPE; d. Utilizing appropriate documentation; and e. Reviewing all Safety and Environmental concerns with regard to hydraulic systems.	25	25			
TP 5 Inspecting hydraulic controls, to include, but not limited to:					
a. Valves: (1) Recognizing flow-control markings on physical valves. b. Pumps; c. Motors; d. Cylinders; e. Piping; f. Actuators; g. Accumulators; h. Filters; i. Coolers; j. Reliefs; k. Safety controls and interlocks; l. Remote controls; m. Pilot-operated piggyback valves; and n. Recognizing symptoms of hydraulic faults.	100	200			TEL resources in the form of working animations (3-D cutaways with real-time flow) are to be used in support of TPs 1-4. e.g. pump animations, valve operation, DCV spool operation, solenoid-activation, and piggy-back valve inter-relationship Instructor to describe operating characteristics, benefits and disadvantages of different

					hydraulic component types in different operations.
TP 6 Inspecting hydraulic directional control valves and hydraulic actuating-control, to include, but not limited to:					
a. Multi-position, multi-direction control valves; b. Manually-operated; c. Remotely-operated, to include, but not limited to: (5) Electrically; (6) Hydraulically; and (7) Electro-hydraulic. d. Adjusting proportional integral derivative controller (AUT0230); (1) Calibrating power distribution control systems; (2) Adjusting proportional integral derivative controller	100	300			
TP 7 Repairing hydraulic steering controls by:					
a. Discussing unique characteristics of the steering controls.	50	100			Instructor to introduce the concept of hydraulic feedback.
TP 8 Repair hydraulic power unit by:					
a. Auto-centering b. Testing hydraulic steering control (BAT0350)	50				Instructors to emphasize the differences between the hydraulic power units, focussing on the rudder centering unit.

PO 007 Service Fluid Power Systems

EC 007.03

The content of EO 007.03 will be assessed by an in-class activity with feedback provided by instructors.

PO 007 SERVICE FLUID POWER SYSTEMS

EO 007.04 Troubleshoot Electro-Pneumatics

1. Performance: Troubleshoot Electro-Pneumatics.

2. Conditions:

   a. Given:

      (1) Electro-Pneumatic system;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall troubleshoot Electro-pneumatics independently, IAW refs by:

   a. Troubleshooting HVAC system (AOT1940) to include:
   b. Inspecting smoke evacuation system components (AOT0630); and
   c. Inspecting electro-pneumatic valves (ATT0370).
   d. Troubleshooting propulsion diesel engine low pressure control air (ALT2200);
   e. Assembling and disassembling electro pneumatic valves (ATT0040) by:
      (1) Installing HVAC/NBCD air distribution components (AOT0720); and
      (2) Adjusting smoke evacuation system components (AOT0080).
   f. Assembling and disassembling pneumatic control system components (AKT0180) by:
      (1) Installing electro-pneumatic valves (ATT0630);
      (2) Repairing pneumatic-operated valves (AKT2150);
      (3) Repairing electro-pneumatic valves (ATT0970);
      (4) Repairing variable-position pneumatic-controlled valves (AKT2550); and
      (5) Repairing pneumatic actuators (AKT2110);
   g. Trialing HVAC/NBCD air distribution systems (AOT1790) by:

(1) Testing pneumatic-control components (ALT1660); and

(2) Testing electro-pneumatic valves (ATT1170).

4. Time Required.

a. EO – 550 minutes;
b. EC – 400 minutes; and
c. PC – 300 minutes (50 min per student x 12).

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids. Pneumatic trainer.

7. Test Details. EO 007.04 will be assessed by PC 007 part 2.

8. References. A8, A43 and C35, C41, C42, C43,C44, C45 and C55.

9. Remarks. Nil.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP 1 Troubleshooting HVAC system by:							
a. Inspecting smoke evacuation system components (AOT0630); and b. Inspecting electro-pneumatic valves (ATT0370).		25	100		IL DP	C43	Instructor to discuss the differences with regards to working fluids (hydraulic vs. pneumatic). Requires a pneumatic trainer.
TP 2 Troubleshooting propulsion diesel engine low pressure control air, to include, but not limited to:							
a. Troubleshooting gearbox pneumatic control system to include, but not limited to: (1) Clutch; and		25	100		IL DP	C43	Trainees to learn about pneumatic actuators.

(2) Governor.					
TP 3 Assemble and disassemble electro pneumatic valves by					
a. Installing HVAC/NBCD air distribution components to include, but not limited to: (1) Indications; (2) Pilot valves; (3) Actuators; and (4) Louvres. b. Adjusting smoke evacuation system components.	50	150	IL DP	C43	
TP 4 Assembling and disassembling pneumatic control system components					
a. Installing electro-pneumatic valves; b. Repairing pneumatic-operated valves; c. Repairing electro-pneumatic valves; d. Repairing variable-position pneumatic-controlled valves; and e. Repairing pneumatic actuators. f. Repairing pneumatic motors.	50	150	IL DP	C43	Trainees to participate in training lab to confirm knowledge.
TP 5 Trial HVAC/NBCD air distribution systems by					
a. Testing pneumatic-control components to include, but not limited to: (1) Actuators; (2) Valves; (3) Dampers; b. Testing electro-pneumatic valves to include, but not limited to: (1) Smoke-control.	50	50	IL FT	C43	Field trip will be conducted in concert with hydraulic field trip.

PO 007 SERVICE FLUID POWER SYSTEMS

EC 007.04

The content of EO 007.04 will be assessed by an in-class activity with feedback provided by instructors.

PERFORMANCE CHECK - PO 007

PC 007

PC Time: 50 mins per trainee to be completed in 2, 25 minute parts.

ASSESSMENT ACTIVITY: The trainee will complete hydraulic instruction then complete PC part 1 sequentially. The trainee will then complete the pneumatic instruction and PC part 2 sequentially. The average of the 2 part PC will be their overall PO 007 Service Fluid Power Systems mark.

The trainee will proceed to repair the fault(s) and confirm proper functionality of the system with the instructor present. An example of a PC would be as follows: The trainee receives a notification that the hydraulic/pneumatic system has a fault. Utilizing drawings and technical manuals, along with their understanding of troubleshooting techniques, the trainee will troubleshoot the hydraulic system and identify a list of potential faults to the instructor. The trainee will identify means of eliminating the potential sources, progressively honing their hypothesis and subsequently rectifying the discovered fault(s). The instructor will have the opportunity to ask probing questions and fully develop the trainee's solution.

Pass/fail criteria: Passing 60% and SAT on all critical items.

ASSESSMENT RESOURCES:

- Hydraulic Trainer
- Pneumatic Trainer
- Schematic
- Multi-Meter

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt. Each part of the PC will have a 60% minimum requirement to pass PO 007.

PERFORMANCE CHECKLIST			
Date:	SN/PRI:	Rank:	Name:
TASKS: Successfully and safely complete practical assessment checklist.			

Administrative Instructions:

1. This practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from PO 007.

2. **Standard: The Marine Technician Electrical Specialist shall independently demonstrate the ability to Maintain Electro- Hydraulics and Pneumatics.**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment the following should be considered;

3. Ensure the trainee is given an opportunity for the maintenance routine to progress to its conclusion;
4. If the trainee experiences difficulty, provide minimal assistance to maintain the standard through PC;
5. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either;
   a. Paused and corrective action taken before continuing, or halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

6. Various Components;
7. PPE (glasses, gloves, ear protection);
8. Test Tools if required
9. Supervision.
10. Instructor to pre-build the PC circuit in the Hydraulics lab Part 1 and Pneumatics Lab for Part 2

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Familiarization with Circuit (5 MINS) Time:			
Confirm circuit operation			Circuit must be function tested
Observe safety at all times			Failure to follow safe procedures constitutes an automatic fail
Instructor comments:			

Hydraulic FAULT (25 MINS) Time 1st Fault:

Student picks faults from envelope, instructor inserts the fault, no guidance from instructor.

Fault chosen:	Fault details (exact location of fault in circuit):

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Test circuit to observe fault			

Test meter			
Use meter correctly			
Explain logic behind chosen troubleshooting technique			
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail
Follow proper course of action to repair Fault			Bench test replacement part
Prove faults corrected			
Instructor comments:			

[redacted]

[redacted]

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Familiarization with Circuit (5 MINS) Time:			
Confirm circuit operation			Circuit must be function tested
Observe safety at all times			Failure to follow safe procedures constitutes an automatic fail
Instructor comments:			

Pneumatic FAULT (25 MINS) Time 1st Fault:

Student picks faults from envelope; instructor inserts the faults, no guidance from instructor.

Fault chosen:	Fault details (exact location of fault in circuit):

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Test circuit to observe fault			
Test meter			
Use meter correctly			
Explain logic behind chosen troubleshooting technique			
Correctly identify fault			Failure to identify fault within time limit constitutes an automatic fail

Follow proper course of action to repair Fault			Bench test replacement part
Prove faults corrected			
Instructor comments:			
Overall Assessment	**Result (instructor to circle final grade)** **PASS / FAIL**		
To be completed by Instructor / Invigilator			
Strengths:	Area of improvement:		
Date:	**Trainee's Signature:**	**Instructor's Signature:**	

PO 008 MAINTAIN CONTROL NETWORKS

EO 008.01 Troubleshooting Control Networks

1. Performance. Troubleshooting Control Networks.

2. Conditions.

   a. Given:

      (1) Test Equipment and Tools;
      (2) PPE;
      (3) References; and
      (4) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall troubleshoot control networks independently, IAW refs by:

4. Troubleshooting control access network buses by:

   a. Identifying potential network equipment failures to include:
   b. Stars;
   c. Rings;
   d. Peer to Peer vs. Client servers;
   e. Mesh;
   f. LAN/WAN;
   g. Ethernet; and
   h. Point to point and point to multipoint.
   i. Identifying network hardware component failures to include:
      (1) Network interface cards;
      (2) Hubs;
      (3) Switches;
      (4) Bridges;
      (5) Routers;
      (6) Gateways; and
      (7) Servers.

5. Time required.

a. EO – 200 minutes.
b. EC – 50 minutes

6. Instructional Methods. Interactive lecture.

7. Training Aids.

8. Operational network
9. Network cables

10. Test Details. EO 008.01 will be assessed by written EC upon completion of EO 008.01 IAW Annex K.

11. References. C47, and C48.

12. Remarks. Nil.

Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Investigate Communication Networks by:							
a. Troubleshooting control access network buses by: (1) Identifying network types to include but not limited to: (a) Stars; (b) Rings; (c) Peer to Peer vs. Client servers; (d) Mesh; (e) LAN/WAN; (f) Ethernet; (g) Point to point and point to multipoint. (2) Identifying network components to include but not limited to: (a) Network interface cards;		200			IL	C47 C48	Instructor to provide quizzes to support instructed material focus on identifying network components.

(b) Hubs; (c) Switches; (d) Bridges; (e) Routers; (f) Gateways; and (g) Severs.						

PO 008 MAINTAIN CONTROL NETWORKS

EC 008.01

EC Time: 50 minutes.

ASSESSMENT ACTIVITY: The trainee will perform a written assessment, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- EC

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PO 008 Maintain Control Networks

EO 008.02 Repair Fiber Optic Networks

1. Performance. Repair Fiber Optic Networks

2. Conditions.

   a. Given:

      (1) Test Equipment and Tools;
      (2) PPE;
      (3) References; and
      (4) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard. The Marine Technician Electrical Specialist shall repair fiber optic networks independently, IAW refs by:

   a. Troubleshooting fiber optic networks (ALT1930) to include:

      (1) Connecting/disconnecting CCTV networks and equipment (ALT0400); and
      (2) Testing CCTV network and equipment (ALT1500).

4. Time required.

   a. EO – 1600 minutes
   b. CEC – 350 minutes

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstrate – performance.

6. Training Aids.

a. Fusion splicer;
b. Fibre optic meter; and
c. Fiber optic cable.

7. Test Details. EO 008.02 will be assessed by EC 008.02 IAW Annex K.

8. References. C62, C63, C59, and C60.

9. Remarks. TEL trainer would be ideal. These are instructed at Camosun / Dalhousie

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)		Method	Refs	Comments
		T	D | P			
TP1 Investigate Fiber Optic Networks by:						
a. Troubleshooting fiber optic networks by: (1) Identifying fiber cable types. b. Repairing fiber optic networks to include but not limited to: (1) Connecting/disconnecting CCTV networks and equipment by: (a) Investigating TCP/IP protocols to include but not limited to: 1. Key architectural principles; 2. Abstraction layers; 3. Comparison of TCP/IP & OSI layering; 4. Addressing including: a. IPv4; and b. IPv6. 5. Error checking tools for connectivity problems. (2) Analyzing fiber optic cabling to include but not limited to:	200	1400	IL DP	C50 C52 C59	

(a) Uses and principal of operation; (b) Mechanisms of operation; (c) Manufacturing processes; and (d) Troubleshooting and repair. (3) Analyzing fiber optic communications to include but not limited to: (a) Transmitters; (b) Receivers; (c) Digital predistortion; (d) Amplifiers; (e) Wavelength-division multiplexing; (f) Parameters; and (g) Comparison with electrical transmissions.					

PO 008 Repair Fiber Optic Networks
Critical Enabling Check – EO 008.02

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilitator Resources and Instructions

1. Classroom.
2. CEC 008.02 will be a written assessment based on content from EO 008.02
3. Pass/Fail IAW Annex K.
4. Time required: 350 minutes.
5. References are not required.

PART II – ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

WRITTEN ASSESSMENT

1. CEC 008.02 will be a written assessment based on content from EO 008.02
2. During this written assessment, you will be required to repair fiber optic networks. You must pass this test to pass EO 008.02.
3. References will not be provided.
4. Pass mark is 60%.
5. You will have 350 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Determined by academic instructor.

PO 008 MAINTAIN CONTROL NETWORKS

EO 008.03 Repair Control Networks

1. Performance. Repair Control Networks.

2. Conditions.

   a. Given:

      (1) Test Equipment and Tools;
      (2) PPE;
      (3) References; and
      (4) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist shall repair control networks independently, IAW refs by:

   a. Inspecting Network Communication Cables (ALT0660) by:

      (1) Troubleshooting Cat 5e cable networks (ALT1830); and
      (2) Testing network communication cables (ALT1640).

   b. Repairing Cat 5e cable networks (ALT0960) by:

      (1) Testing Cat 5e networks (ALT1490); and
      (2) Testing network communication cables (ALT1640).

4. Time required.

   a. EO - 700 minutes;
   b. EC - 100 minutes; and
   c. PC - 600 minutes.

5. Instructional Methods.

a. Interactive lecture; and
b. Demonstration – performance.

6. Training Aids.

a. Operational network;
b. CCTV camera; and
c. Network cables.

7. Test Details. EO 008.03 will be assessed practically in PC 008

8. References. C47, and C48.

9. Remarks. EC is to be assessed through an in-class activity.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP1 Repair Communication Networks							
a.	Inspecting Network Communication Cables by: (1) Troubleshooting Cat 5e cable networks by: (a) Identifying network cable types to include but not limited to: 1. Co-Axial; 2. Twisted pair (CAT5e/CAT 6); and	50	200		IL DP	C47 C48	Trainees to build a working Cat 5 network.
b.	Repairing Cat 5e cable networks, to include but not limited to: (1) Examining Ethernet including: (a) Varieties of Ethernet; (b) Frame structure; and	600	300		IL DP	C47 C48	Practical to consist of trainee building a small functional communication network.

(c) Error conditions. (2) Repairing twisted pairs including: (a) Cat 5; (b) Cat 5e; and (c) Cat 6. (3) Testing Cat 5e networks; and (4) Testing network communication cables.					TEL trainer will be used to allow trainees to construct a virtual communicati on network.

PO 008 MAINTAIN CONTROL NETWORKS

EC 008.03

The content of EO 008.03 will be assessed by an in-class activity with feedback provided by instructors.

PO 008 MAINTAIN CONTROL NETWORKS

PC 008

PC Time: 50 mins per trainee

ASSESSMENT ACTIVITY: The trainee will build CAT5e cable which is then tested using a network meter and plugged into the Maintenance Trainer's IPMS network between the POU and any network switch to prove system function.

Pass/fail criteria: Passing 60% and SAT on all critical items.

ASSESSMENT RESOURCES:

- Cat5E Cable and components
- Cat5E Crimper
- Network Tester
- Maintenance Trainer

IMPACT OF FAILURE: Remedial Training / Re-test / TRB after second unsuccessful attempt

PERFORMANCE CHECKLIST			
Date:	SN/PRI:	Rank:	Name:

TASKS: Successfully and safely complete practical assessment checklist.

Administrative Instructions:

3. This practical assessment checklist has been created to limit subjectivity during the assessment on each of the performance elements listed. The trainee is to be measured according to the checklist which has been derived from the standard statements from PO 008.

4. **Standard: The Marine Technician Electrical Specialist shall independently demonstrate the ability to Repair Control Networks**

In determining the level of assistance an instructor or assessor may provide a trainee during their assessment, the following should be considered;

4. Ensure the trainee is given an opportunity for the maintenance routine to progress to its conclusion;
5. If the trainee experiences difficulty, provide minimal assistance to maintain the standard through PC;
6. **Physical safety shall be observed throughout for both the trainee and the instructor**. If a safety infraction is committed or about to be committed the testing should be either;
   a. Paused and corrective action taken before continuing, or halted if the infraction is serious in nature.

CONDITIONS and EQUIPMENT:

6. Cat5E Wire and RJ45 plugs;
7. Cat5E Crimper
8. Network Tester
9. IPMS Maintenance Trainer with Network Switch and POU

TEST PROCEDURES	SAT	UNSAT	INSTRUCTOR COMMENTS
Build a Cat5E patch cable to connect a POU into an IPMS Network:			

Measure and Cut the length of Cat5E patch cable needed			
Prepared the ends of Cat5E cable for crimping			
Correctly orient the wires into the RJ45 Plug			
Crimp RJ45 plug on to Cat5E cable			
Test Cat5E patch cable with network tester			
Test Cat5E patch cable by connecting a POU into a working IPMS network			
	/10		Student must achieve 60% to Pass

Instructor comments:

Overall Assessment	**Result** **(instructor to circle final grade)** **PASS / FAIL**

To be completed by Instructor / Invigilator

Strengths:	Area of improvement:

Date:	**Trainee's Signature:**	**Instructor's Signature:**

PO 009 MAINTAIN HALON 1301 EXTINGUISHING SYSTEM

1. Performance. The Performance Objective associated with PO 009 Maintain Halon, and associated tasks, shall be trained IAW Halon 1301 Extinguishing System Maintainer QSP, MITE ID 109149.

2. Time Required.

MODULE 2:

Total Minutes (Instruction, Data Collection & Assessment)	2650 minutes
Actual Training Days (ATT) (based on 400 minutes per day)	7 days

3. Test Details. The Performance Check is Part 2 (Practical) of PC 401 inside Halon QSP, and will be conducted in two parts:

a. Isolate power in order to conduct planned maintenance on Halon bottles:

(1) Marked in accordance with a checklist.
(2) Time limit = 30 minutes; and
(3) Pass/Fail.

b. Recognize faults and troubleshoot the Fire Protection System:

(1) Marked in accordance with a checklist;
(2) Time limit = 50 minutes; and
(3) Pass/Fail.

(a) the Halon mock-up is to be pressurized with compressed air prior to conducting the practical PC;
(b) any accidental release of gas by the trainee during the conduct of the practical PC will result in an automatic failure, with the exception of trapped gas released during uncoupling the valves;
(c) accidental release of gas to the atmosphere (with the exception of the release of tapped gas when disconnecting the actuators) will result in an automatic failure

4. Remarks. Any considerations or special instructions (i.e. Instructor requirements, Advance scheduling, etc.) are IAW Halon 1301 Extinguishing System Maintainer QSP.

PO 009 MAINTAIN HALON 1301 EXTINGUISHING SYSTEM

PC 009

CHECKLIST	PASS	FAIL	Comment
Prepare a Halon zone for maintenance. (Pg 5-430) **Ref: Damage Control System C-77-231-000/MS-001**			
Inform appropriate personnel that zone is being taken out of service and will not be available until the maintenance is complete. IAW Procedure 1			
Place a Halon Zone in Maintenance Mode (Bk 2 Pg 216)			
Establish Zone Status. Interrogate IPMS pages: BAM overview, DCIB and Halon Status to determine status of all devices associated with the HRC (Bk 2 Pg 215) IAW Procedure 1			
Isolate power from UPS to the appropriate HRC (UPS 28v Breaker off). IAW procedure 2			
Isolate loop power to the HRC by removing J1 and J2 IAW Procedure 3			
Remove solenoids in accordance to procedure 4			Critical
Inspect Halon solenoid in accordance to procedure 5			
Carry out maintenance.			
Re-install solenoids for the return of Halon Zone to service (pg 5-455**)** Ref: Damage Control System C-77-231-000/MS-001			
Visually inspect zone hardware. Ensure all parts are assembled with the exception of cylinder solenoids.			
Re-connect loop to HRC in accordance to Procedure 2.			
Inspect Halon cylinder solenoids in accordance to Procedure 3			
Restore UPS power to the zone by making breaker to the HRC. In accordance to Procedure 4. **The solenoids are to be left disconnected until the checks in this section are complete. (Take Voltage reading at J9 pin 5 and 6 Primary, Pin 7 and 8 Reserve. Ensure 0V between pins)**			Critical
Replace the solenoids in accordance to Procedure 5			
Take a Halon Zone out of Maintenance Mode (Bk 2 Pg 217)			
Inform appropriate personnel that the zone is back in service.			Critical
Sub-Total			
Total - Student must complete 60% pass rate with Critical components achieved.			

PO 010 PERFORM ROUTINE MOTOR MONITORING

EO 010.01 Test electric motors by Operating vibration monitor suite

1. Performance: Test electric motor by the operating vibration monitor suite.

2. Conditions:

   a. Given:

      (1) Motor;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist perform scheduled motor monitoring routines using specialty tools, IAW refs to include:

   a. Testing electric motors by operating vibration monitor suite (HT0160).

4. Time Required.

   a. EO – 1000 minutes;
   b. EC – 50 minutes; and
   c. PC – 300 minutes.

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstration – performance.

6. Training Aids.

   a. One motor per trainee; and
   b. Pertinent vibration analysis equipment.

7. Test Details. EO 010.01 will be assessed through a practical PC.

8. References. A3, A4, and A5.

9. Remarks.

   a. Training in conjunction with vibration monitor (Mite 108503); and
   b. Train sequence PO 002 before PO 010.

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP 1 Test electric motors to include but not limited to:							
a. Utilizing vibration analysis (HT0160 Operate vibration monitor suite) to troubleshoot electric motors		400		600		A3 A4 A5	

PO 010 PERFORM ROUTINE MOTOR MONITORING

EC 010.01

EC Time: 50 mins

ASSESSMENT ACTIVITY: The trainee will conduct a written assessment on theory of Vibration analysis.

Pass/fail criteria: Passing 60%

ASSESSMENT RESOURCES:

- EC

IMPACT OF FAILURE: Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

PO 010 Perform Routine Motor Monitoring

EO 010.02 Test electrical systems with thermal imaging equipment.

1. Performance: Testing electrical systems with thermal imaging equipment.

2. Conditions:

   a. Given:

      (1) Motor;
      (2) PPE;
      (3) Testing equipment and tools;
      (4) References; and
      (5) Supervision;

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist perform scheduled motor monitoring routines using specialty tools, IAW refs to include:

   a. Testing electrical systems with thermal imaging equipment, to include:

      (1) Reading Heat Signatures.

4. Time Required.

   a. EO – 100 minutes.

5. Instructional Methods.

   a. Interactive lecture; and
   b. Demonstration – performance.

6. Training Aids.

   a. One motor per trainee;
   b. Thermal imaging camera

7. Test Details. EO 010.02 will be assessed via in-class activity.

8. References. A3, A4, A5 and TBD by course instructors.

9. Remarks. Training in conjunction with Vibration Monitor course (Mite 108503). Train sequence PO 002 before PO 010

10. Teaching Points.

Teaching Point	Sub Teaching Points	Instructional Technique (Time in min)			Method	Refs	Comments
		T	D	P			
TP 1 Utilize equipment health monitoring to troubleshoot motors, to include, but not limited to:							
a. Training in conjunction with thermal imaging.		50		50		A3 A4 A5	

PO 010 Perform scheduled motor monitoring routine

EC 010.02

The content of EO 010.02 will be assessed by an in-class activity with feedback provided by instructors.

PO 010 Perform scheduled motor monitoring routine

PC 010

PC Time: 50 minutes per trainee

ASSESSMENT ACTIVITY: The trainee will practically assess the trainee's ability to Test electrical systems using the vibration monitoring suite, IAW refs.

Pass/fail criteria: Passing 60%.

ASSESSMENT RESOURCES:

- Vibration analysis suite
- VA Motor Trainer

IMPACT OF FAILURE: TED / Remedial Training / Re-test / TRB after second unsuccessful attempt

PO 011 PERFORM HIGH RELIABILITY SOLDERING AND TERMINATION TECHNIQUES

EO 011.01 Conduct high reliability soldering, wire wrapping and connector terminations

1. Performance: Conduct high reliability soldering, wire wrapping and connector terminations.

2. Conditions:

   a. Given:

      (1) PPE;
      (2) Specialty tools;
      (3) Appropriate tools, equipment and spare parts;
      (4) References;
      (5) Assistance; and
      (6) Supervision.

   b. Denied: Nil;

   c. Environmental:

      (1) Classroom; and
      (2) Workshop.

3. Standard: The Marine Technician Electrical Specialist perform soldering and termination techniques using specialty tools, IAW refs to include:

   a. Perform high reliability soldering techniques; and

   b. Perform connector terminations and wire wrapping techniques .

4. Time Required.

   a. EO – 1300 minutes;
   b. EC – Conducted as in-class activity; and
   c. PC – 100 minutes.

5. Instructional Methods.

   a. Interactive lecture; and

b. Demonstration – performance.

6. Training Aids.

a. Soldering station;
b. Soldering tools; and
c. Solder equipment Materials.

7. Test Details. EO 011.01 will be assessed through a practical PC.

8. References. TBD by contractors.

9. Remarks. Lab experiments must support knowledge.

10. Teaching Points.

Teaching Points	Comments
TP 1: Introduction to PCB Repair:	
a. Safety; and b. Hand tools and hand tool safety.	
TP 2: Basic Soldering Techniques:	
a. types of solder; b. types of flux; c. lead preparation; d. lead termination; e. forming leads; f. soldering iron and iron preparation; g. thermal linkage; h. workplace indicators (visual inspection of work in progress); i. dwell time; j. good solder connections; k. basic rules for soldering; and l. common solder faults;	
TP 3: Printed Circuit Boards:	
a. PCB types and materials; b. PCB soldering techniques; c. lead termination; d. heat sensitive devices; e. discrete component insertion; f. component soldering; g. discrete component extraction; and	

h. printed circuit board cleaning;	
TP 4: PCB Coatings:	
a. reasons for coating; b. types of coating; c. identification of coatings; d. removal methods; and e. application methods;	
TP 5: Basic Connector Termination Techniques:	
a. wire splicing techniques (Western Union); b. heat shrink sleeving; c. termination banana plugs and test probes; d. terminating crimp connectors; e. terminating BNC connectors; f. terminating multi-pin connectors; g. connector assemblies and strain relief; and h. multi-pin connector pin insertion/extraction techniques;	
TP 6: PCB Repairs:	
a. run and pad repair; b. solid wire equivalent; c. lifted runs; d. terminal area; e. missing pad; f. plated through interface connection; g. edge connector repair; and h. heat damage	
TP 7: Basic Wire Wrapping Techniques:	
a. wire wrap materials and tools; and b. wire wrap process.	

ANNEX A – MAIN REFERENCES

A = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other publications

Refs. Code	Publication Number	Publication Title
CF or other official Canadian government publication		
A1	MA-3170	Applied Mathematics Master Lesson Plan.
A2	D-03-003-005/SF-000	General Electric Specifications for Canadian Forces Ships
A3	D-03-002-006/SG-Z01	Inspection and Run-in Tests, Shipboard Electric Motors
A4	C-03-005-012/AM-001	Naval Material Management System Manual.
A5	C-03-010-106/NY-082	Naval Preventive Maintenance Schedule (Electrical) - Motors and Controllers - Applicable to Halifax Class
A6	D-01-400-001/SG-000	Engineering Drawing Practice
A7	D-01-003-001/SG-000	Standard Requirements for the Engineering Content of Ship Drawings and Associated Lists
A8	C-27-778-000/MS-001	Operation and Maintenance Instructions With Parts Identification List Steering Gear Control System Applicable To Halifax Class
A9	C-27-890-000/MS-002	Sewage Treatment Plant
A10	C-24-548-000/MS-001	Operation and Maintenance Instructions Propulsion Cruise Diesel Engine (Pielstick) 20 Pa6 V 280 Mpc 20 Cyl - Books 1-4, Applicable To Halifax Class
A11	C-27-A51-000/MS-001	Operating and instructions for vacuum collection holding and transport (VCHT) system
A12		Small Systems Specialist Support – Lab Book, Canadian Forces School of Communication and Electronics, CFB Kingston (2000)

A13	CD300-100812-01	Software User's Manual for the HCI HMI
A14	CD300-102873-01	Halifax Class IPMS Operation & Maintenance Manual
A15	ED899-100650-02	RTU Schematic Diagram
A16	CD300-101305-01	Halifax Class IPMS Interconnect Document
A17	A31 - C-24-673-000/NY-001	Naval Preventive Maintenance Schedule (Electrical / Engineering (IMCS)) Instruments and Safety Systems (Pielstick Engine) Local Control Cubicle Automatic Controls Applicable to Halifax Class
A18		Qualification Standard & Plan Pre-Academic
A19		Canadian Forces Naval Engineering School Standing Orders 423
A20		Canadian Forces Naval Engineering School E&T DIV Handbook
A21		Canadian Forces Naval Engineering School Class Leader Handbook
A22	C-03-010-556/MS-001	Vortex Flow Meter
A23	C-03-010-571/MS-001	Gems Tank Level Switches
A24	C-03-010-570/MD-F01	Accelerometers
A25	C-03-010-573/MS-F01	Operation and Maintenance Instructions with Parts - identification List - Liquid Level Switches for Marine Service "Gems
A26	C-03-010-572/MS-F01	Flow Switches and Flow Controls (Gems)
A27	C-01-010-572/MS-001	Pressure Transducer
A28	C-26-412-000/MS-001	850KW Diesel Generator
A29	C-26-383-F00/TB-001	Technical Bulletin (Marine) Sabt-Switches Leakage Hcm-Csi-Lm-Tb-001-1408001-9 (9 Sep 14) Applicable to Halifax Class (330)
A30	C-03-010-510/NG-001	Shipboard Equipment Test Sheet for Air Circuit Breakers Applicable To All Ships
A31	C-03-010-510/TB-001	Technical Bulletin (Marine)- Test Procedures Type ACB Circuit Breakers
A32	C-26-383-B00/MS-001	Technical Manual - Automatic Change-Over Switch, ASCO 942, Catalog No. 435a3d, Type

		ACOS, 450 Volts, 60Hz, Appl.2, 250 Amperes, Drip proof ENCLOSURE
A33	C-23-050-001/PT-001	Investigations in Electric Power Technology [Lab Volt Student's Manual]
A34	C-26-321-000/MS-001	Equipment for DG Set AEG 279 995 755 Vol. 1
A35	C-38-224-000/MS-000	Secondary Electrical Power Distribution and Lighting System for CPF
A36	C-26-428-000/MS-001	CPF Switchboard Manual, Vols. 1-8
A37	C-26-383-A00-MS-001	Technical Manual Automatic Change-Over Switch Type ACOS, 450 Volts, 660Hz, Appl. 2, 100 Amperes
A38	C-26-383-C00/MS-001	
A39	C-26-383-B00/MS-001	
A40	C-77-778-000/MS-001	Steering control system
A41		Ships Standing Orders AL10.1
A42	C-77-231-000/MS-001	DCC Vol 1-5
A43		CAPS SDD Vol 10 Damage Control System
A44	C-03-005-012/AM-001	MARCORD EHM program;
A45	RDIMs #302654 v39	MARPAC- Halifax Class SEMS Manual
Official Allied Force or Allied government publication		
B1	ISBN 978-1387965533	The Navy Electricity and Electronics Training Series: Module 15 Principles of Synchros, Servos, and Gyros, United States. Navy.
Commercial publications		
C1	ISBN: 0-13-173044-4	Introductory Circuit Analysis, Robert L. Boylestad, 11th ed.
C2	ISBN 0-13-238351-9	Principles of Electric Circuits, Conventional Current Version, Thomas L. Floyd, 8th ed.
C3	ISBN: 978-0-07-352957-8	Engineering Circuit Analysis, William H. Hayt, Jack Kemmerly, Steven M. Durbin, 8th ed.
C4	ISBN: 978-1-118-07887-7	Principles of Electric Machines and Power Electronics, DR. P. C. SEN, 3rd ed.

C5	ISBN: 0-07-2465239	Electric Machinery Fundamentals: Stephen J. Chapman, 4th ed.
C6	ISBN 978-0321131942	Basic Technical Mathematics with Calculus, 8th Edition, Allyn J. Washington, Pearson Education, Inc., 2005.
C7	ISBN 978-0137157099	College Mathematics, 2009 Update with Mymathlab, Cheryl Cleaves and Margie Hobbs, Pearson Education, Inc., 2009.
C8	ISBN 9781337798686	Understanding Motor Controls – Stephen Herman
C9	ISBN 978-0071833950	Printed Circuit Cards – Clyde F. Coombs & Happy Holden
C10	LCCN 65-028814	Engineering Drawings 10th Ed. (French and Vierick, McGraw Hill)
C11	ISBN 9780870062834	Blue Print Reading for Industry (Walter C Brown, The Goodheart-Willcox Company Inc., 1979
C12	ASIN B0066AJ0ZO	Developing Electrical Troubleshooting Skills (Tpc Training Systems), Edward J. Amrein - Loose Leaf, 1998
C13		EasyPIC User's Guide
C14		EasyPIC User's Schematics
C15	ISBN: 978-0-12-382036-5	Power electronics handbook, devices, circuits and applications, Muhammad H. Rashid, 3rd Ed
C16	ISBN: 0-306-48048-4	Fundamentals of Power Electronics, Robert W. Erickson, Dragan Maksimovic, 2ndEd
C17	ISBN: 978-0-07-338067-4	Power Electronics, Daniel W. Hart
C18	ISBN: 0-13-093083-0	Electrical machines, drives and power systems, Theodore Wildi, 5th edition
C19		HCI Operation and Maintenance Manual
C20	ISBN 978-0-13-226275-0	Introduction to Control System Technology, 5th Ed. (Robert Bateson)
C21	ISBN 9780131194588	Process Control Instrumentation Technology (Curtis Johnson)
C22	ISBN 978-1-4081-7559-0	Instrumentation and Control Systems (Reeds Marine Engineering Series)
C23	ISBN 9780071135412	Power Electronics, Principles and Applications, 1st Ed., Joseph Vithayathil, McGraw-Hill, 1995
C24	ISBN 9780130167620	Modern Electronic Communication, 7th Ed., Gary M. Miller, Prentice-Hall (2001)

C25		Woodward Governor Company 2301 A Power Generation
C26		Electro-Hydraulic Servo Systems, Johnson Merrill, 1989
C27	ISBN 978-0130617613 (conventional flow) OR 978-0130617507 (electron flow)	Introductory Electronic Devices and Circuits, 6th Edition, R. Paynter, 2003.
C28	ISBN 978-0028008455	Electronic Principles, 5th Edition, Albert P. Malvino, McGraw-Hill, (1993). Chapter 17.
C29	ISBN 0-13-262226-2	Electronic Devices And Circuit Theory, RobertL.Boylestad,11th ed.
C30	ISBN 0-13-254985-9	Electronic Devices, Thomas L.Floyd, 9th ed.
C31	ISBN 9780827358256	Industrial Electronics (Humphries and Sheets)
C32		SABT book from L3 Mapps
C33	ISBN 9780130231208	Applied Fluid Mechanics. Robert L. Mott, 5th edition
C34	ISBN 9780978802202	Eaton Industrial Hydraulics Manual 5th edition
C35	ISBN 9780130615084	Basic Fluid Power Dudley A. Pease, John J. Pippinger 2nd edition.
C36	ISBN 978-0963416216	Eaton closed loop electro-hydraulic systems manual 2nd edition
C37	ASIN B004ICFSJE	Vickers Industrial Hydraulics Manual, 2nd Ed., 1989
C38	ISBN 9780132484695	Electronic Instruments and Measurements, Jones and Chin, John Wiley and Sons Inc.
C39	ISBN 9780132499545	Electronic Instruments and Measurement, David A. Bell, Prentice-Hall
C40		Hydraulic Trainer, Vols. 1, 2, 3, 4, 5 and 6 Mannesman, Rexroth
C41	ISBN 978-0960564453	Industrial Fluid Power, Vol.1, Womack Educational Publishers.
C42	ISBN 9780943719016	Industrial Fluid Power, Vol.2, Womack Educational Publishers.
C43	ISBN 9780943719009	Industrial Fluid Power, Vol.3, Womack Educational Publishers.
C44		Industrial Fluid Power, Vol.4, Womack Educational Publishers.
C45		Industrial Fluid Power, Vol.5, Womack Educational Publishers.

C46	ISBN 978-0135103821	Digital systems, principals and applications (11th Edition), Tocci, Ronald J., Prentice Hall (2011)
C47	ISBN 978-0-07-149730-5	CCNA Cisco certified network associate study guide, Richard Deal
C48	ISBN 978-0-782-14014-9	Network & Study guide 3rd edition exam N100-002, Davig groth, sybex publishing
C49	ISBN 978-0317579123	Electronic Assembly, 2nd Ed., Ryan
C50	ISBN 978-1511445658	Understanding Fibre Optics (5th Edition), Hecht, Jeff, Creatspace independent Publishing Platform (29 Apr 2015)
C51		GTL-7-151.52-TR3, Rev 0, L3 IPMS Training Manual (interm)
C52		Ascent Commtest VA software reference guide, Copyright 1999-2004;
C53		Ascent Commtest VA VB5 instrument reference guide, Copyright 199-2004; and
C54	ISBN 0-920855-26-1	IPT's Rotating machinery Equipment Handbook, Bruce M Basaraba & James A Archer, Copyright 1995
C55	DLQD-DP401	Dolang - Double Sided advanced electric pneumatic trainer – Instruction Manual
Other publications		
D1	PS804-108393-01	Procurement Specification PS
D2		Student Handout – Ground Testing RTUs
D3		Student Handout – VRTU Bite
D4		74LS32 Quad 2-Input OR Gate Datasheet
D5		74LS08 Quad 2-Input AND Gate Datasheet
D6		7407 Hex Buffer/Driver Datasheet
D7		74LS04 Hex Inverting Gate Datasheet
D8		74LS00 Quad 2-Input NAND Gate Datasheet
D9		74LS02 Quad 2-Input NOR Gate Datasheet
D10		74LS266 Quad 2-Input Exclusive-NOR Gate Datasheet

D11		74F86 Quadruple 2-Input Exclusive-Or Gate Datasheet
D12		74LS74 Dual D-Type Positive Edge-Triggered F/F Datasheet
D13		74HC76 Dual J-K Flip-Flop Datasheet
D14		74LV595 8-Bit Shift Register Datasheet
D15		74LS85 4-Bit Magnitude Comparator Datasheet
D16		74LS173 4-Bit D-Type Register Datasheet
D17		74LS541 Octal Buffer/Line Driver Datasheet
D18		1N4148 High Conductance Fast Diode Datasheet
D19		1N700 Series Zener Diode Datasheets
D20		1N900 Series Zener Diode Datasheets
D21		1N4700 Series Zener Diode Datasheets
D22		1N5200 Series Zener Diode Datasheets
D23		1N4000 Series Rectifier Diode Datasheets
D24		Aluminium Electrolytic Capacitors Datasheets
D25		Low Voltage - P.C. Mount Transformer Datasheets
D26		Schottky Diode Datasheets
D27		LED Datasheets
D28		2N2222 Transistor Datasheet
D29		2N3904 Transistor Datasheet
D30		2N5772 Transistor Datasheet
D31		TIP31 Complementary Silicon Power Transistor Datasheets
D32		2N5684 Complementary Silicon Power Transistor Datasheets

D33		2N1711 Transistor Datasheet
D34		BC107 Transistor Datasheet
D35		2N5458 Transistor Datasheet
D36		2N5486 RF Amplifier Datasheet
D37		2N7000 Field Effect Transistor Datasheet
D38		20N10L MOS Transistor Datasheet
D39		MTM12N10 Power FET Datasheet
D40		IRL540 HEXFET© Power MOSFET Datasheet
D41		LM741 Op Amp Datasheet
D42		LM759 Power Op Amp Datasheet
D43		LM7372 High Speed Op Amp Datasheet
D44		LF351 JFET Op Amp Datasheet
D45		LF355 JFET Op Amp Datasheet
D46		LM112 Op Amp Datasheet
D47		LM311 Voltage Comparator Datasheet
D48		OPA234 Low-Power, Precision Single-Supply Operational Amplifier Datasheet
D49		LTC2051 Dual/Quad Zero-Drift Operational Amplifier Datasheet
D50		OPA111 Low Noise Precision Operational Amplifier Datasheet
D51		LM158 Dual Op Amp Datasheet
D52		MC34060A Precision Switch Mode Pulse Width Modulator Control Circuit Datasheet
D53		LM386 Low Voltage Audio Power Amplifier Datasheet
D54		M7805 Three-Terminal Positive Voltage Regulator Datasheet

D55		LM309 5-Volt Regulator Datasheet
D56		LM317 1.2V to 37V Voltage Regulator Datasheet
D57		LM340-5 Three-Terminal Positive Voltage Regulator Datasheet
D58		LM340-15 Three-Terminal Positive Voltage Regulator Datasheet
D59		Handouts HO-15 and 16
D60		Handout HO – 12
D61		PowerPoint – 8, PP – 08A
D62		PowerPoint shows 9 and 9 A.
D63		Handout - PLC Programming guide and Lab manual

ANNEX B – TASK LIST

Task No	Task Statement	PO
AMT0160	Interpreting drawings/schematics	001
AFT0470	Reviewing manufacturer specifications	001
AFT1000	Complete environmental documentation	001
AUT4060	Troubleshooting motor controllers	002
AUT0920	Inspecting circuit cards	002
AUT0930	Inspecting control relays	002
AKT0660	Inspecting inductors	002
AUT1730	Inspecting visually, motherboards	002
ALT0650	Inspecting motherboards	002
AUT0890	Inspecting capacitors	002
ALT2110	Troubleshooting printed circuit boards	002
BAT0280	Repairing steering controller	002
AOT1270	Repairing steering controller	002
AOT1050	Repairing black water collection controller	002
AOT1330	Repairing refrigeration electrical	002
BAT0190	Repairing centering unit	002
ALT1120	Repairing helo fuel/de-fuel control panel (HFCP)	002
AWT0020	Assemble/disassemble main refrigeration system components	002
AUT2910	Repairing whistle speed controllers	002
AOT1070	Repairing calorifiers	002
AUT0640	Calibrating whistle speed controllers	002
ART0130	Calibrating stores lift	002
AOT0270	Calibrating refrigeration control systems	002
ART0120	Calibrating crane controllers	002
AUT0470	Calibrating circuit cards	002
AIT0630	Repairing aux steam generator electronic safety control	002
ALT1030	Repairing control transformer	002
AUT0210	Adjusting overload relay trip setting	002
AUT0220	Adjusting overload relay	002
AUT2140	Repairing control relays	002
AOT0120	Assemble/disassemble black water collection system	002
AUT0270	Adjusting timing relays	002
AUT3250	Testing electrical controllers	002
AOT1750	Trialing black and grey water system	002
AWT0210	Testing refrigeration electrical system	002
ALT1740	Testing propulsion diesel engine starter box	002
AUT3950	Testing whistle speed controller	002

ART1060	Testing stores lift	002
AUT3160	Testing control transformers	002
AUT3870	Testing timing relays	002
AUT3140	Testing control relays	002
AUT3530	Testing overload relays	002
AUT2130	Repairing circuit cards	002
AUT2630	Repairing rectifiers	002
AKT2480	Repairing timing relays	002
AUT0700	Connecting/disconnecting power electronics	002
AUT2120	Repairing circuit card cage assembly	002
AUT3130	Testing circuit cards	002
AUT3120	Testing circuit card cage assembly	002
BTO0220	Troubleshooting electric propulsion system	002
AOT1950	Troubleshooting machinery space ventilation system	002
AUT4080	Troubleshooting ship's whistles	002
AIT0980	Troubleshooting low pressure air compressors	002
AIT0960	Troubleshooting high pressure air compressors	002
ART0380	Investigating system operation of king post	002
AWT0060	Inspecting chiller units	002
ART1340	Troubleshooting windlass	002
ART1310	Troubleshooting kingpost	002
ART1270	Troubleshooting capstan	002
BTO0260	Troubleshooting propulsion motor	002
AUT1180	Inspecting governor stepper motor	002
HT0160	Operate vibration monitor suite	002
AUT2490	Repairing motor impeller	002
ART0550	Repairing capstan	002
ART0530	Repairing boat davit	002
AKT2540	Repairing vane axial fan	002
AKT1460	Repairing centrifugal pump	002
ART0650	Repairing kingpost	002
AOT0070	Adjusting sewage macerator	002
AUT2320	Repairing enclosure fans	002
ART0510	Repairing windlass	002
ART0600	Repairing dumbwaiter	002
AIT0830	Repairing LP air compressor	002
AUT2360	Repairing governor stepper motor	002
AKT1450	Repairing centrifugal fan	002
ART0020	Adjusting capstan	002
ART0730	Repairing stores lift	002
AIT0700	Repairing HP air compressor	002
AUT0250	Adjusting stepper motor	002
AOT0290	Calibrating ship's ventilation system	002
AUT3390	Testing governor stepper motor	002
AOT1800	Trialing sewage macerator	002

BAT0390	Troubleshooting steering control system	003
BAT0100	Inspecting electronic steering system	003
BAT0130	Inspecting helm unit	003
BAT0380	Troubleshooting auto pilot system	003
BAT0060	Inspecting auto steering unit	003
AUT1360	Inspecting power electronics	003
ALT1920	Troubleshooting engine control systems	003
AVT1390	Troubleshooting gas turbines	003
ALT2160	Troubleshooting propulsion diesel engine controls	003
ALT2190	Troubleshooting propulsion diesel engine local control cubicle	003
ALT2140	Troubleshooting propulsion diesel engine preheat control system	003
ALT2180	Troubleshooting propulsion diesel engine fire detection control	003
ALT1970	Troubleshooting human/machine interface (HMI)	003
ALT1980	Troubleshooting integrated platform control system (IPCS) man	003
ALT1990	Troubleshooting integrated platform management system (IPMS)	003
ALT1900	Troubleshooting diesel generator local control panel (LCP)	003
ALT2010	Troubleshooting local operating panel	003
ALT0620	Inspecting local operating panel	003
AYT0420	Troubleshooting plastic compress melt unit	003
AOT1960	Troubleshooting reverse osmosis desalination (ROD) unit	003
AIT0910	Troubleshooting auxiliary steam generator	003
AWT0220	Troubleshooting chiller units	003
AOT1830	Troubleshooting black & grey water treatment system	003
AOT1840	Troubleshooting black water system	003
AOT0450	Inspecting black water treatment system	003
ALT2000	Troubleshooting internal integrated platform management system	003
ALT0080	Analyzing digital data	003
AFT0970	Collecting equipment health monitoring data	003
AUT0810	Inspecting backplanes	003
AUT0650	Inspecting motherboards	003
CT0420	Troubleshooting damage control systems	003
ALT2210	Troubleshooting remote terminal unit (RTU)	003
AUT0970	Inspecting digital input/output processor (WAGO)	003
ALT1820	Troubleshooting bus access modules (BAMS)	003
ALT0480	Inspecting bus access modules (BAMS)	003
ALT1800	Troubleshooting analog digital acquisition modules (ADAM)	003
BAT0180	Repairing auto steering unit	003
BAT0220	Repairing helm units	003
AUT2590	Repairing power supplies	003
AUT2570	Repairing power electronics	003
AUT0570	Calibrating power electronics	003
AUT0440	Calibrating automatic control systems	003
AUT3580	Testing power electronics	003
AUT3010	Testing amplifiers	003
AUT3220	Testing diodes	003

ALT1060	Repairing DG local control panel	003
ALT1540	Testing DG local control panels	003
ALT1340	Repairing QM local operating panel	003
ALT1750	Testing QM local operating panels	003
ALT1670	Testing portable operating units (POU)	003
ALT1650	Testing onboard training systems (OBTS	003
AUT4100	Uploading/downloading PLC program	003
ALT0570	Inspecting DIP switches settings	003
AOT1580	Testing chillers units	003
AOT1760	Trialing black and grey water treatment systems	003
AYT0290	Repairing plastic compress melt unit	003
AYT0060	Calibrating plastic compress melt unit	003
ALT0950	Maintaining Bus Access modules (BAMS)	003
ALT0380	Connecting/disconnecting BAMS and associated devices	003
AUT0560	Calibrating power distribution control systems	003
AUT0230	Adjusting proportional integral derivative controller	003
ALT1470	Testing bus access modules (BAMs)	003
ALT1760	Testing remote terminal units (RTU)	003
ALT1520	Testing components with built-in test equipment (BITE)	003
ALT0910	Performing analog/digital ground checks	003
ALT1710	Testing printed circuit boards	003
ALT1640	Testing network communication cables	003
ALT1680	Testing power data distribution boxes (PDDB)	003
ALT1780	Testing signal conditioners	003
ALT1790	Testing WAGO modules	003
ALT1690	Testing power level angle actuators	003
HT0170	Operating virtual remote terminal units (VRTUs)	003
AVT1140	Repairing main diesel engine electronic governor	003
ALT1730	Testing propulsion diesel engine controls	003
AVT1230	Testing diesel engine electronic governor	003
AUT1850	Installing propulsion control system components	003
AVT0300	Assembling/disassembling diesel engine electric starting system	003
AUT0510	Calibrating electrical propulsion control system components	003
ALT1630	Testing momentum modules	003
ALT1870	Troubleshooting control sensors	004
AUT2680	Repairing speed sensors	004
AUT2750	Repairing tachogenerator	004
ATT0170	Assembling/Disassembling pressure sensors	004
ATT0730	Installing pressure switches	004
ATT0180	Assembling/disassembling pressure switches	004
ATT0720	Installing pressure sensors	004
AUT2530	Repairing optical sensors	004
AKT2080	Repairing photo sensors	004
ATT0270	Assembling/Disassembling vacuum sensors	004
ATT0840	Installing vacuum sensors	004

AUT2820	Repairing thermistors	004
AUT2840	Repairing thermostats	004
ATT0250	Assembling/disassembling temperature sensors	004
AUT2780	Repairing Temperature transducer	004
AUT2290	Repairing electrically operated thermostatic control valve	004
AUT2830	Repairing thermocouples	004
AVT0970	Repairing diesel engine pyrometer	004
ATT0830	Installing temperature switches	004
ATT0820	Installing temperature sensors	004
ANT0160	Calibrating halon fire extinguishing system	004
ANT0060	Assembling/disassembling fine water spray components	004
AUT1830	Installing damage control components	004
ANT0860	Repairing explosive gas detectors	004
ATT0790	Installing solenoids	004
ANT0960	Repairing manual pull stations	004
BAT0270	Repairing rudder angle indicator	004
BAT0210	Repairing feedback transmitter	004
BAT0290	Repairing steering feedback electrical control unit	004
AKT1890	Repairing level indicators	004
ATT0980	Repairing flow meter	004
ATT0060	Assembling/disassembling flow meters	004
BBT0390	Installing ship’s fluid tanks elements	004
AST0080	Repairing fuel metering unit	004
AOT1360	Repairing salinity cells	004
AUT2610	Repairing pressure transducer	004
AUT2790	Repairing tension transducers	004
ANT0250	Evaluating halon fire extinguishing system to include	004
ANT1300	Testing fine water spray system	004
ANT1340	Testing Halon fire extinguishing system	004
ALT1770	Testing shaft knot indicators	004
AOT1700	Testing salinity cell	004
AOT0280	Calibrating salinity cell	004
AUT3520	Testing optical sensors	004
AUT3550	Testing photo sensors	004
AT0240	Testing diesel engine pyrometer operation	004
AUT3860	Testing thermostats	004
AUT3810	Testing temperature transducers	004
AUT3790	Testing temperature sensors	004
ANT1370	Testing leak/flood detection system	004
AOT0240	Calibrating fitted refrigeration leak detection system	004
AOT1640	Testing fitted refrigeration leak detection system	004
AUT3620	Testing pressure sensors	004
AUT0600	Calibrating pressure sensors	004
AUT3640	Testing pressure transducers	004
AUT3720	Testing speed sensors	004

AUT3780	Testing tachogenerator	004
ANT1290	Testing explosive gas detector	004
AUT3820	Testing tension transducers	004
CT0440	Troubleshooting power generation	005
AUT0800	Inspecting automatic voltage regulators	005
AUT4030	Troubleshooting generators	005
AUT4050	Troubleshooting motor alternator sets	005
AUT0380	Assembling/disassembling generator components	005
AUT2000	Repairing automatic voltage regulators	005
AUT0100	Adjusting automatic voltage regulators	005
AUT0320	Assembling/disassembling automatic voltage regulators	005
AUT3380	Testing generator main field windings	005
AUT1900	Measuring electrical parameters on generator windings	005
AUT3670	Testing voltage regulators	005
CT0440	Troubleshooting power generation and distribution system	006
ALT1910	Troubleshooting electrical switchboard control (ECS)	006
AUT1510	Inspecting synchroscope	006
AUT1080	Inspecting electrical switchboard transducers	006
AUT1490	Inspecting switchboard tangle box	006
AUT1060	Inspecting electrical switchboard breakers	006
AUT0750	Inspecting 440 VAC distribution breakers	006
AUT0760	Inspecting 440 VAC distribution panels	006
AUT0810	Inspecting backplane	006
AUT0850	Inspecting bus bars	006
AUT1020	Inspecting electrical disconnect links	006
AUT1500	Inspecting switching unit auto (SABT)	006
AUT0580	Calibrating power generation control system	006
AUT0520	Calibrating electrical switchboard parameters	006
AUT3060	Testing bus bars	006
AUT3260	Testing electrical disconnect links	006
AUT2040	Repairing breakers	006
AUT0130	Adjusting circuit (cct) breakers	006
AUT2270	Repairing electrical switchboard breakers	006
AUT3300	Testing electrical switchboard breakers	006
AUT3050	Testing breakers	006
AUT2990	Testing 440 VAC distribution panels	006
AUT0080	Adjusting automatic change-over switching unit	006
AUT0430	Calibrating automatic change-over switching unit	006
AUT0260	Adjusting switching unit auto (SABT)	006
AUT0170	Adjusting manual change-over switching unit	006
AUT0550	Calibrating manual change-over switching unit	006
AUT2710	Repairing switching unit auto	006
AUT3750	Testing switching unit auto	006
AUT2720	Repairing switching unit auto (SABT	006
AUT3760	Testing switching unit auto (SABT	006

AUT2730	Repairing switching unit manual	006
AUT3770	Testing switching unit manual	006
ANT1400	Testing portable casualty power breakers	006
ANT1390	Testing multi-taps	006
ANT1420	Testing riser terminals	006
BAT0140	Inspecting hydraulic controls	007
AQT0030	Inspecting hydraulic directional control valves	007
BAT0240	Repairing hydraulic steering controls	007
AKT1840	Repairing hydraulic power unit	007
BAT0350	Testing hydraulic steering control	007
AOT1940	Troubleshooting HVAC system	007
AOT0630	Inspecting smoke evacuation system components	007
ATT0370	Inspecting electro-pneumatic valves	007
ALT2200	Troubleshooting propulsion diesel engine low pressure control air	007
ATT0040	Assembling and disassembling electro pneumatic valves	007
AOT0720	Installing HVAC/NBCD air distribution components	007
AOT0080	Adjusting smoke evacuation system components	007
AKT0180	Assembling and disassembling pneumatic control system	007
ATT0630	Installing electro-pneumatic valves	007
AKT2150	Repairing pneumatic-operated valves	007
ATT0970	Repairing electro-pneumatic valves	007
AKT2550	Repairing variable-position pneumatic-controlled valves	007
AKT2110	Repairing pneumatic actuators	007
AOT1790	Trialing HVAC/NBCD air distribution systems	007
ALT1660	Testing pneumatic-control components	007
ATT1170	Testing electro-pneumatic valves	007
ALT1860	Troubleshooting control access network buses	008
ALT1840	Troubleshooting CCTV Servers	008
ALT0660	Inspecting Network Communication Cables	008
ALT1830	Troubleshooting Cat 5e cable networks	008
ALT1930	Troubleshooting fibre optic networks	008
ALT0960	Repairing Cat 5e cable networks	008
ALT1490	Testing Cat 5e networks	008
ALT1640	Testing network communication cables	008
ALT0400	Connecting/disconnecting CCTV networks and equipment	008
ALT1500	Testing CCTV network and equipment	008

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 5 Aug 2021 – 13 Aug 2021

Location: WP1020 NTDC(P) Campus, Esquimalt, BC--(Haida –Venture)

Board Members:

Surname	Initials	Rank	MOSID	Unit	Role
Siska	D	CPO2		NTDC(P)	Board Chairperson
Salmon	B	PO2		NTDC(P)	Board vice-chair
Timbers	AR	LT	00211	NTDC(P)	TDO
Swinton		MS		NTDC(P)	SME/Secretary
Perkins	A	MS		NTDC(P)	SME
Malone	S	PO2		NFS(P)	SME
Quin	P	PO2		NTDC(P)	SME
Musteca	W	Civ		Elms contract	SME
Gagnon	F	Civ		Elms Contract	SME
Cane	M	PO2		NFS(A)	SME
Valentine	S	Civ		NTDC(P)	IDO
McConnell		Maj			DCO
Sutton		CPO2		TM	SME
Campbell	A	Civ		Calian contract	SME

Board Objective:

1. The primary purpose of this QSRB is to assess and action the recommendations submitted by the MTAT review of Martech training with the overall goal of reducing overall training time.

2. A secondary goal of the QSRB is to develop a road map outlining the academic knowledge and practical skills required at various rank qualifications to ensure there are no training gaps nor redundant training.

3. In concert with this road map the master task list will be reviewed to ensure that the division of tasks mirror operational realities. This board is working under the assumption that the Martech training will be common at the S3 level and split into unique streams at the S2 level.

4. Lastly the board will review the QSP to ensure the content reflects the operational realties and current technology available to the RCN.

Interpretation of Board Mandate:

5. The purpose of the MTAT review was to identify potential time savings to ensure an adequate supply of qualified sailors to support fleet operations.

6. This board aims to develop effective and efficient training that ensures that the right person receives that right training at the right time.

Specification Task – No Train Decisions:

7. Nil.

Discussions/Decisions:

Day 1 – 5 AUG 21

8. Upon review of the MTAT review EO 001.03 it was decided that redundant teaching points from S3 be removed. Teaching points E & F should be moved to EO 002.03. This removes 1200 minutes from EO 001.03.

9. EO 001.04 Gas Free will be moved to its own PO to facilitate PLARs and assessment. The board concluded that the S1E does not require the full course or the gas free qualification. Parts of the Gas Free course will be covered with the intention being to provide the S1 with the knowledge required for them to safely do their job but no MITE code will be awarded. This removes 2100 minutes from the course.

10. The Gas Free standalone course will not be affected and could be offered if the qualification is required. The S1 mech course will continue to offer the full course and award the qualification.

11. EO 002.01 NFS-A is running new academic material, which requires 10,000 less minutes than the legacy material. Camosun timings are set as per an existing contract and will remain the same. Adjustments will be considered for NFS-A it is estimated they will require 12,000 minutes to deliver this material.

12. Time cuts TBD from NFS-A. NFS-P will continue to deliver through Camosun.

13. CPO1 Budden (MTAT): QSP/contract compliance between coasts needs to be addressed to ensure sailors are receiving the right instruction while understanding the realities of conducting courses at multiple locations does require some flexibility.

14. Contracts are currently awaiting approval, no changes will be made until new contracts are in place.

15. This board acknowledges that because the differences between NFS-A and NFS-P specifically in how the academic portion are delivered each coast will have its own timings and schedule. Steps have been made to capture these differences for example having two timings for a lesson or have two different schedules.

16. EO 002.05 VA/Thermal Stand-alone course with N-Qual/MITE will remain for NFS-P and be embedded within the S1 QSP. These embedded courses will be scheduled strictly through FMF for delivery.

17.Recommend making EO 002.05 its own PO separated into 3 troubleshooting ECs/

18.VA and thermal imaging were moved to a separate PO (troubleshooting using specialty tools) this makes instruction and PLARs easier. Specific timings for all sections of this lesson needs to be identified. Relocate the timings from EO 002.05 to the new PO 009.

19.EO 003.05 will need to be piloted by both coasts and instructors will need to assess instructional material and courseware.

20.EO 007.03-08 new to S1 Elec QSP needs to be piloted

21.A NFS-A trial has identified redundant material that can be removed from 007.03-05 and 007.06-08.

22.24V DC system courseware owned by L3 for faultfinding. Should/can this be taught at the S1 level.

23.PO 008 was expanded into four EOs, courseware previously developed for networking lessons. This material needs to be piloted for any significant conclusions can be made.

24.QSP timings should reflect NFS-A requirements for academic lessons. A table will reflect the differences between how each coast delivers the training. The board recommends including two timing tables to the QSP to justify timing differences between the two coasts.

<u>Day 2 – 6 AUG 21</u>

25. Recommends that OJT packages be developed to provide sailors with an opportunity to consolidate skills/knowledges covered during the residential portion of the course until an adequate TEL or training aid can be fully implemented.

26.RaceRocks has developed digital training aids for this course and others. These need to be located and made available on SharePoint so instructors can use these tools. There was some discussion as to what exactly had been built, what had been delivered and where it had been delivered to.

27.It was confirmed that NFS-A academics requires 20,000 minutes less. Timings for EO 008.02-04 have been confirmed.

28.PO 002 assessments will include a practical PC faultfinding – 2x double faults 100min per trainee (15 min intro, 2x 40 min faults and 5 min debrief) EC 002.03 & EC 003.04 will be 50 minute written assessments.

29.Timings for PO 003 are undetermined due to a lack of training aides and testing materials. Pilot will need to be completed and recommendations can be made. Time will remain unchanged in the QSP until this can take place.

30.EC 003.03 is a 50 min written assessment. EO 003.04 OBTS 50 min (requires review: new EC to be developed. West coast to confirm synchro-servos at NFS-P to be used for assessment purposes to mimic RAI. NO PC for PO 003

31.JTAR needs to be reviewed to ensure division of tasks between Elec and Mech streams reflects operational realities. Should also review to ensure tasks in the JTAR are assigned to appropriate rank/position.

32.West coast requires thyristor package for LabVolt testing.

33.Recommend block review of IPMS to familiarize trainees with HMI.

34.Roadmap S1 Elec PO 003 to USQ Ms Elec.

35.Recommend re-evaluation and adjustment of PO 003 to include controls introduction to controls non-class specific training. Need to be simplified to reflect that of an S1 level of training.

36. Recommends that a tack home package be developed as a review of S2 material.

37.PO 005 will have a 50 min/ trainee practical PC LabVolt (thyristor and PID module required. EO 005.01-03 will be assessed by written in-class ECs.

38.Test is in-line with instructional material but not with what is in the fleet (MWM vs CAT).

39.PO 006 will have a 50min written PC and EC 006.01 and EC 006.02 will be a take home quiz/assessment yet to be created.

40.ACOS/MCOS/SABT training aides are unavailable to NFS-P.

41.ECs 008.01, 008.02 & 008.04 will be assessed by take home quiz/assignments. EC 008.03 will be a 100min practical assessment conducted as an in-class activity. The PC will be developed and administered by an outside agency.

Day 3 – 9 AUG 21

42.EO 001.02 – DRMIS Maintainer timings for course are 3 days (2100 min) not 7 days

43.EO 001.03 Recommend to remove the overlap in S1 from S3 training Teaching points 5 and 6 from PO 001 and place lesson in PO 002 EO 002.03.

44.Gas Free certification is not required for the S1 Elec. Theory/academic lessons to remain, practical lessons to be removed from S1 electrical.

45.Make PO 009 for VA and Thermal Imaging lessons.

46.PO 004 Investigate procurement channels through enterprise management for direct lines to LCMM's. NFS-A and NFS-P to develop comprehensive list of missing equipment by PO/EO.

47.Reallocation of timings in PO 007 EO 007.01-.03 to become EO 007.01 with the EC from EO 007.03. EO 007.04-.06 to become EO 007.02 with EC from EO 007.06. PC will remain the same.

Day 4 – 10 AUG 21

48.Recommendations to use legacy content from the QL4 course in the new USQ-S2 Elec:

a. Academics
    i. EO 001.01 – Mathematics
    ii. 97x60 min periods
    iii. Train

b. EO 002.01 – Physics
    i. 83x60 min periods
    ii. Train

c. EO 003.01 – Circuit analysis
    i. 100x60 min periods
    ii. 12.01 of USQ S2 ver. 1.1 – 2800 min + 100 min EC
    iii. Standard H-L omitted from QL4 when translated to S2
    iv. Train

d. EO 003.02 – Rotating Machinery
    i. 93x60 min periods
    ii. 12.01 of USQ S2 ver. 1.1 3700 min + 100 min EC + 50 min Review
    iii. Standard I added to S2 QSP
    iv. Train

e. EO 004.01/004.02 – Gather written and graphic information
    i. 44x50 min periods
    ii. EO 001.01 of USQ S2 ver. 1.1 Tech Com Mod A 44x50 min periods
    iii. Standard d3 need to be modified – not trained
    iv. Train as per Mod A

f. Recommend later review of entire Tech Com QSP if timings need to be cut.

49. Academic phase of training would be beneficial to be taught at Camosun College as a preferred method of delivery as a stepping stone for future training and employment. NFS-P does not have the infrastructure to support the instructional requirements for training the academic phase of S2 Elec training.

50. Common Equipment from the legacy QL4 QSP

a. PO 005 – Maintain Motors and Controllers
    i. EO 005.01
        1. 8x60 min periods
        2. No Train

    ii. EO 005.02+5.03 – Understand/Identify Motor control schematics
        1. 18x50 min period + 24x50 min period

    iii. EO 012.03 of USQ S2 ver. 1.1 EO 012.03 1400 min total

    iv. Train (captured in EO 12.03 of AB QSP – Review timings)

51. 12.03 USQ S2 ver.1.1 – No train Bearings (850 min) for electrical. Recommend being looked at for being implemented to mechanical S2 training. 3rd line maintenance is not done on board ships.

52. Disassemble/reassemble motor = add 4x 50 min periods.
53. Reallocate timings from the bearing lesson to be 2 days for 12.03 of the S2 QSP ver. 1.1 for fault finding phase.

54.EO 006.01 – Maintain PG&D Recommend class specific package/OJPR to cover teaching points covered for PO 006 though PO 010.

Post-board Activities:

55.Work QSPRB decisions through QSP.

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

Board Objective:

1.The purpose of this writing board was to develop the QSP for the USQ LS Electrical Specialist in the overall context of the Marine Technician trade development MIP. This was the fifth QSPWB to be convened in the process.

Interpretation of Board Mandate:

2.Specific instructions delivered to the board in the fulfillment of the board's mandate were focused primarily on course duration and the incorporation of technology-enabled learning (TEL). At a strategic level, institutional leadership within the RCN has recognized the requirement to deliver training which is aimed at a demographic whose familiarity with daily technological integration is such that it is unrealistic and inefficient to not leverage this knowledge in a training capacity. A significant return on investment can be realized by actively incorporating this technology. Multiple peer-reviewed research studies examining the benefits of TEL-based course delivery has been released by allied Navies as well as industrial vocational occupational classifications that are congruent with RCN STEM trades. The known benefits include, but are not limited to: faster comprehension and acquisition of both cognitive and hand

skills; better retention of skills over extended periods (reduced skill fade); increased active learning and engagement among trainees; reduced requirements for physical material and resources; and, a reduced physical footprint (real property) required to deliver training.

3.The board was provided a directive to incorporate or suggest TEL resources to enhance the learning experience at any, and all, available opportunities within the QSP. In many cases, TEL has subsequently been suggested as either the primary means of reinforcing or testing skills. As such, the QSP reflects the attitude that the procurement of TEL resources is not an impediment to the delivery of the course content. There are ECs and PCs developed within the content of this QSP that reflect this intention i.e. ECs and PCs require alternative testing methods if the indicated TEL is unavailable. Alternatively, in a number of teaching points, TEL is used as a visual or kinesthetic learning delivery vehicle that is designed to augment and enhance legacy teaching objectives that anecdotal experience from the board SMEs suggested might be improved for future trainees.

Specification Task – No Train Decisions:

4.In instances where the board decided not to train (NT) a task that is present in the JTAR, this decision is reflected in the JTAR master list comments field. The decision to NT tasks was decided based on either redundancy as compared with a like task in the QSP, class specificity, or discounted because the relevant skills were covered at a previous qualification level.

The NT tasks determined during the board are as follows:

Task Number	Task Description	Reason
KT0340	Repair chilled water system (subs)	Submarine class
KT0660	Troubleshoot main diesel engine start control panel (Sub)	Submarine class
KT0520	Test generator rectifier (subs)	Submarine class
ANT1630	Troubleshoot electronic gas detector	Submarine class
ANT0370	Inspect electronic gas detector	Submarine class
ANT1260	Test electronic gas detector	Submarine class
ANT0820	Repair electronic gas detector	Submarine class
AUT3400	Test hydrogen detector sensor	Submarine class
AUT0450	Calibrate battery charging equipment	Submarine class
AUT3040	Test battery charging system	Submarine class
AUT4000	Troubleshoot battery charging system	Submarine class
AVT1300	Troubleshoot auxiliary diesel engine exhaust system	Submarine class
AVT1280	Troubleshoot auxiliary diesel engine combustion air system	Submarine class
AVT1320	Troubleshoot auxiliary diesel engine fresh water system	Submarine class
AVT1330	Troubleshoot auxiliary diesel engine internal components	Submarine class
AVT1310	Troubleshoot auxiliary diesel engine fuel system	Submarine class
AVT1290	Troubleshoot auxiliary diesel engine control system	Submarine class
AVT1270	Troubleshoot auxiliary diesel engine	Submarine class
AVT0800	Repair auxiliary diesel engine governor	Submarine class
AVT1350	Troubleshoot auxiliary diesel engine starting air system	Submarine class

AVT1360	Troubleshoot auxiliary diesel engine sea water cooling system	Submarine class
AUT3490	Test main battery charging system (electric propulsion)	Submarine class
AUT2460	Repair main battery charging system (electric propulsion)	Submarine class
AUT4040	Troubleshoot main battery charging system (electric propulsion)	Submarine class
AHT0010	Repair the propulsion motor (AOPS)	AOPS class
AHT0020	Test the propulsion motor (AOPS)	AOPS class
AHT0040	Clean the propulsion motor (AOPS)	AOPS class
ANT1620	Troubleshoot deluge system	280 class
AUT1860	Investigate operation of 24 volt distribution system	280 class
ART1290	Troubleshoot high line winch	Tanker class
ART1300	Troubleshoot inhaul/outhaul winch	Tanker class
AVT1370	Troubleshoot auxiliary gas turbine engine control system	280 class
ANT0810	Repair deluge system	280 class
CT0300	Operate torsion meters	280 class
ART0040	Adjust stores elevator	Tanker class
AUT0780	Inspect AC to DC converter	280 class
AUT0940	Inspect DC to DC converter	280 class
AKT1090	Investigate system operation logs	Handled by IPMS
AKT1080	Investigate system maintenance logbooks	Handled by IPMS
ART0720	Repair stores elevator	Tanker class
ART0590	Repair crane safety system	Tanker class
ART0630	Repair inhaul/outhaul antislack device (ASD)	Tanker class
AOT1240	Repair hot fresh water heater	Tanker/280 class
AKT0320	Calibrate stores elevators	Tanker class
ART0110	Assemble/disassemble stores elevator components	Tanker class
AUT2150	Repair DC to DC converter	280 class
AVT0680	Investigate system operation of auxiliary gas turbine	280 class
AUT3180	Test DC to DC converter	280 class
ART0070	Assemble/disassemble crane components	Tanker class
AHT0050	Troubleshoot the propulsion motor (AOPS)	AOPS class
AHT0080	Repair the propulsors (AOPS)	AOPS class
AHT0090	Test the propulsors (AOPS)	AOPS class
AHT0110	Clean the propulsors (AOPS)	AOPS class
AHT0120	Troubleshoot the propulsors (AOPS)	AOPS class
AHT0180	Clean silencer (AOPS)	AOPS class
AHT0190	Troubleshoot silencer (AOPS)	AOPS class
AHT0220	Repair selective catalytic reduction (AOPS)	AOPS class
AHT0250	Clean selective catalytic reduction (AOPS)	AOPS class
AHT0260	Troubleshoot selective catalytic reduction (AOPS)	AOPS class
AHT0320	Clean urea system (AOPS)	AOPS class
AHT0330	Troubleshoot urea system (AOPS)	AOPS class
AHT0360	Repair variable frequency reversible drives (AOPS)	AOPS class
AHT0370	Test variable frequency reversible drives (AOPS)	AOPS class
AHT0380	Adjust variable frequency reversible drives (AOPS)	AOPS class
AHT0390	Clean variable frequency reversible drives AOPS)	AOPS class

AHT0400	Troubleshoot variable frequency reversible drives AOPS)	AOPS class
AHT0430	Repair the fin stabilizer bonding system (AOPS)	AOPS class
AHT0440	Test the fin stabilizer bonding system (AOPS)	AOPS class
AHT0450	Adjust the fin stabilizer bonding system (AOPS)	AOPS class
QT0080	Operate the heeling system (AOPS)	AOPS class
QT0090	Operate the active roll fin stabilization system (AOPS)	AOPS class
QT0100	Operate the thermal oil dump cooler (AOPS)	AOPS class
QT0110	Operate the deaerator (AOPS)	AOPS class
QT0120	Operate the propulsion couplings (AOPS)	AOPS class
QT0130	Operate variable frequency reversible drives (AOPS)	AOPS class
QT0140	Operate the fin stabilizer bonding system (AOPS)	AOPS class
AUT0010	Repair the breaking resistors (AOPS)	AOPS class
AUT0020	Test the breaking resistors (AOPS)	AOPS class
AUT0030	Adjust the braking resistors (AOPS)	AOPS class
AUT0040	Clean the braking resistors (AOPS)	AOPS class
AUT0050	Troubleshoot the breaking resistors (AOPS)	AOPS class
AVT0010	Repair the propulsion converter (AOPS)	AOPS class
AVT0020	Test the propulsion converter (AOPS)	AOPS class
AVT0030	Adjust the propulsion converter (AOPS)	AOPS class
AVT0040	Clean the propulsion converter (AOPS)	AOPS class
AVT0050	Troubleshoot the propulsion converter (AOPS)	AOPS class
AVT0090	Test the propulsion couplings (AOPS)	AOPS class
AVT0100	Adjust the propulsion couplings (AOPS)	AOPS class
AVT0110	Clean the propulsion couplings (AOPS)	AOPS class
AVT0120	Troubleshoot the propulsion couplings (AOPS)	AOPS class
QT0010	Operate the propulsion converter (AOPS)	AOPS class
QT0020	Operate the braking resistors (AOPS)	AOPS class
QT0030	Operate the propulsion motor (AOPS)	AOPS class
QT0040	Operate the propulsors (AOPS)	AOPS class
QT0050	Operate silencer (AOPS)	AOPS class
QT0060	Operate selective catalytic reduction (AOPS)	AOPS class
QT0070	Operate urea system (AOPS)	AOPS class
AIT0140	Repair the heeling system (AOPS)	AOPS class
AIT0150	Test the heeling system (AOPS)	AOPS class
AIT0160	Adjust the heeling system (AOPS)	AOPS class
AIT0170	Troubleshoot the heeling system (AOPS)	AOPS class
AIT0200	Repair the active roll fin stabilization system (AOPS)	AOPS class
AIT0210	Test the active roll fin stabilization system (AOPS)	AOPS class
AIT0220	Adjust the active roll fin stabilization system (AOPS)	AOPS class
AIT0230	Troubleshoot the active roll fin stabilization system (AOPS)	AOPS class
AIT0300	Troubleshoot the thermal oil dump cooler (AOPS)	AOPS class
AIT0370	Troubleshoot the deaerator (AOPS)	AOPS class
AHT0460	Troubleshoot the fin stabilizer bonding system (AOPS)	AOPS class
AIT0010	Operate jacking system (AOPS)	AOPS class
AIT0020	Repair jacking system (AOPS)	AOPS class
AIT0040	Adjust jacking system (AOPS)	AOPS class
AIT0050	Troubleshoot jacking system (AOPS)	AOPS class

AIT0080	Repair the bow thruster and control system (AOPS)	AOPS class
AIT0090	Test the bow thruster and control system (AOPS)	AOPS class
AIT0100	Adjust the bow thruster and control system (AOPS)	AOPS class
AIT0110	Troubleshoot the bow thruster and control system (AOPS)	AOPS class
KT0470	Test alarm relay outputs	Submarine class
AJT0270	Lead repair of flight deck systems	Mechanical specialty
AOT0350	Evaluate HVAC/NBCD air distribution system	Mechanical specialty
AOT0340	Evaluate black & grey water treatment system	Mechanical specialty
AOT0360	Evaluate sewage macerator function	Mechanical specialty
AOT0370	Evaluate smoke evacuation system function	Mechanical specialty
AOT0330	Evaluate black & grey water collection system	Mechanical specialty
AKT1430	Repair brake pads	Mechanical specialty
BT0050	Evaluate gearbox function	Mechanical specialty
BBT0200	Inspect hangar door lifting assemblies	Mechanical specialty
AVT0690	Investigate system operation of main propulsion gas turbine	Mechanical specialty
BBT0080	Assemble/disassemble hangar door lifting assemblies	Mechanical specialty
AMT0080	Assemble/disassemble seating mount arrangements	Mechanical specialty
ANT1470	Test welding machine	Mechanical specialty
CT0430	Troubleshoot main propulsion system	Mechanical specialty
BT0060	Evaluate shaftline function	Mechanical specialty
AKT1280	Measure machinery mount wear	Mechanical specialty
AOT1810	Trial smoke evacuation system	Mechanical specialty
AKT0780	Inspect shock mounts	Mechanical specialty
AOT0590	Inspect HVAC/NBCD air distribution system	Mechanical specialty
AKT0820	Inspect visually, chip detector	Mechanical specialty
AUT2110	Repair chip detector	Mechanical specialty
AKT0970	Inspect visually, machinery mount	Mechanical specialty
AVT1050	Repair gas turbine exhaust mounts	Mechanical specialty
AKT1950	Repair magnetic brake	Mechanical specialty
UT0240	Operate electronic gas detector	Mechanical specialty
BBT0590	Repair hangar door lifting assemblies	Mechanical specialty
AMT0520	Investigate seating/hanging/shock mount arrangements	Mechanical specialty
AMT0690	Repair seating/hanging/shock mount arrangements	Mechanical specialty
AMT0830	Troubleshoot seating/hanging/shock mount arrangements	Mechanical specialty
AMT0480	Install seating/hanging/shock mount arrangements	Mechanical specialty
ATT0850	Install valves	Mechanical specialty
ANT0170	Calibrate twin agent units	Mechanical specialty
ANT1700	Troubleshoot magazine flood and spray system	Trained as AB
ANT1350	Test heat detector	Trained as AB
ANT1430	Test smoke detector	Trained as AB
ALT1015	Repair detector	Trained as AB
AUT0880	Inspect cannon plugs	Trained as AB
APT0160	Inspect cathodic protection twin controller	Trained as AB
AOT0530	Inspect fan coil units	Trained as AB
AOT0530	Inspect fan coil units	Trained as AB

AOT0220	Calibrate calorifiers	Trained as AB
AUT1550	Inspect temperature transducer	Trained as AB
AUT1130	Inspect field devices	Trained as AB
AFT1300	Draft system/equipment drawings (supports training)	Trained at lower level
AUT1650	Inspect vacuum sensors	Trained as AB
AUT1050	Inspect electrical motor	Trained as AB
AUT1400	Inspect pressure transducer	Trained as AB
AUT1300	Inspect motor windings	Trained as AB
AUT1770	Inspect visually, tension transducer	Trained as AB
AUT1090	Inspect electronic controller enclosure	Trained as OS
ATT0390	Inspect flow meters	Trained as AB
ALT0640	Inspect MOB system	Trained as OS
ALT0630	Inspect man Locked In Fridge alarm system	Trained as OS
AKT0750	Inspect pressure switches	Trained as AB
AKT0520	Inspect 12 volt Alternator	Trained as AB
AZT0310	Repair trim control system	Trained as AB
AUT2620	Repair push buttons	Trained as AB
AUT2560	Repair potentiometer	Trained as AB
AUT2050	Repair cable insulation	Trained as OS
AUT2210	Repair electrical connections	Trained as OS
AUT2220	Repair electrical connectors	Trained as OS
AUT2180	Repair electrical cable shielding	Trained as OS
AUT2800	Repair terminal strip	Trained as OS
AUT2770	Repair temperature switches	Trained as AB
AUT2760	Repair temperature sensors	Trained as AB
AKT2220	Repair pressure switches	Trained as AB
BAT0200	Repair drum switch	Trained as AB
AKT1900	Repair level switch	Trained as AB
AUT2330	Repair field devices	Trained as AB
AKT2210	Repair pressure sensors	Trained as AB
AKT1440	Repair bushings	Trained as OS
AUT3960	Test window heaters	Trained as AB
AUT3340	Test field devices	Trained as AB
AUT3800	Test temperature switches	Trained as AB
AT0040	Conduct phase sequence test	Trained as AB
AUT3090	Test capacitors	Trained as AB
PT0050	Configure degaussing gyro control units	Trained as OS
AKT0170	Assemble/disassemble motor	Trained as AB
AUT3710	Test solenoid	Trained at lower level
AZT0380	Test trim control system	Trained as AB
AUT3420	Test inductors	Trained as AB
AT0010	Conduct flow test	Trained as AB
AUT3840	Test thermistors	Trained as AB
AUT3560	Test pin connectors	Trained as AB
APT0170	Inspect degaussing deperming cables	Performed ashore
ANT0910	Repair fitted fire suppression manual activation stations	Performed ashore
AUT0420	Balance electrical motors	Performed ashore
ALT0130	Calibrate analog measuring test equipment	Performed ashore

AUT0530	Calibrate electrical/electronic/analog/digital/measuring equipment	Performed ashore
ANT0130	Calibrate electronic gas detector	Performed ashore
ANT0580	Inspect thermal imaging camera	Performed ashore
AUT0530	Calibrate electrical/electronic/analog/digital/measuring equipment	Performed ashore
APT0460	Test degaussing/deperming cables	Performed ashore
APT0340	Repair degaussing deperming cables	Performed ashore
AUT2500	Repair motor windings	Performed ashore
AUT2160	Repair degaussing coils	Performed ashore
CT0460	Repair torsion meter system	Performed ashore
CT0450	Troubleshoot torsion meter system	Performed ashore
AUT0540	Calibrate high voltage circuit breakers	AOPS class
ANT0330	Inspect casualty power cables routes	On-the-job experience
ALT1810	Troubleshoot B/W and colour	On-the-job experience
ALT0470	Inspect B/W and colour printers	On-the-job experience
ALT0270	Configure B/W and colour printers	On-the-job experience
ANT0340	Inspect casualty power riser terminal cables	Move to OS level
PT0040	Configure degaussing equipment	Move to AB level
AUT1170	Inspect generator windings	Move to OS level
APT0100	Inspect anti-fouling controller	Trained as OS
LT0030	Lead operation of hull protection systems	Move to MS level
AFT1240	Draft engineering change proposal	Move to MS level
LT0030	Lead operation of hull protection systems	Move to MS level
AFT1240	Draft engineering change proposal	Move to MS Mechanical Specialist
ANT1360	Test insulation on casualty power runs	On-the-job experience
ALT0220	Clean control sensors	On-the-job experience
AVT0340	Assemble/disassemble diesel engine pre-heat system components	On-the-job experience
ALT1290	Repair printed circuit boards	On-the-job experience
ART0180	Inspect hand held stores lift remote controller	On-the-job experience
PT0300	Operate primary switch gear mechanical links locally	On-the-job experience
ANT1220	Test casualty power breakers	On-the-job experience
ANT1230	Test casualty power runs	On-the-job experience
AUT0160	Adjust fan impeller	On-the-job experience
AYT0460	Assemble/disassemble black water treatment system components	On-the-job experience
AOT0160	Assemble/disassemble HVAC/NBCD air distribution system components	On-the-job experience
AUT1880	Measure electrical switchboard parameters	On-the-job experience
AMT0510	Install welding equipment (arc)	On-the-job experience
ANT0990	Repair portable casualty power breaker	On-the-job experience
UT0230	Monitor confined space entry	On-the-job experience
JT0170	Operate black & grey water treatment system	On-the-job experience
ANT0330	Inspect casualty power cables routes	On-the-job experience
AKT2010	Repair motor seals	On-the-job experience
ART0930	Test hand held stores lift remote controller	On-the-job experience
BAT0080	Inspect drum switch	On-the-job experience

ART0170	Inspect dumbwaiter	On-the-job experience
AOT0600	Inspect machinery space air distribution supply and exhaust fan	On-the-job experience
ALT0610	Inspect jumper cables	On-the-job experience
AUT1820	Inspect xenon searchlight	On-the-job experience
AUT2080	Repair capacitors	On-the-job experience
AUT2030	Repair breaker dust covers	On-the-job experience
AUT3570	Test potentiometer	On-the-job experience
AUT3900	Test uninterruptible power supply	On-the-job experience
AFT1280	Draft pre-installation failure (PIF) reports	Learned at AB level via experience
ANT1370	Test leak/flood detection system	Learned at AB level via experience
AUT3970	Test workshop equipment	On-the-job experience
AUT2810	Repair test panel	On-the-job experience
AKT2900	Troubleshoot test panel	On-the-job experience
AOT1120	Repair convection oven	In-ship support contract
AOT1100	Repair commercial laundry appliances	In-ship support contract
AOT1230	Repair griddles	In-ship support contract
AOT1200	Repair galley equipment keypad	In-ship support contract
AOT1140	Repair deep fat fryers	In-ship support contract
AOT1170	Repair food mixer	In-ship support contract
AOT1540	Test galley equipment digital display	In-ship support contract
AOT1210	Repair galley equipment temperature controls	In-ship support contract
AOT1320	Repair refrigeration condenser fan	In-ship support contract
AOT0580	Inspect griddles	In-ship support contract
AOT0130	Assemble/disassemble domestic refrigeration system components	In-ship support contract
AOT0570	Inspect galley equipment temperature controls	In-ship support contract
AOT0410	Inspect galley equipment digital display	In-ship support contract
AOT0640	Inspect electric steam table	In-ship support contract
AOT0680	Inspect warming drawer	In-ship support contract
AOT0560	Inspect galley equipment keypad	In-ship support contract
BAT0170	Investigate system operation of steering system	On-the-job experience

AOT0560	Inspect galley equipment keypad	In-ship support contract
AOT0490	Inspect convection oven	In-ship support contract
AOT0510	Inspect deep fat fryers	In-ship support contract
AOT1030	Repair galley equipment digital display	On-the-job experience
AOT0710	Install hotel service system components	In-ship support contract
AOT0540	Inspect food mixer	In-ship support contract
AOT0480	Inspect commercial laundry appliances	In-ship support contract
AOT0520	Inspect domestic laundry appliances	In-ship support contract
AOT1870	Troubleshoot domestic refrigeration equipment	In-ship support contract
ATT1110	Repair water cooler	In-ship support contract
AOT1920	Troubleshoot galley equipment keypad	In-ship support contract
AOT1910	Troubleshoot galley equipment	In-ship support contract
AVT1080	Repair gas turbine igniter	On-the-job experience
AVT1260	Test diesel engine system operation	Move to LS Watchkeeper level
AYT0050	Assemble/disassemble solid waste handling equipment	Not fitted
AKT0240	Calibrate freon detection unit	Not fitted
ALT0240	Clean remote terminal unit (RTU)	Not fitted
AYT0440	Troubleshoot solid waste handling equipment	Not fitted
AOT0610	Inspect potato peeler	Not fitted
AUT3610	Test power supply	Redundant (see AUT3600)
AOT1880	Troubleshoot fresh water chemical injection system	On-the-job experience
AVT1340	Troubleshoot auxiliary diesel engine lube oil system	On-the-job experience
AOT0470	Inspect chilled water and hydronic system components	On-the-job experience
AOT0700	Install chilled water and hydronic system components	On-the-job experience
AOT1080	Repair chilled water and hydronic system components	On-the-job experience
AOT1570	Test chilled water/hydronic system components	On-the-job experience
AIT0570	Inspect auxiliary systems components	On-the-job experience
ANT1250	Test damage control alarm system	On-the-job experience
ANT0470	Inspect Halon fire extinguishing system	Trained as AB
ANT1660	Troubleshoot fine water spray system	Trained as AB
AKT1430	Repair brake pads	On-the-job experience
AKT1950	Repair magnetic brake	On-the-job experience
CT0430	Troubleshoot main propulsion system	Already trained @ mechanical specialty

BT0060	Evaluate shaftline function	Already trained @ mechanical specialty
AVT0690	Investigate system operation of main propulsion gas turbine	Already trained @ mechanical specialty
BT0050	Evaluate gearbox function	Already trained @ mechanical specialty
ATT0850	Install valves	Already trained @ mechanical specialty
BBT0080	Assemble/disassemble hangar door lifting assemblies	Already trained @ mechanical specialty
AOT0330	Evaluate black & grey water collection system	Already trained @ mechanical specialty
BBT0200	Inspect hangar door lifting assemblies	Already trained @ mechanical specialty
UT0240	Operate electronic gas detector	Already trained @ mechanical specialty
BBT0590	Repair hangar door lifting assembles	Already trained @ mechanical specialty
AOT1810	Trial smoke evacuation system	Already trained @ mechanical specialty
AOT0360	Evaluate sewage macerator function	Already trained @ mechanical specialty
AVT0970	Repair diesel engine pyrometer	Duplicate task from Mechanical
ANT0960	Repair manual pull stations	Duplicate task from Mechanical
AUT3680	Test rheostat	On-the-job experience
AKT2500	Repair ultra-violet emitter	On-the-job experience
BAT0270	Repair rudder angle indicator	Duplicate task from Mechanical
AVT1140	Repair main diesel engine electronic governor	Duplicate task from Mechanical
AUT2280	Repair electrical switchboard meters	On-the-job experience
ANT1610	Troubleshoot damage control console	Redundant with ALT0560
AUT1990	Repair 440 VAC step-down transformers	On-the-job experience
AUT0770	Inspect 440 VAC step-down transformers	On-the-job experience
AUT2850	Repair transformer-rectifiers	On-the-job experience
AUT3890	Test uninterruptible power supplies wiring harness	On-the-job experience
ANT0260	Evaluate thermal imaging camera function (in support of PM)	On-the-job experience
AUT2390	Repair internal wiring	On-the-job experience
AUT2310	Repair electronic controller enclosure	On-the-job experience
AUT1910	Measure generator air gaps	Repair facility
AUT2200	Repair electrical connection box	On-the-job experience
ALT2170	Troubleshoot propulsion diesel engine enclosure carbon monoxide sensor	Performed by LS Mechanical
ALT1360	Repairing shaft knot indicators	Redundant, covered by other tasks

ALT1850	Troubleshoot central alarm monitoring system	Redundant, covered by other tasks
ANT1690	Troubleshoot Halon extinguishing system	Redundant, covered by other tasks
IT0270	Investigate operation of fire detection system	Redundant, covered by other tasks
ANT0470	Inspect Halon fire extinguishing system	Redundant, covered by other tasks
AYT0410	Troubleshoot oil pollution abatement treatment system	Redundant, covered by other tasks
ANT1670	Troubleshoot fire detection system	Redundant, covered by other tasks
AHT0570	Inspect ancillary systems components	Redundant, covered by other tasks
AYT0390	Troubleshoot dirty oil system	Redundant, covered by other tasks
AYT0210	Repair oil content monitor for oil pollution abatement treatment system	Redundant, covered by other tasks
AT0100	Conduct signal conditioning unit test	Redundant, covered by other tasks
ALT1510	Test central alarm monitoring system	Redundant, covered by other tasks
AWT0070	Inspect fitted refrigeration leak detection system	Redundant, covered by other tasks
ALT1320	Repair propulsion diesel engine sensors	Redundant, covered by other tasks
AWT0130	Repair fitted refrigeration leak detection system	Redundant, covered by other tasks
ANT0090	Assemble/disassemble halon system components	Redundant, covered by other tasks
ANT0930	Repair Halon fire extinguishing system	Redundant, covered by other tasks
AKT1520	Repair combustion engine acoustic enclosure fitted fire suppression system	Redundant, covered by other tasks
AYT0260	Repair oil pollution abatement treatment unit	Redundant, covered by other tasks
AUT1730	Inspect visually, motherboard	On-the-job experience
AUT 3660	Test rectifier	Redundant, covered in another PO
AUT3880	Test transformer-rectifiers	Redundant, covered in another PO
AUT1410	Inspect rectifier	Redundant, covered in another PO
AUT1610	Inspect transformer-rectifiers	Redundant, covered in another PO
AUT0480	Calibrate degaussing coils	On-the-job experience
APT0060	Calibrate active shaft grounding power controller	On-the-job experience
APT0250	Repair active shaft grounding power controller	On-the-job experience
APT0360	Repair degaussing rectifier controllers	On-the-job experience
APT0180	Inspect degaussing gyro control unit	On-the-job experience

PT0080	Investigate system operation of degaussing system	On-the-job experience
APT0490	Troubleshoot active shaft grounding system	On-the-job experience
APT0090	Inspect active shaft grounding power controller	On-the-job experience
APT0350	Repair degaussing gyro control unit	On-the-job experience
AUT3190	Test degaussing coils	On-the-job experience
APT0190	Inspect degaussing rectifier controllers	On-the-job experience
AUT0950	Inspect degaussing coils	On-the-job experience
AUT4020	Troubleshoot degaussing system	On-the-job experience
APT0500	Troubleshoot cathodic protection system	On-the-job experience
APT0140	Inspect cathodic protection power supplies (CAPAC)	On-the-job experience
APT0110	Inspect cathodic protection anodes	On-the-job experience
APT0130	Inspect cathodic protection cathodes	On-the-job experience
APT0150	Inspect cathodic protection reference electrodes	On-the-job experience
APT0050	Assemble / disassemble cathodic protection system components	On-the-job experience
AUT0460	Calibrate cathodic protection system components	On-the-job experience
APT0320	Repair cathodic protection reference electrodes	On-the-job experience
APT0280	Repair cathodic protection anodes	On-the-job experience
APT0300	Repair cathodic protection cathodes	On-the-job experience
APT0310	Repair cathodic protection power supplies (CAPAC)	On-the-job experience
APT0330	Repair cathodic protection twin controller	On-the-job experience
APT0370	Repair main cathodic protection shaft slip rings	On-the-job experience
APT0260	Repair anti-fouling controller	On-the-job experience
AUT1840	Install PG & D system components	Redundant
AUT1980	Repair 440 VAC distribution panel	Repair facility
AUT2010	Repair backplane	On-the-job experience
AUT3030	Test backplane	On-the-job experience
AUT3600	Test power supplies	On-the-job experience
AUT3980	Troubleshoot 400 Hz system	On-the-job experience
AUT0590	Calibrate power supplies	On-the-job experience
AJT0360	Repair helicopter start system	On-the-job experience
AUT2340	Repair final resource system charger (MCDV PLC UPS battery charger)	On-the-job experience
AUT0400	Assemble / disassemble static frequency converter	On-the-job experience
AUT2690	Repair static frequency converter	On-the-job experience
AUT1640	Inspect uninterruptible power supply	On-the-job experience
AUT0630	Calibrate uninterruptible power supply	On-the-job experience
AUT2880	Repair uninterruptible power supply	On-the-job experience
AUT0410	Assemble / disassemble uninterruptible power supply	On-the-job experience
ALT1950	Troubleshoot grounding detector	Kit no longer installed
BT0210	Troubleshoot controllable reversible pitch propeller system	Redundant task
AOT0430	Inspecting black & grey water collection system	On-the-job experience
AOT0440	Inspecting black water collection system	On-the-job experience
AOT1850	Troubleshooting calorifiers	On-the-job experience
ALT1960	Troubleshooting helo fuel/de-fuel control panel (HFCP)	On-the-job experience
AUT1810	Inspecting whistle controllers	On-the-job experience
AUT1800	Inspecting whistle automatic cycle timer controllers	On-the-job experience
AUT2160	Troubleshooting propulsion diesel engine controls	On-the-job experience

AWT0080	Inspecting main refrigeration system	On-the-job experience
BAT0400	Troubleshooting steering system	On-the-job experience
ART0390	Investigating system operation of stores lift	On-the-job experience
ALT0670	Inspecting PDE starter box	On-the-job experience
AOT0460	Inspecting calorifiers	On-the-job experience
ART0370	Investigating system operation of crane	On-the-job experience
ART1280	Troubleshooting crane	On-the-job experience
BAT0070	Inspecting centering unit	On-the-job experience
ALT2150).	Troubleshooting propulsion diesel engine auxiliary starter box	On-the-job experience
ALT0560	Inspecting DG local control panel	On-the-job experience
AVT1390	Troubleshooting gas turbines	On-the-job experience
ALT2160	Troubleshooting propulsion diesel engine controls	On-the-job experience
ALT2190	Troubleshooting propulsion diesel engine local control cubicle (LCC)	On-the-job experience
ALT2140	Troubleshooting propulsion diesel engine preheat control system	On-the-job experience
ALT2180	Troubleshooting propulsion diesel engine fire detection control system	On-the-job experience
AVT1390	Troubleshooting gas turbines	On-the-job experience
ALT2160	Troubleshooting propulsion diesel engine controls	On-the-job experience
AUT1580	Inspecting thermistors	Task redundant with ALT1870
AKT2890	Troubleshooting tension transducer	Task redundant with ALT1870
BAT0150	Inspecting rudder angle indicator	Task redundant with ALT1870
BAT0110	Inspecting feedback transmitter	Task redundant with ALT1870
ALT2220	Troubleshooting shaft knot indicators	Task redundant with ALT1870
NT0410	Inspecting explosive gas detector	Task redundant with ALT1870
AUT1460	Inspecting speed sensors	Task redundant with ALT1870
AUT1520	Inspecting tachogenerator	Task redundant with ALT1870
AUT1330	Inspecting optical sensors	Task redundant with ALT1870
AKT0730	Inspecting photo sensors	Task redundant with ALT1870
AUT1070	Inspecting electrical switchboard meters	Task redundant with ALT1870
AUT0370	Assembling/disassembling generators	On-the-job experience
ANT1580	Troubleshoot casualty power system	On-the-job experience
AUT0320	Inspecting casualty power breakers	
ANT0500	Inspecting multi-taps	On-the-job experience
ANT0310	Inspecting bulkhead terminals	On-the-job experience
AUT2100	Repairing casualty power breakers	On-the-job experience

ANT0980	Repairing multi-taps	On-the-job experience
ANT0780	Repairing bulkhead terminals	On-the-job experience
ANT1020	Repairing riser terminals	On-the-job experience
ATT0450	Inspecting hydraulic actuating control valves	On-the-job experience
AUT2300	Repairing electro-hydraulic valves	On-the-job experience
ALT1940	Troubleshooting gearbox pneumatic control system	On-the-job experience
AOT0260	Calibrating HVAC/NBCD air distribution systems	On-the-job experience
AIT0840	Repairing LP air dryer	On-the-job experience
AOT0330	Calibrating smoke evacuation systems components	On-the-job experience
AVT0980	Repairing diesel engine starter	On-the-job experience
AKT2140	Repairing pneumatic motors	On-the-job experience
AVT1160	Repairing main diesel engine pneumatic governors	On-the-job experience
ART0570	Repairing crane control unit	On-the-job experience
ALT1330	Repairing propulsion diesel engine starter box	On-the-job experience
ART0050	Adjusting vertical stores lift	On-the-job experience
AKT1890	Repairing level indicators	Redundant, covered by other tasks
ATT0980	Repairing flow meter	Redundant, covered by other tasks
ATT0060	Assembling/disassembling flow meters	Redundant, covered by other tasks
BBT0390	Installing ship’s fluid tanks elements	Redundant, covered by other tasks
AST0080	Repairing fuel metering unit	Redundant, covered by other tasks
AOT1360	Repairing salinity cells	Redundant, covered by other tasks
AUT2610	Repairing pressure transducer	Redundant, covered by other tasks
AUT2790	Repairing tension transducers	Redundant, covered by other tasks
AUT2750	Repairing tachogenerator	Redundant, covered by other tasks
ATT0170	Assembling/Disassembling pressure sensors	Redundant, covered by other tasks
ATT0730	Installing pressure switches	Redundant, covered by other tasks
ATT0180	Assembling/disassembling pressure switches	Redundant, covered by other tasks
ATT0720	Installing pressure sensors	Redundant, covered by other tasks
AUT2530	Repairing optical sensors	Redundant, covered by other tasks
AKT2080	Repairing photo sensors	Redundant, covered by other tasks
ATT0270	Assembling/Disassembling vacuum sensors	Redundant, covered by other tasks
ATT0840	Installing vacuum sensors	Redundant, covered by other tasks

AUT2600	Repairing pressure sensing temperature control valves	Redundant, covered by other tasks
AUT2820	Repairing thermistors	Redundant, covered by other tasks
AUT2840	Repairing thermostats	Redundant, covered by other tasks
ATT0250	Assembling/disassembling temperature sensors	Redundant, covered by other tasks
AUT2780	Repairing Temperature transducer	Redundant, covered by other tasks
AUT2290	Repairing electrically operated thermostatic control valve	Redundant, covered by other tasks
AUT 2830	Repairing thermocouples	Redundant, covered by other tasks
AVT0970	Repairing diesel engine pyrometer	Redundant, covered by other tasks
ATT0820	Installing temperature sensors	Redundant, covered by other tasks
ANT0160	Calibrating halon fire extinguishing system	Redundant, covered by other tasks
ANT0060	Assembling/disassembling fine water spray components	Redundant, covered by other tasks
AUT1830	Installing damage control components	Redundant, covered by other tasks
ANT0860	Repairing explosive gas detectors	Redundant, covered by other tasks
ATT0790	Installing solenoids	Redundant, covered by other tasks
ANT0960	Repairing manual pull stations	Redundant, covered by other tasks
ANT0250	Evaluating halon fire extinguishing system to include	Redundant, covered by other tasks
ANT1300	Testing fine water spray system	Redundant, covered by other tasks
ANT1340	Testing Halon fire extinguishing system	Redundant, covered by other tasks
ALT1770	Testing shaft knot indicators	Redundant, covered by other tasks
AOT1700	Testing salinity cell	Redundant, covered by other tasks
AOT0280	Calibrating salinity cell	Redundant, covered by other tasks
AUT3520	Testing optical sensors	Redundant, covered by other tasks
AUT3550	Testing photo sensors	Redundant, covered by other tasks
AT0240	Testing diesel engine pyrometer operation	Redundant, covered by other tasks
AUT3860	Testing thermostats	Redundant, covered by other tasks

AUT3810	Testing temperature transducers	Redundant, covered by other tasks
AUT3790	Testing temperature sensors	Redundant, covered by other tasks
ANT1370	Testing leak/flood detection system	Redundant, covered by other tasks
AOT0240	Calibrating fitted refrigeration leak detection system	Redundant, covered by other tasks
TAOT1640	Testing fitted refrigeration leak detection system	Redundant, covered by other tasks
AUT3620	Testing pressure sensors	Redundant, covered by other tasks
AUT0600	Calibrating pressure sensors	Redundant, covered by other tasks
AUT3640	Testing pressure transducers	Redundant, covered by other tasks
AUT3720	Testing speed sensors	Redundant, covered by other tasks
AUT3780	Testing tachogenerator	Redundant, covered by other tasks
ANT1290	Testing explosive gas detector	Redundant, covered by other tasks
AUT3820	Testing tension transducers	Redundant, covered by other tasks

Discussions/Decisions:

23 Oct 17

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

5.The board convened at 1300 on 23 Oct 2017. The lack of available board chair persons resulted in a late assignment of a board chair for the LS USQ. Consequently, there was a delay in commencing the board resulting from necessary time to complete administrative details in support of the board chair's arrival. On commencement of the board, an introductory brief was conducted by the TDO to familiarize board personnel with the QSWB expectations and identify

proposed sequences and timings for the conduct of the board. This brief was truncated when it was noted that the board composition was comprised of a number of board members who had previously participated in the RQ LSW QSWB. The Board Chair addressed the board and provided introductory comments and a brief background on his career before facilitating introductions amongst board members. The board chair then disseminated intentions regarding the method for dispute resolution, the disposition of administrative periods during the course of the board, and the daily itinerary to be followed. Technical equipment was tested for functionality prior to board adjourning for the day.

24 Oct 17

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

6. The board commenced this day's work with a review of the USQ-AB ROP to provide a foundation of the knowledge, skills and abilities (KSAs) which will have been formally addressed before trainees reach the USQ-LS course. Advantages and disadvantages of the KSAs as well as the process used to build the USQ-AB were evaluated with the intent to align wherever possible with practices from the USQ-AB board to maintain similar terminology, nomenclature, etc. The discussion concluded that a similar way of interpreting the tasks should be utilized on the LS ELEC SPEC board to ensure that trainees were presented with familiar terms. Co-operation with the USQ-LS MECH SPEC QSPWB would be sought to maintain a common look.

7. Lessons Learned from previous boards were presented by the TDO in order to emphasize that the total testing time for POs needs to be properly captured, in minutes, with the focus centred on communicating the total testing time required to process an entire class. Errors in this calculation in previous boards resulted in compounding scheduling errors during previous course implementation.

8. The board en masse, expressed multiple concerns regarding the lack of consultation with Camosun College and NFS(A) academic staff (providers of legacy QL5 Electrical Technician academic training for the West and East coast, respectively). Direction provided prior to the formation of the board emphasized that academic staff and NFS(A/P) instructors would be present for a minimum of one (1) week of the board's assigned production time to contribute to QSP design. The board confirmed that these resources would not be available on-site, and this led to follow-on discussion about the best means to conduct academic training in the future. In

particular, much discussion was focused on the ability to deliver academic training via e-learning in consideration of the soon-to-expire contract with Camosun College.

9.SMEs who recently attended the legacy course at Camosun College explicated how certain classes at the college were modified for military consumption. The board discussed the accreditation acquired at Camosun College, and the value in continuing to provide military education which meets or exceeds similar civilian standards. For example, legacy courses (particularly academic training) has traditionally been instructed to a depth and breadth that allowed senior personnel to challenge the Certified Electrical Technologist professional designation. There are myriad opinions expressed in the RCN regarding the return on investment of such rigorous training. Conflicting information during the course of the board, including information presented at the Admiral--Pacific Fleet brief delivered in October did not clarify whether the strategic vision of the RCN is to continue the depth of such training or align with, but not subscribe to standardized international operational training such as the International Maritime Organization (IMO) 708 standard. After heated debate, the board consensus was that academic training should be reined-in to approach the level of a journeyman technician—neither reduced to the level of an operator (as per IMO 708) nor so intense as to prepare trainees to challenge the CET qualification. TDO put the “academic instructor” discussion in the Parking Lot and indicated that they would investigate the issue.

10.The decision was made to ban the use of the term PO and EO to describe the preliminary groupings of JTAR tasks to avoid prematurely organizing tasks under legacy conventions. Avoiding such conventions would, in theory, maximize the effectiveness of the TDO-directed scalaring process. The importance of Critical ECs was brought up by the TDO. The latter re-iterated that it’s important to break lessons down into chunks of time allocated to each part of a subject.

11.Legacy pre-requisites for LS Electricians such as fleet examinations, minimum at-sea experience, time-in-service, etc. was discussed at length by the board. The consensus opinion recognized that there must be a mechanism to ensure a sufficient degree of deck-plate experience servicing equipment is acquired before commencing technician training. A series of recommendations for establishing this threshold experience was devised and recorded in Chapter 1 of the QSP.

12.Senior SMEs tabled the notion that under the legacy training delivery scheme, junior personnel are being sent on-course with insufficient experience applying their skills on in-service machinery. This is a result of attempts to fill the billets necessary to maintain the Fleets’ operational tempos. The SMEs highlighted the disadvantages this pressure is creating, foremost that the most recent generation of technicians can display difficulty with basic hand skills as a result of being loaded on subsequent training without a reasonable skills consolidation period due to a lack of available platforms for training.

13.Bearing these concerns in mind, the board (under TDO-direction) generated a list identifying the board’s conception of the nature of KSAs a LS Electrical Specialist could expect to be tasked with onboard a ship upon their arrival. The generalized description is as follows:

- Maintain all electrical equipment, including:
- Corrective Maintenance (CM)
    - Preventive Maintenance (PM)
    - Network Maintenance in support of platform management systems such as the Integrated Platform Management System (IPMS)
- Respond to MCR emergencies

- Utilize an enterprise resource system (ERP) such as the Defence Resource Management Information System (DRMIS) as a system of record for:
    - CM
    - PM
    - Materiel support needs (Stores)
    - Raising notifications ashore
- Performance management of subordinate personnel:
- Use of the CFPAS system to support subordinates' performance management:
    - Production and use of less-formalized performance management mechanisms i.e. divisional notes. *
- Advise & Brief the Chain of Command at appropriate levels of detail
- Naval Environmental Tasks, such as:
- Section Base Teams (SBTs)
- Damage Control Responsibilities
- Routine duties in support of sailing evolutions
- Watch-keeping duties
- Integration with other MAR TECHS

NB : *SLt Valentine suggested that the new Leadership, Respect and Honour course being implemented would meet this need independently of direct teaching points within the LS Electrical Specialist QSP.

15.The proposed KSAs were checked against the JBOS job description of the LS Electrical Specialist for congruency. The job description is as follows:

"The primary responsibility is to perform planned and corrective maintenance including adjusting, cleaning, operating, fault diagnostics, troubleshooting, repair, refurbishment of faulty electrical and electronic components; to maintain the operational effectiveness and availability of all flight deck electrical and electronic systems and the testing and installation of equipment and systems.

"The main tasks are the execution of planned and corrective maintenance in order to maintain the operational effectiveness of all power generation and distribution equipment and systems and the maintenance of calibration of precision electronic and pneumatic control systems/components related to propulsion, ancillary, auxiliary, HVAC, electrical, damage control, liquid level management, sanitary treatment, and power generation and distribution machinery on board HMC Ships.

"The Elect shall read/interpret technical drawings, schematics, and specifications in performance of their duties; complete all associated maintenance administration; have detailed skills and knowledge on electrical/electronic, power generation and distribution systems and refrigeration systems, and a general knowledge of mechanical systems. The Elect may be called upon as Engineering Watchkeeper dependent on class requirement."

16.Board consensus found that the KSAs identified were consistent The consensus was that both lists were in agreement.

17.Based on the JTAR task list provided by the Occupational Advisor (OA), LS Electrical Specialist tasks were clustered into preliminary categories, based on the first word of each task according to the following divisions:

- Troubleshoot
- Repair
- Inspect
- No train
- Calibrate
- Connect
- Install
- Assemble
- Clean
- Collect
- Test
- Draft

18.When the board encountered the task “Calibration of Puckmaker” a significant decision point was reached by the board. The consensus recognized that of the 766 tasks provided by the OA, several of the tasks were redundant or no-train by experience in their natures. This stems from the fact that although ships are fitted with specific pieces of kit, the conceptual operation of electrical aspects that support many pieces of equipment are universal in their nature. For example, if a technician is taught calibration methods in general, they should possess the necessary critical thinking skills, experience, and education to utilize resources in an appropriate fashion to apply these skills to a variety of calibration tasks. Going forward, the decision was made to stop teaching ‘electrical systems’ and instead use systems-level thinking in the organization of electrical training. Consequently, tasks would be grouped by the relationships between them and the technical knowledge required to perform work on these tasks.

19.Following a similar line of though as presented in para 18, the board recognized that high-reliability soldering is no longer utilized onboard ship. The repair-by-replacement philosophy has removed the requirement and opportunity to repair equipment that traditionally requires this skill. The decision was made to remove high reliability soldering from the QSP.

20. The board filtered all JTAR tasks that pertained to equipment that is no longer in-service (such as equipment particular to IROQUOIS- and PROTECTEUR-Class vessels) or is equipment that is most appropriately learned through experience.

Oct 25

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME

Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

21.All galley equipment tasks were determined NT, as *Key Foods* and its Atlantic equivalent are now contracted to deal with all galley equipment.

22,The board used the JTAR as a reference for certain task code prefixes in order to facilitate sorting tasks. The coded reference list can be found as Annex 1 to Appendix C.

23. The board discussed the need for consistency in terminology, its application and the intended meaning. It was noted that many of the performance verbs utilized by the OA have virtually identical meanings. The OA was asked to develop discrete and distinct meanings and intents for the following task verbs:

- Inspect
- Repair
- Test
- Trial
- Troubleshoot
- Adjust

24.It was discussed that revisions of the OS/AB QSP, interim development of the LS Electrical Specialist USQ & LS Mechanical Specialist USQ, as well as development of the MS QSP should reflect a consistency in the use and definition of terminology to ensure future trainees are instructed to convey briefs in a common vernacular.

25.Based on the decision rendered in para 24, it was agreed that when the term “repair” is mentioned in the name of a task, “replace” is assumed, if applicable.

26.The subject of degaussing was debated. Specifically, the board attempted to determine which aspects of using/maintaining the system should be taught at a level subordinate to LS. Review of the OS QSP revealed that service members trained to this level understand how to to maintain and operate the degaussing system. Similar knowledge requirements were consequently stripped from the LS Electrical Specialist task list.

27.As discussed in para 23, certain terminology is ill-defined. A consultation with the concurrently sitting LS Mechanical Specialist board was conducted to try to establish consensus. This discussion was terminated when the respective board chairs determined that neither the Mechanical, nor Electrical parties had solidified their positions enough. The Electrical board then re-convened on their own to determine the usage of the following terms:

- Inspect:
    - Looking for potential issues
    - PM
    - Visual
    - Examine

- Test:
    - Testing for operation
    - Trials (Set parameters)

    - Measure

- Troubleshoot:
    - Assemble/Reassemble/Disconnect
    - Fault finding
    - Investigate (low level)
    - Detailed inspection

- Repair:
    - Replace
    - Calibrate (set to a known setting)
    - Reset
    - Adjust

- Evaluate = Recommend

28.After conferring with the LS Mechanical Specialist board, it was observed that both sides had independently arrived at reasonable approximations of matching definitions. The final consensus was recorded as follows:

- Troubleshoot: The inspection, investigation, evaluation and analysis of a piece of equipment or component to determine its functionality.
- Repair: To bring back to service after a failure, including the disassembly, reassembly, installing, adjusting and replacement.
- Test: To calibrate, trial, monitor, measure, review and operate to bring back to a specified set of parameters.

29.Based on information provided by the NTDC MSE Division Commander, all tasks referring exclusively to AOPS-related tasks were removed from the task list as they are beyond the mandate of this QSP.

Oct 26

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

30.The board reviewed the task list provided by the LS Mechanical Specialist board. These tasks were assigned to the board because the Mechanical Specialist board believed they were electrical tasks in nature. The board audited the tasks, identifying any relevant tasks for inclusion, no-training tasks as appropriate, and returning any erroneously identified tasks to the LS Mechanical Specialist board.

31.Thirty-two (32) tasks that were originally in the LS Electrical Specialist JTAR were re-directed to the LS Mechanical Specialist as apposite per para 30.

32.The board partook in a TDO-provided brief informing board members about PO creation, the job process organization and its associated steps. Following this, the TDO scalared a hypothetical PO to provide a worked example. On completion of the brief, the board commenced with PO organization and creation.

33.After the initial PO creation, the board reviewed PO and EO titles and associated tasks. After consulting the JTAR, “Initiate work request (AFT1500)” was moved from NT to Trained (Technical Admin). It was then decided to make “Technical Admin” into an EO and to move it under another PO.

Oct 27

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

34.The Board chair pointed out that the POs needed more consistency in their titles to address the fact that the JTAR-derived tasks mix platforms, systems and specific equipment. For example, the Motors PO contained several different pieces of kit, and a mix of Marine Systems. It was agreed that the standard would be designed to instruct on the general topic of motors and will not address each individual onboard piece of equipment as a separate topic that must be individually instructed.

35.All preliminary POs were reviewed by the board as a whole for titling and task categorization:

- Preliminary titles were agreed upon for the POs.
- Power generation was separated from power distribution for ease of instruction and assessment.
- Casualty Power was made a part of power distribution.

- An idea was raised suggesting a joint training module focused on admin with the concurrently run MARMECH board. Further discussion will be required.
- Power controllers and electronic controllers were separated for ease of training and due to the complex nature of electronics as a subject.
- The decision was made to include Ozone Depleting Substances (ODS) and Vibration Analysis (VA) in LS Electrical Specialist training.

Parking Lot tasks at end of day:

AKT 0510 – Inspect regulator
ART0010 – Adjust boat davit
AUT4070 -- Troubleshoot navigation lighting
AUT3270 – Test electrical filters
AUT2240 – Repair electrical filters
AUT0500 – Calibrate domestic equipment sensors
ALT1890 – Troubleshoot controllable reversible pitch propeller supervisory panel
ALT1880 – Troubleshoot controllable reversible pitch propeller arbitration boxes
ALT0390 – Connect/disconnect breakout boxes
AUT0280 – Adjust variable speed drive controller
AUT2890 – Repair variable speed drive controller
AUT3930 – Test variable speed drive controller
ANT0020 – Assemble/disassemble aqueous film forming foam components
ANT1480 – Trial aqueous film forming foam system
ANT0280 – Inspect aqueous film forming foam system
ANT1550 – Troubleshoot aqueous film forming foam system
ANT0280 – Inspect aqueous film forming foam system
ANT0700 – Repair aqueous film forming foam system

Oct 30

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

36.PO titles were refined. The repeated use of the word “maintain” for many of the POs was debated. Although several other words were considered, in the end it was agreed that “maintain” best describes most of an electrician’s duties. Some members of the board requested that a section of the QSP mention definitions for “maintain” and other descriptive words for the

POs. These definitions should be identified for consistency across all QSPs and retroactively inserted into Appendix 2 of Annex C.

37.The board discussed where an ODS course should be placed and whether the unit technical inspector course should be taken by all LS technicians. Investigation revealed that the certification for ODS occurred at the OS RQ and therefore does not need to be included.

38.After some discussion regarding inclusion of general technical administration as a separate EO, including which PO was most appropriate to encapsulate this EO, it was decided that it should remain a separate PO. EO content was then discussed. The placement of the “analyze air in confined spaces” course was considered for inclusion. Although this course is presently instructed at the MS level, board SMEs asserted that this responsibility could be effectively handled by a LS. At the suggestion of the Board chair, the Admin PO was renamed Complete Technical Information to support the EOs Analyze TI, Implement TI and Evaluate Technical Administration. It was decided that VA should be created as a task within the Admin PO.

39.The POs were then placed in order of instruction. There was debate over whether Power Generation should be taught before or after Control Systems. Although PG&D has historically always been taught before Control Systems and Field Devices, everyone agreed that this doesn’t necessarily make it the best strategy.

40.The TDO briefed the board regarding how to compose POs, then went over the Chapter 2 template. The board commenced writing the POs.

41.Both the Electrical and Mechanical boards met to update each other and discuss certain points of contention:

- The Mechanical ROP up to the current date was reviewed.
- QSP Board end dates were brought up and whether the Electrical board’s end date will be different due to having started a week later than the Mechanical board. An ETA on that date is not yet known.
- An email was sent informing the boards that academic instructors will be available to consult on the QSP for the week of 13-17 November for the Electrical board.
- Tasks returned to the Mechanical board by the Electrical board were explained.
- The term “evaluate” was discussed and it was agreed that its use in tasks containing the word “system” will be associated with a MS level and above.
- The Admin PO was discussed:
    - Electrical will re-add TI (with only a few dissenting votes on the matter) after consultation with East Coast subject matter experts.
    - Drafting (Mech) and Schematic Drawing (Elec) will be removed from Admin and will be put into separate EOs, respective of the specialty.
    - Overall agreement on associated tasks:
        - Naval Technical Communication
        - DRMIS
        - TI
        - Analyze Technical Documentation

42.Vibration Analysis was removed from the Admin PO and moved into Service Electric Motors (PO 008), to align with the JTAR task regarding “HT0160 Operate Vibration Monitor Suite.” The board sought approval from the NPTG HQ Training Manager to move task down to the LS level from the MS RQ.

43.Initial PO drafts were reviewed by the board and modified, where necessary. Draft POs were sent off to NPTG HQ Training Manager.

Oct 31

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

44.The board debated the deviating from the verb definitions previously identified by the Mechanical Specialist board. After much discussion, the decision was made to move the term “calibrate” from the Test definition into the Repair definition. The sub-division of task verbs was adjusted to reflect the following categories:

- Troubleshoot:
    - Inspect
    - Investigate
    - Analyze
- Repair:
    - Replace
    - Adjust
    - Calibrate
    - Assemble/disassemble
    - Install
- Test:
    - Trial
    - Monitor
    - Measure
    - Operate

45.Placed in the Parking Lot today:

- The board made inquiries with NPTG HQ and NFS with respect to any regulations or policy that control PIP creations to the reference for creating a PIP for threshold testing, in addition to what constitutes a Leading Seaman acting lacking.
- The board will inquire further as to Yesterday’s discussion regarding the minimum, maximum and optimal number of trainees in the LS Mar Tech Electrical Specialist course, in

addition to the pre-requirements for said course, e.g. LS rank with accumulated 180 sea days, Chapter 1 (Para. 1-5) special instructions, etc.

Nov 1

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
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Eustace	TAS	LS	00379	HMCS OTTAWA	SME
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46.The board commenced with a brief from the CO NTDC(P) Cdr Koester regarding the potential opportunity for inclusion of TEL into the QSP development. The board members confirmed the mandate of the board and the direction to design training for the “IDEAL” learning environment, including the equipment for EC/PC testing. All QSP design decisions will be made to reflect this directive.

47.Cdr Koester’s question period provided an opportunity for the Board Chair to ask about the lead time for equipment acquisition to support the introduction of TEL into the QSP. Specifically, the board wished to know if they could design the QSP to align with training capabilities provided by an omnibus electrical solution i.e. electrical simulation suite. Cdr Koester instructed the board to develop the QSP as if there were no equipment-derived restrictions to the conduct of training.

48.Upon completion of the brief the board continued with the process of scalaring out the EOs to facilitate training design.

49.The following changes were made to Chapter 2:

- Servicing SCRs was added as a first EO to PO 003.
- The following tasks were moved from PO 003 to PO 004:
    - ALT1360 Repair shaft knot indicators
    - ANT0060 Assemble/disassemble fine water spray components
    - ALT1770 Test knot indicators
    - ANT1300 Test fine water spray system
    - ANT0250 Evaluate halon fire extinguishers system
    - ALT2220 Troubleshoot shaft knot indicators
- The following tasks were moved from PO 005 to PO 007:
    - AUT1980 Repair 440 VAC distribution panel
    - ANT1020 Repair riser terminals

    - AUT2990 Test 440 VAC distribution panel
- Other tasks moved:
    - “AUT2590 Repair power supplies”, from PO 006 to PO 003.
    - “AUT0700 Connect/disconnect power electronics”, from PO 003 to PO 002.
    - “AUT0580 Calibrate power generation control system”, from PO 002 to PO 007.
    - “ALT1720 Test PLC”, from PO 002 to PO 003.
    - “AUT2740 Repair synchroscope”, from PO 005 to PO 004

50.Placed in the Parking Lot today:

- The board decided to consult with the Mechanical Specialist QSP board on the subject of the Unit Technical Inspector course and if there would be any mandatory required training for Mar Techs. As the Mechanical board is waiting to receive a response NPTG HQ Training Manager.

Nov 2

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

51.The Board commenced the day with a discussion of where to place the essential academic knowledge. EOs. The following list was generated:

- PO 002:
    - *Apply a detailed knowledge of the concepts of circuit analysis*
    - *Apply a detailed knowledge of essential mathematic skills*
    - *Apply a detailed knowledge of theoretical concepts of control systems*

- PO 003:
    - *Apply a detailed knowledge of theoretical concepts of control systems*
    - *Apply a detailed knowledge of current microcomputers*
- PO 004:
    - *Apply a detailed knowledge of electronic circuit fundamentals*
    - *Apply a detailed knowledge of electrical circuit fundamentals*
- PO 009:
    - *Apply fluid power theory*
    - *Apply a detailed knowledge of pneumatic systems and electrically operated and controlled fluid power*

52.The TDO provided guidance on the construction of the Enabling Objectives, specifically paras 1-3 and conducted a demonstration for the board. Board members then individually commenced drafting paras 1-3 for all of the EOs. The paras 1-3 completed by the end of day were reviewed as a group by the board.

Nov 3

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

53.The Board continued paras 1-3 development for the remaining EOs.

54.The Electrical and Mechanical boards attended a brief by Lt(N) Sprang on the TEL products currently available to the CF. There was much discussion on whether or not it was practical to incorporate requirements for TEL that may not be available when the courses being created are actually being run. Lt(N) Sprang encouraged the attending QSP boards to put everything they felt was required, useful and desirable into the QSPs, regardless of current availability. She also insisted that the boards contact her with questions, requests and requirements for TEL, as this is the best way to get new products created and implemented.

55.Paras 1-3 which were drafted during today's board work were reviewed as a group with the board.

Nov 6

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME

Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

56.The Board continued paras 1-3 development for the remaining EOs.

57.The task “ALT1930 Troubleshoot fibre optic network” was added to the Parking Lot, as the board debated whether or not a fibre-related EO should be created. The consensus was to determine what fibre skills are included in the WEngTech QSP regarding fibre. After some research, the board created PO 012 Maintain Control Networks. The relevant tasks from PO 003 were removed and re-scalared into the newly created PO, under new EOs, as detailed:

- ALT0960 Repair Cat 5e cable network
- ALT0400 Connect/disconnect CCTV network and equipment
- ALT1500 Test CCTV network and equipment
- ALT1860 Troubleshoot control access network bus
- ALT1840 Troubleshoot CCTV server
- ALT1830 Troubleshoot Cat 5e cable network
- ALT0660 Inspect network communication cables
- ALT1490 Test Cat 5e network
- ALT1640 Test network communication cables
- ALT1930 Troubleshoot fibre optic network (removed from the Parking Lot)

58.Some of the board debated the structure of EO 003.02 and how it was scalared. It appeared to separate the components into marine-based systems, an organizational style the board had agreed to move away from. The EO was re-structured to reflect the actual instructional needs to be provided to teach controllers, not specific onboard marine systems.

59.The most recently completed EO paras 1-3 which had been completed were reviewed as a group with the board.

Nov 7

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
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60.Discussion was sparked by the PO 004 EOs over the consistency of the Standard section, specifically as it relates to sensors/devices listed and overall paragraph structures. The board agreed that the way tasks are introduced in one EO should carry on consistently throughout the rest of the EOs. The board reached a consensus to review all EOs later and either resolve the inconsistencies or instead create Para 10s accordingly, but then later decided on alignment of paragraphs between EOs with regards to structure and items mentioned (systems, sensors and devices).

61.Task “AUT1730 Inspect visually, motherboard” was removed from POs 002 and 003 and added to the NT list, as it is a skill acquired through job experience.

62.The board declared that it could not advance the training points as approval to develop was not yet received by the NPTG HQ Training Manager. The Board chair overruled this decision and advised the board to continue working on the teaching points as if approval had been gained.

63.The scalaring and order of tasks in PO 003 Service Electronic Controllers was re-examined. After discussion, it was decided to rename the PO *Service Electronic Control Systems* to better capture the nature of the tasks and ensure that the sequence of EOs made logical sense.

64.As the ROP was being reviewed, the NPTG HQ Training Manager arrived to touch base with the board, providing an opportunity to raise and discuss any outstanding issues:

- PO structure
- PO justifications, such as how the board has removed mentions of certain systems when teaching a common subject is preferable.
- Rationalizing that the UTI course does not need to be taught to all LS Mar Techs.

65.The NPTG HQ Training Manager approved the preliminary POs and progress-to-date of the board.

Nov 8

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
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Eustace	TAS	LS	00379	HMCS OTTAWA	SME
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66.The Board chair reviewed some re-structured POs and EOs, as well as some initial drafts of para 10 development. Para 10 discussion led to the subject of training aids, specifically the need for 3D simulators of generators. It was agreed that the TDO would investigate procuring a list of the available 3D simulators created by the Army Learning Support Centre (ALSC). The Board chair suggested that the most efficient solution would be to liaise with manufacturers in order to acquire the 3D simulators of their own products. The Board continued to draft Para 10s.

67.The possibility of removing “PO 010 Stabilize Impressed Current Systems” entirely was raised as all of the tasks are either accomplished through job experience or otherwise taught via various POs. It was suggested that much of the system-specific training could be abolished by instead ensuring that technicians are properly instructed concerning methods for using and researching publications.

68.In response to the discussion of para 67, PO 010 was removed from the QSP development.

69.The Board continued writing para 10s, expressing their difficulty with generating teaching points and appropriate training aids to enhance learning. The TDO impressed upon the board that they should express in detail the type of training aids they wish to have used, whether or not they know of their current existence.

70.Two of the board’s SMEs attended a portion of the LS Mechanical Specialist board’s discussion with the NFS(P) academic instructor regarding mathematics instruction. The consensus was that the level of math required for Mecanical Specialist training is significantly lower than what the Electrical Specialist QSP should and would include. Consequently, the decision was made that it is not possible to have a common LS Technician academics curriculum.

71.While reviewing the Troubleshooting Secondary Power Generation Systems EO para 10, the board contentiously debated the necessity of investing additional teaching time towards electrical safety (in addition to UPSs and 400 Hz systems, in general) when the fundamentals of these subjects will have already been instructed earlier in the course. The fate of the Secondary Power Generation Systems PO was then put to a vote. It was decided to transfer the applicable tasks (below) into PO 005:

- AUT1640 Inspect uninterruptible power supply
- AUT3980 Troubleshoot 400 Hz system
- AUT0590 Calibrate power supplies
- AJT0360 Repair helicopter start system
- AUT2690 Repair static frequency converter
- AUT0400 Assemble / disassemble static frequency converter
- AUT2340 Repair final resource system charger (MCDV PLC UPS battery charger)
- AUT0410 Assemble / disassemble uninterruptible power supply
- AUT2880 Repair uninterruptible power supply
- AUT0630 Calibrate uninterruptible power supply
- AUT3600 Test power supplies

Nov 9

Surname	Initials	Rank	MOSID	Unit	Role

Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

72.The TDO reiterated the board timeline, emphasizing that ten days remained to meet te board mandate. Re-focused, the board Board continued drafting Para 10s.

73.There was discussion over moving “EO 003.06 Service SCRs” to PO 004, as this PO includes electronic circuit fundamentals.

74.Board reviewed Para 10 drafts completed to date.

Nov 14

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

75.The academic instructor who was to be available via teleconference informed the board that he was sick and would not be reviewing the course today.

76.In place of the scheduled academic consultation, the Board continued drafting Para 10s.

77.All tasks from EO 007.03 were either removed and placed in EO 007.02, or moved to NT as detailed below:

- AUT1840 Install PG & D system components

- AUT1980 Repair 440 VAC distribution panel
- AUT2010 Repair backplane
- AUT3030 Test backplane

78.Primary and secondary power were amalgamated into “PO 005 - Maintain Power Generation Systems,” thereby making “PO 006 – Maintain Secondary Power Generation Systems” obsolete.

79.One of the senior SMEs presence was required at the MARE council meeting taking place in Victoria and was excused from the board for the following two days.

Nov 15

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

80.“EO 012.03 Test Control Networks” was removed, its associated tasks (listed below) moved into “EO 012.01 - Troubleshoot Control Networks” and “EO 012.02 Repair Control Networks:”

- ALT1490 Test Cat 5e network
- ALT1640 Test network communication cables
- ALT1500 Test CCTV network and equipment

81.The TDO, Board Chair and one of the board’s SMEs contacted Mr. Ian Dawson, a long-standing academic instructor at NFS(A) to discuss the depth of legacy academic content. He agreed that certain academic classes could be optimized. The board forwarded the para 10 academic element drafts for estimation of course timelines.

82.The board continued drafting Para 10s. The MarMech EOs for “PO 001 Complete Technical Information” were received and reviewed.

Nov 16

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson

Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

83.After reviewing the administration PO (001) provided by the LS Mechanical Specialist board, it was decided that an additional EO should be created to encompass the task of reviewing manufacturer specifications (AFT0470).

84.The subject of using current Lab Volt equipment assets which are native to NFS(A/P) was discussed. The consensus was that its adequacy as a troubleshooting practice tool is very limited, as this equipment is not an accurate representation of actual motor controllers. NFS(A) has examples of two industrial multi-fault configurable motor controllers that were built by technicians in MARLANT. The board decided more controllers should be developed in this style to facilitate practical assessments and advanced troubleshooting for technician trainees.

85.The board tentatively decided to leave differential and integral calculus in EO 002.05 due to the fact that the JTAR for MS and above has not been determined, and the level of advanced math required to support future RQ development is unknown. This material is to be reviewed upon completion of said JTAR to finalize requirements for training.

86.EOs continued to be reviewed.

Nov 17

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
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Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

87.The board once again discussed the validity of having all LS tech trainees take the Unit Technical Inspector (UTI) course; it was decided that not all members would need it as only one member of a unit would ever hold the appropriate stamp at a particular point in time. Therefore,

the UTI course would be considered "just in time" training, as the 3.5 days accorded to it could be better spent.

88.The TDO, Board Chair and a board SME spoke to Mr. Ian Dawson regarding the academic EOs. The requirement to include training in microprocessors was discussed, in addition to the potential for a PIP for the LS MarTech Electrical Specialist course.

89.The task "trial Halon fire extinguishing system" was removed from EO 004.05, as the board determined that "trialing" was a verification of operating parameters to a set specification.

90.The task "evaluating halon fire extinguishing system to include (ANT0250)" was understood to mean that a LS technician will not be evaluating the efficiency of the system, but rather its functionality.

Nov 20

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
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Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

91.The Board commenced drafting paras 4-9 for the EOs based on the work produced in para 10. The board discussed the complexity of motor governors, determining that the complexity of the topic was sufficient to necessitate it being its own EO under PO 003 Service Electronic Control Systems. Thusly EO 003.05 Preparing Engine Control Systems was created.

92.It was determined by the board that the teaching points contained in the various EOs under PO 008 Service Electric Motors were not substantial enough to justify their separation into three separate EOs. Additionally, based on the similarity of content, Service Electric Motors became an EO under PO 002 Service Power Controllers.

Nov 21

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson

Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

93.The board continued drafting paras 4-9, transitioning to EC creation. Chapter 3 was reviewed by the collective board. A review of the completed Chapter 4 sections commenced, with a focus on confirming timings and testing details. Task HT 0160 currently present in the MS Electrician task list was determined by the board as critical to be moved to the LS MAR TECH Electrical Specialist USQ due to the nature of the LS technician work.

94.Since the concepts of electrical troubleshooting are ubiquitous to many systems, the following tasks were determined to be NT and are included in the NT task list detailed at the beginning of this ROP.

Inspecting black & grey water collection system (AOT0430);
Inspecting black water collection system (AOT0440);
Troubleshooting calorifiers (AOT1850);
Troubleshooting helo fuel/de-fuel control panel (HFCP) (ALT1960);
Inspecting whistle controllers (AUT1810);
Inspecting whistle automatic cycle timer controllers (AUT1800);
Troubleshooting propulsion diesel engine controls (AUT2160);
Inspecting main refrigeration system (AWT0080);
Troubleshooting steering system (BAT0400);
Investigating system operation of stores lift (ART0390);
Inspecting PDE starter box (ALT0670);
Inspecting calorifiers (AOT0460);
Investigating system operation of crane (ART0370);
Troubleshooting crane (ART1280);
Inspecting centering unit (BAT0070); and
Troubleshooting propulsion diesel engine auxiliary starter box (ALT2150).

Repairing steering controller (BAT0280);
Repairing macerator controller (AOT1270);
Repairing black water collection controller (AOT1050);
Repairing refrigeration electrical systems (AOT1330);
Repairing centering unit (BAT0190);
Repairing helo fuel/de-fuel control panel (HFCP) (ALT1120);
Repairing black water treatment controller (AOT1060);
Assemble/disassemble main refrigeration system components (AWT0020);
Repairing whistle speed controllers (AUT2910);
Repairing calorifiers (AOT1070);
Repairing crane control unit (ART0570);

Repairing propulsion diesel engine starter box (ALT1330);
Adjusting vertical stores lift (ART0050);
Calibrating whistle speed controllers (AUT0640);
Calibrating stores lift (ART0130);
Calibrating refrigeration control systems (AOT 0270);
Calibrating crane controllers (ART0120);
Calibrating circuit cards (AUT0470);
Repairing aux steam generator electronic safety control (AIT0630);
Repairing control transformer (ALT1030);
Adjusting overload relay trip setting (AUT0210);
Adjusting overload relay (AUT0220);
Repairing control relays (AUT2140);
Assemble/disassemble black water collection system components (AOT0120);
Adjusting timing relays (AUT0270);
Testing electrical controllers (AUT3250)
Trialing black and grey water system (AOT1750);
Testing refrigeration electrical system (AWT0210);
Testing propulsion diesel engine starter box (ALT1740);
Testing whistle speed controller (AUT3950);
Testing stores lift (ART1060);
Testing control transformers (AUT3160);
Testing timing relays (AUT3870);
Testing control relays (AUT3140); and
Testing overload relays (AUT3530).

Repairing rectifiers (AUT2630);
Repairing timing relays (AKT2480);
Connecting/disconnecting power electronics (AUT0700)

Troubleshooting electric propulsion system (BTO0220);
Troubleshooting machinery space ventilation system (AOT1950);
Troubleshooting ship's whistles (AUT4080);
Troubleshooting low pressure air compressors (AIT 0980);
Troubleshooting high pressure air compressors (AIT 0960);
Investigating system operation of king post (ART0380);
Inspecting chiller units (AWT0060);
Troubleshooting windlass (ART1340);
Troubleshooting kingpost (ART1310);
Troubleshooting capstan (ART1270);
Troubleshooting propulsion motor (BTO0260); and
Inspecting governor stepper motor (AUT1180).

Repairing motor impeller (AUT2490);
Repairing capstan (ART0550);
Repairing boat davit (ART0530);
Repairing vane axial fan (AKT2540);
Repairing centrifugal pump (AKT1460);
Repairing kingpost (ART0650);
Adjusting sewage macerator (AOT0070);
Repairing enclosure fans (AUT2320);

Repairing windlass (ART0510);
Repairing dumbwaiter (ART0600);
Repairing LP air compressor (AIT0830);
Repairing governor stepper motor (AUT2360);
Repairing centrifugal fan (AKT1450);
Adjusting capstan (ART0020);
Repairing stores lift (ART0730);
Repairing HP air compressor (AIT0700); and
Adjusting stepper motor (AUT0250).

Calibrating ship's ventilation system (AOT0290);
Testing governor stepper motor (AUT3390); and
Trialing sewage macerator (AOT1800).

95.It was decided that material from EO 002.03 – 04 is to be tested together due to the similarity of the troubleshooting/repairing process.

Nov 22

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

96.The Board continued to review the content of Chapter 4. It was determined that to integrate desired technician skills and provide more hands-on development of the technician troubleshooting mindset, VA equipment would be utilized in the testing of EC 002.05. Further discussion was had regarding the timing for testing relative to the number of testing equipment. This prompted the decision to create an appendix to Annex G that would detail the testing time for each of the EC/PC.

Nov 23

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson

Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

97.The Board continued to review Chapter 4.

98.The following tasks were removed from EO 005.02:

- Installing power generation and distribution system components
- Assembling/disassembling generators

98.It was determined that EOs 005.01, 005.02 and 005.03 would have a combined EC. The task Calibrating power generation control system was removed from EO 007.02 (now 006.03).

99.The board incorporated the tasks from EO 006.03 into EO 006.02. EO006.04 was removed due to all of its tasks being redundant.

100.The Board commenced drafting PCs.

Nov 24

Surname	Initials	Rank	MOSID	Unit	Role
Robbins	ML	Lt(N)	00345	ADM(Mat)MEPM/DNPS2	Board Chairperson
Robinson	TN	Lt(N)	00211	NTDC(P)	Training Development Officer
Aubin	DL	CPO2	00379	NTDC(P)	SME
Penney	KG	PO1	00379	NTDC(P)	SME
Blackbourn	SI	PO1	00379	NTDC(P)	SME
Osborne	CG	MS	00379	PCC(P)	SME
Eustace	TAS	LS	00379	HMCS OTTAWA	SME
Wyns	JS	LS	00379	HMCS VANCOUVER	Secretary

101.It was determined that instructional development staff at NFS(A/P) are to be advised that the academic curriculum development shall keep the same breadth of information, while reducing the depth to be contained within a 6-month timeframe. This time is to include instructional, lab and testing time.

102.After some debate and consulting, the board agreed that there was no purpose in modularizing the LS Mar Tech Electrical Specialist course to accommodate reservists, as they are obligated to take the mechanical path. Modularization would still be conducted for the purposes of PLARs.

103.The Board finalized PCs and course timings. This was followed by production of the course history annex. Upon completion, TEL requirements were captured in Annex L and have been sub-divided into two lists. The first list identifies critical training resources which must be acquired before piloting the course. Failure to acquire these resources will create a situation in which teaching points, ECs, and PCs cannot be effectively tested as intended by this QSPWB. Where possible, interim mitigations have been suggested, but the quality of the training will be compromised. The second list identifies 'nice to have' elements which will enhance learning as well as ease of training delivery for trainees and instructors, respectively. Follow-on consultation with the NTDC TEL coordinator and the ALSC will be required to action this TEL annex. Board concluded……

<u>Lessons Learned:</u>

104.Boards composition should be standardized regarding the number of staffed positions to be used as SMEs. This is firmly established by NPTORD 4500-22, and the processes supporting this have clearly been circumvented in the conduct of this board. It is impossible to accurately schedule the time requirements to generate a board of sufficient quality without standardizing the resources available to support the board.

105.As per NPTORD 4500-22 "Subject matter experts—at least one SME shall come from outside NPTG to provide expertise." Failure to incorporate board representation from both coasts, as well as a dedicated academic curriculum developer presented significant obstacles to this board. Trade-wide acceptance and change management of this QSP will be paramount to effectively implement a pilot course due directly to a lack of holistic involvement from key stakeholders.

106.Individuals assigned to act as Board OPIs, support staff, or key personnel in approving board objectives cannot be placed into a position that renders them unreachable or uncommunicative for multiple weeks at a time.

107.IT resources and sufficient PP&S must be available for each member of the board. As experienced during this QSPWB, possessing approximately half the IT assets needed created significant delays in the key activity—writing—of the board.

108.All required administration e.g. CFTPOs, convening orders, etc. must be transmitted to personnel in advance of the board. This will ensure personnel are not removed from the board, or expected to report to home units while in the midst of the secondment.

109.When board personnel are identified and selected, particular attention should be paid to MELs, or pre-existing commitments (e.g. learning agreements) that excuse personnel from the board for extended periods of time on a regular basis. This is not an efficient application of human resources on an already time-limited, rigidly-scoped project.

108.The prevalence of a pervasive attitude has been observed which expresses one of two sentiments. This first statement can be paraphrased as "that's the way we've always done it,"

while the second phrase may be approximated as, “it was good enough for legacy courses.” Neither of these attitudes is conducive to the developmental needs of future trade requirements. Decreased financial resources, unprecedented access to TEL resources, and a changing demographic which constitutes the archetypal technical recruit presents an opportunity to create course content that better addresses the requirements of tomorrow’s electrical specialist. To address this, it is recommended future QSPWBs attempt to frame their courses independently following the guidelines of their TDO, without immediately referencing legacy documentation.

109.The methodology discussed in para 108 should be extended to the creation of testing requirements. An open-minded approach should be employed that encourages testing requirements that as much as possible find innovative, yet comprehensive and efficient ways to examine individuals in ways simulating on-board situations.

110. The next sitting of the JTAR for the MS MAR TECH, scheduled for January/February 2018 should exercise their professional experience to determine the train/no train decisions required for tasks as opposed to generating an exhaustive itinerary of every minute task that may be expected of a MS Electrician.

Guidance to Instructional Staff:

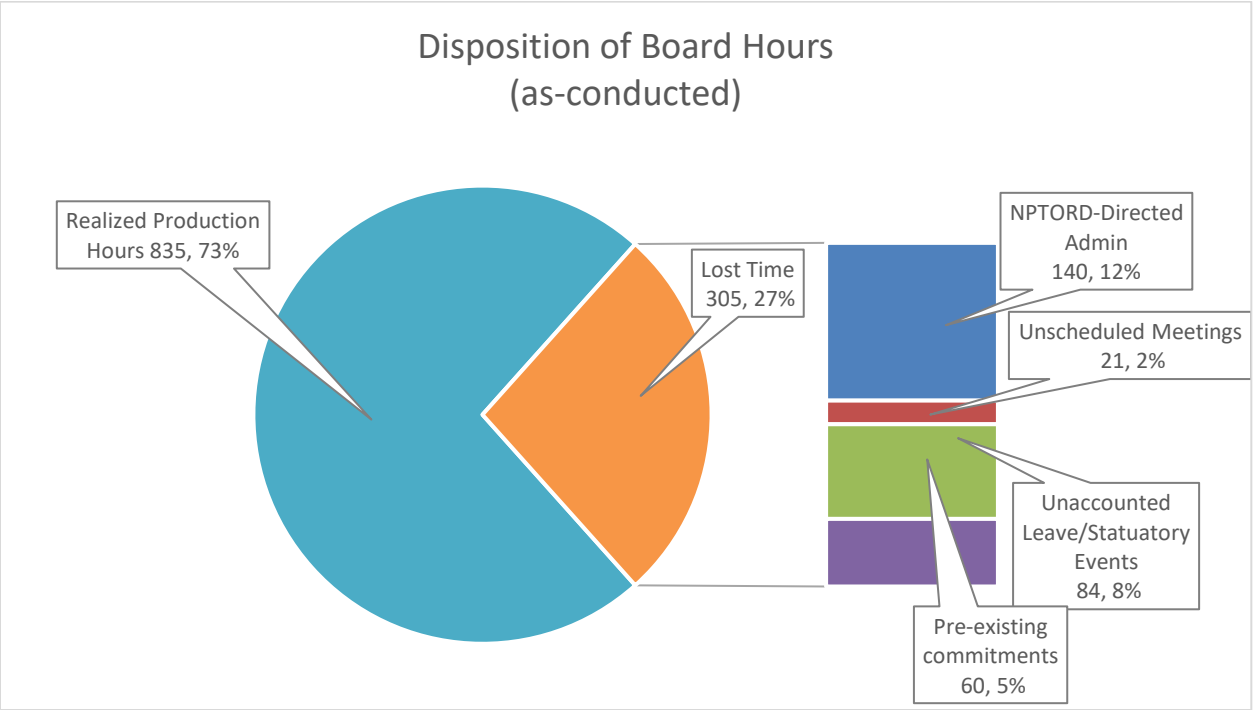
Follow Up Action:

Item	Action	OPI	Expected Date of Completion
1	Instructional checklists to be developed concurrently with MLP development.	NTDC(P)	TBD
2	Critical equipment purchases must be made prior to commencing the course.	NFS(P)	TBD
3	Math academics to be re-examined based on the results of the OS and AB MAR TECH academic review.	NFS(A/P)	TBD
4	Table B3.1 in NPTORD 4500-22 requires updating to reflect 50-minute training periods vs 60-minute training periods e.g. 180 min. for PT vice 150 minutes.	NPTGHQ Training Manager	TBD
5	QSP to be reviewed for changes which are necessary (e.g. academic depth of calculus to be taught) based on the JTAR and QSP developed for MS.	NTDC(P)	TBD
6	Academic instructors to develop a six-month academic curriculum that leverages PIP and DLN needs, as required. This academic portion must be scaled back to reflect a move away from CET certification.	NFS(A/P)	TBD

Item	Action	OPI	Expected Date of Completion
7	ALSC and NFS TEL coordinator to be engaged for development of teaching points	NFS(A/P) MIP content developers	TBD

1.
<u>Other Considerations (if applicable):</u>

The table below identifies the application of time as utilized by this board. Incorporation of many of the guidelines identified previously in this document can create substantial improvements in QSPWB efficiency going forward.



30 May 2023

Changes made to the QSP Post piloting USQS1E serials 0012 and 0014 detailed below;

1. Updated References for each EO.
2. Updated Test Details for each EO.
3. Added High reliability Soldering Training to align with the new Camosun contract and to be taught on the academic phase of USQS1E. This change was made because soldering skill is required onboard HMC ships as parts become more difficult to acquire.
4. Post Pilot timings table(RDIMs# xxxxxxx) were implemented giving reducing total course time from 98925 minutes to 87665 minutes.
5. Updated annex G to align with new EO timings.
6. Updated Ch.1 timing details; BAT+TAT+TTT+PT and ceremonial divisions.

02 Jun 2023

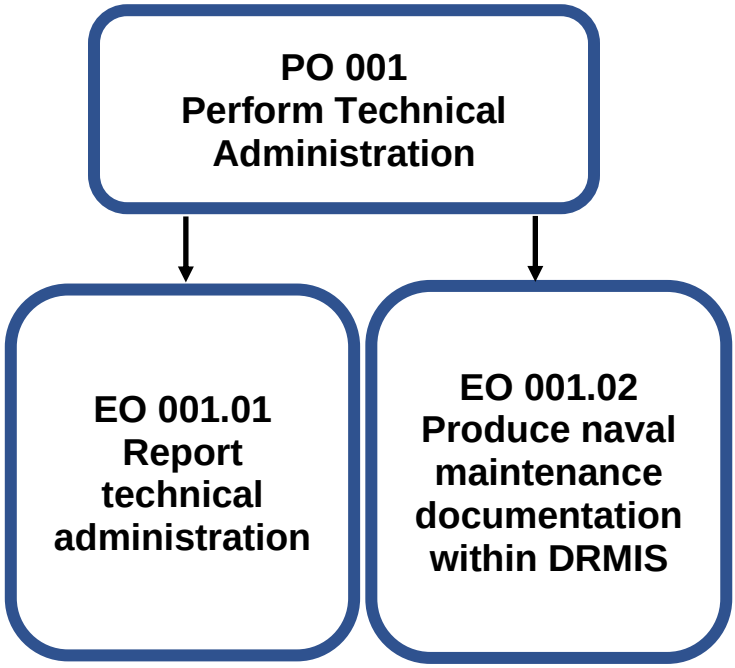
Completed Review of endorsed QSP. Saved as final version in RDIMs for approval.
RDIMS #1262199 v5.

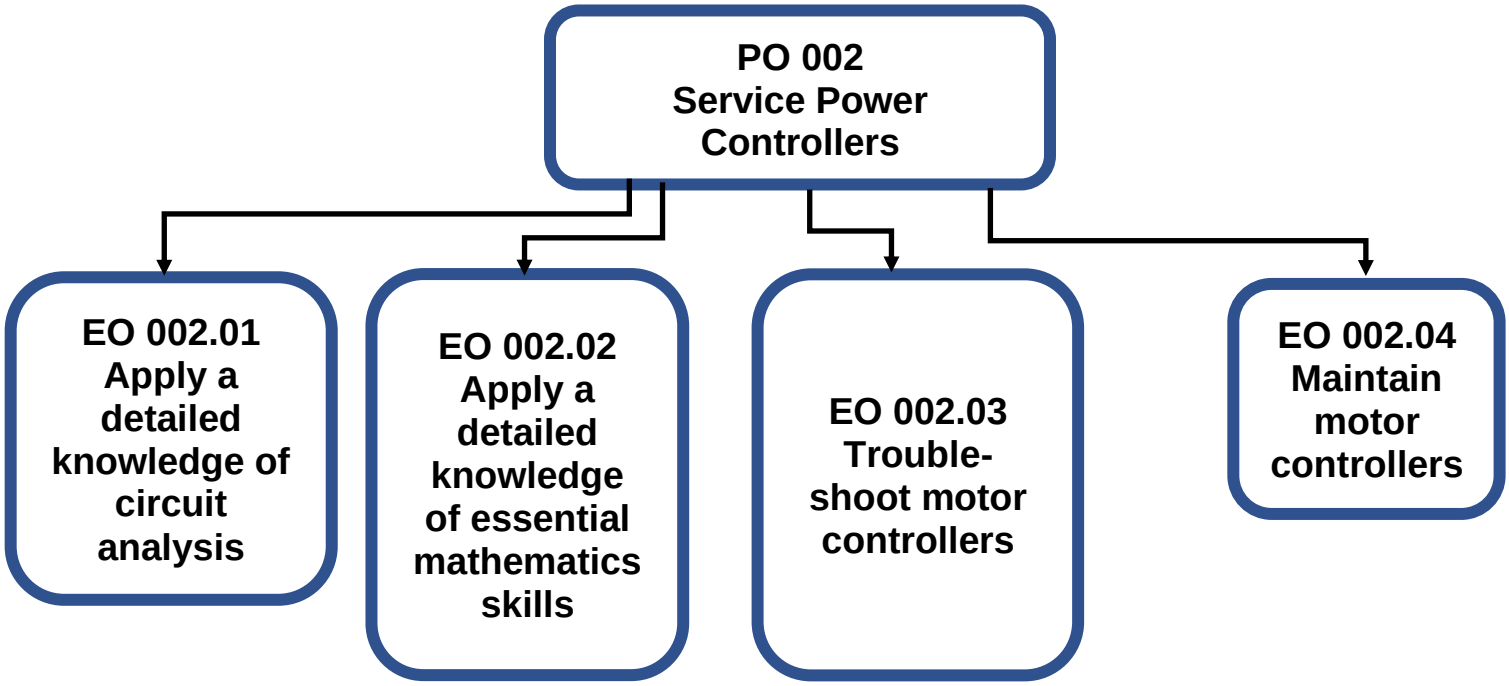
APPENDIX 1 – TASK PREFIXES (ACCORDING TO JTAR – MAR TECH)

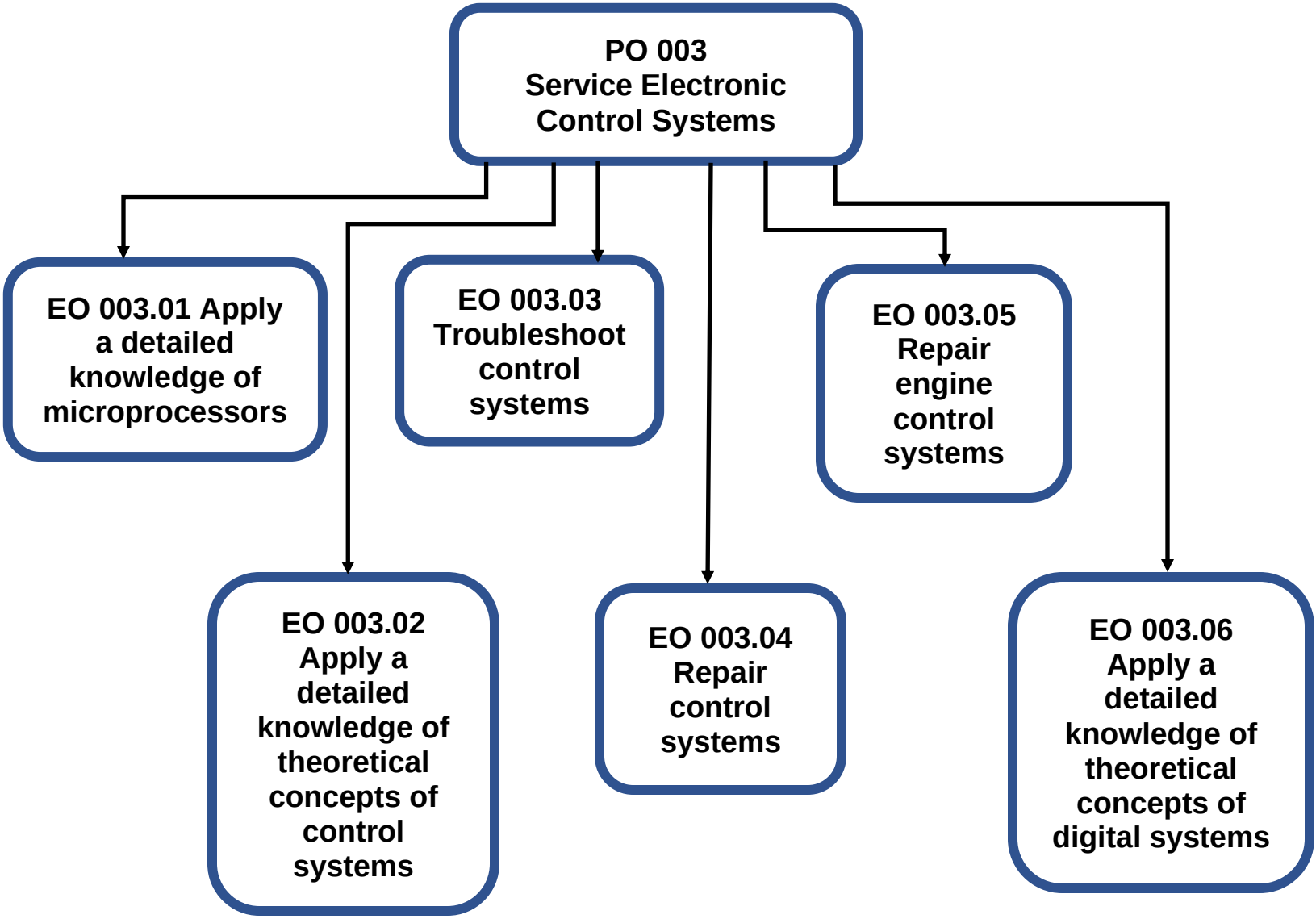
Acronym and Abbreviation	Term or Definition
AAT	Administration, Safety and Environment
ABT	Operations, Transmissions
ACT	Training, Administration
ADT	Training, Collective
AET	Training, Individual
AFT	Administration, Technical
AGT	Maintenance
AHT	Maintenance, Ancillary Systems
AIT	Maintenance, Auxiliary Systems
AJT	Maintenance, Aviation Support
AKT	Maintenance, Common
ALT	Maintenance, Control Systems
AMT	Maintenance, Fabrication
ANT	Maintenance, Firefighting/Damage Control
AOT	Maintenance, Hotel Services
APT	Maintenance, Hull Protection
AQT	Maintenance, Hydraulics
ART	Maintenance, Lifting Appliances
AST	Maintenance, Petroleum/Oils/Lubricants (POL)
AT	Maintenance, Technical Administration
ATT	Maintenance, Plumbing
AUT	Maintenance, Power Generation & Distribution
AVT	Maintenance, Prime Mover
AWT	Maintenance, Refrigeration and A/C
AXT	Maintenance, Replenishment at Sea Systems
AYT	Maintenance, Safety and Environment
AZT	Maintenance, Small Boats
BAT	Maintenance, Steering
BBT	Maintenance, Structures
BT	Maintenance, Transmissions
CT	Operations
DT	Operations, Ancillary Systems
ET	Operations, Auxiliary Systems
FT	Operations, Aviation Support
GT	Operations, Common

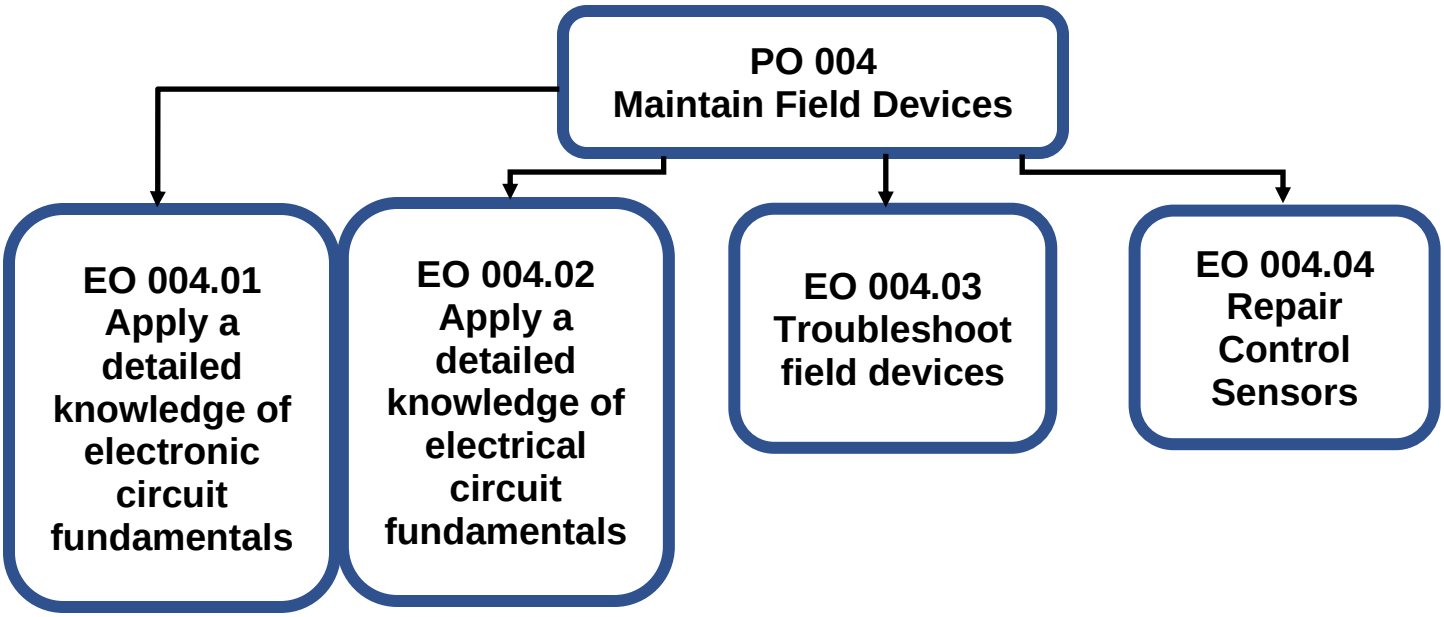
HT	Operations, Control Systems
IT	IT, Operations, Firefighting/Damage Control
JT	Operations, Hotel Services
KT	Maintenance, Submarine
LT	Operations, Hull Protection
MT	Operations, Hydraulics
NT	Operations, Lifting Appliances
OT	Operations, Petroleum/Oils/Lubricants (POL)
PT	Operations, Power Generation & Distribution
QT	Operations, Prime Mover
RT	Operations, Propulsion
ST	Operations, Refrigeration and A/C
TT	General Administration
UT	Operations, Safety and Environment
VT	Operations, Smalls Boats
WT	Operations, Stability
XT	Operations, Steering
YT	Operations, Submarine
ZT	Operations, Technical Administration

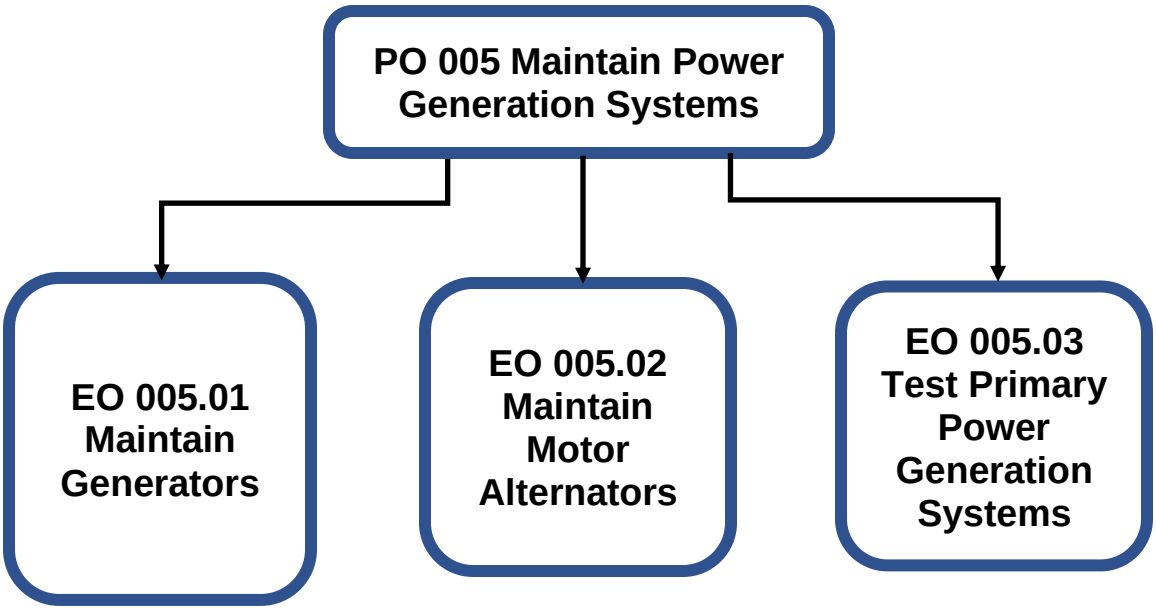
ANNEX D – QSPWB SCALAR

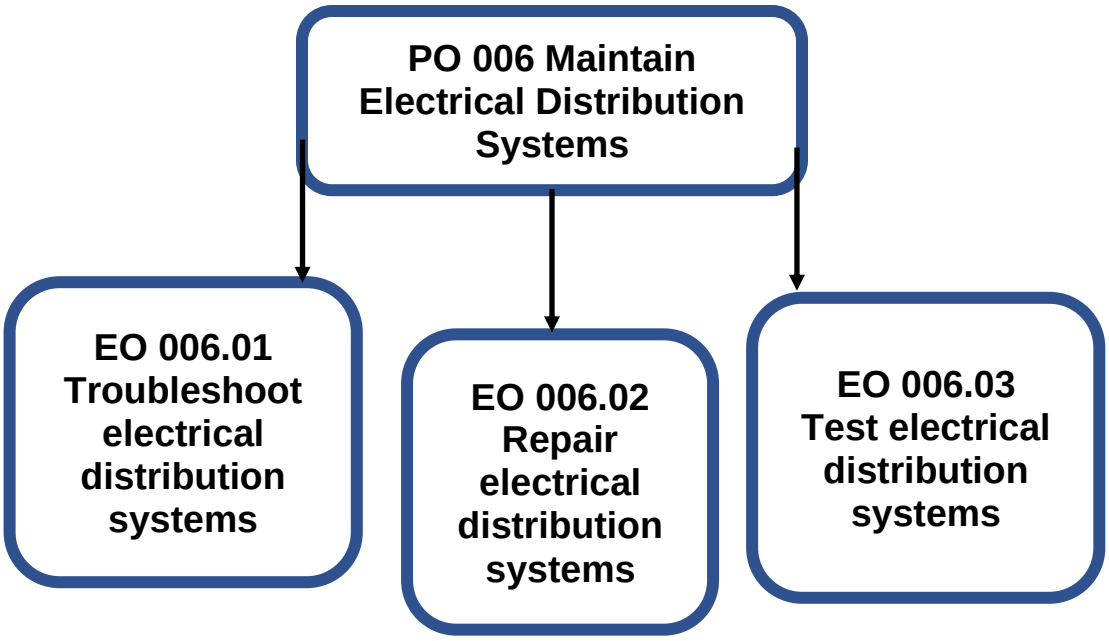


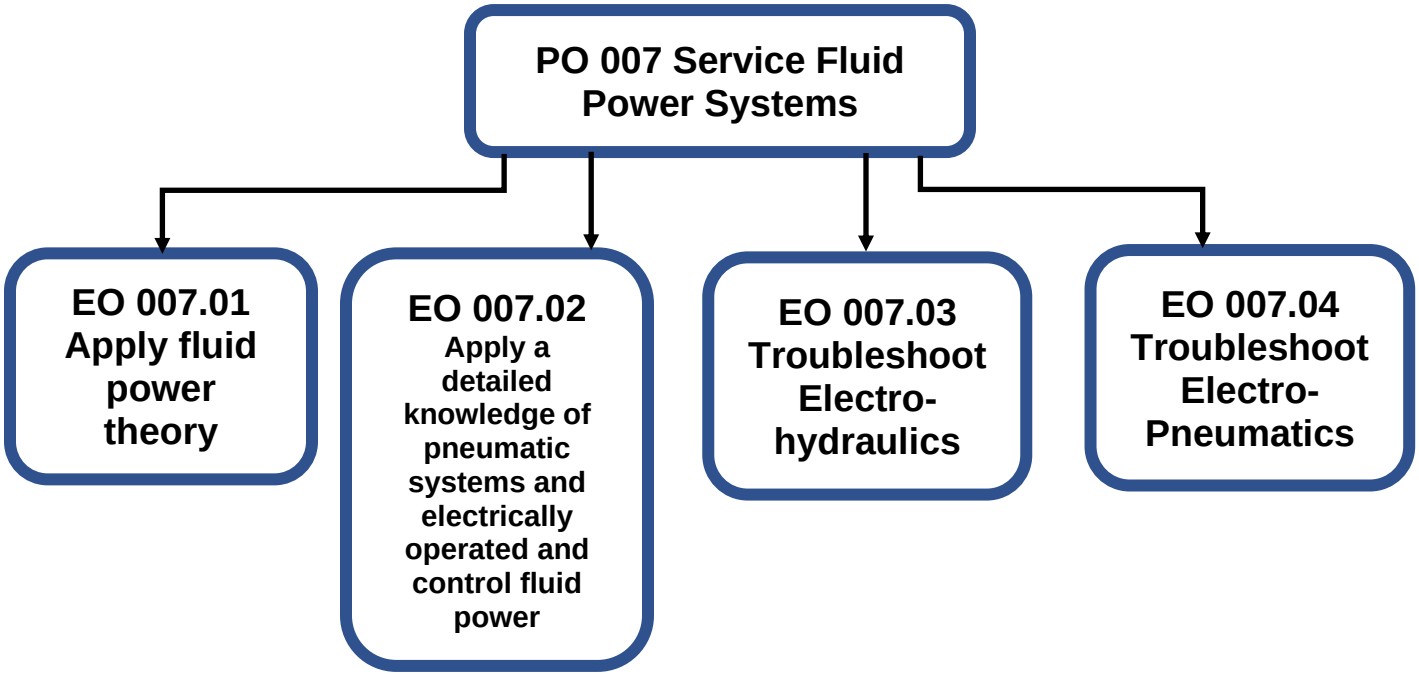


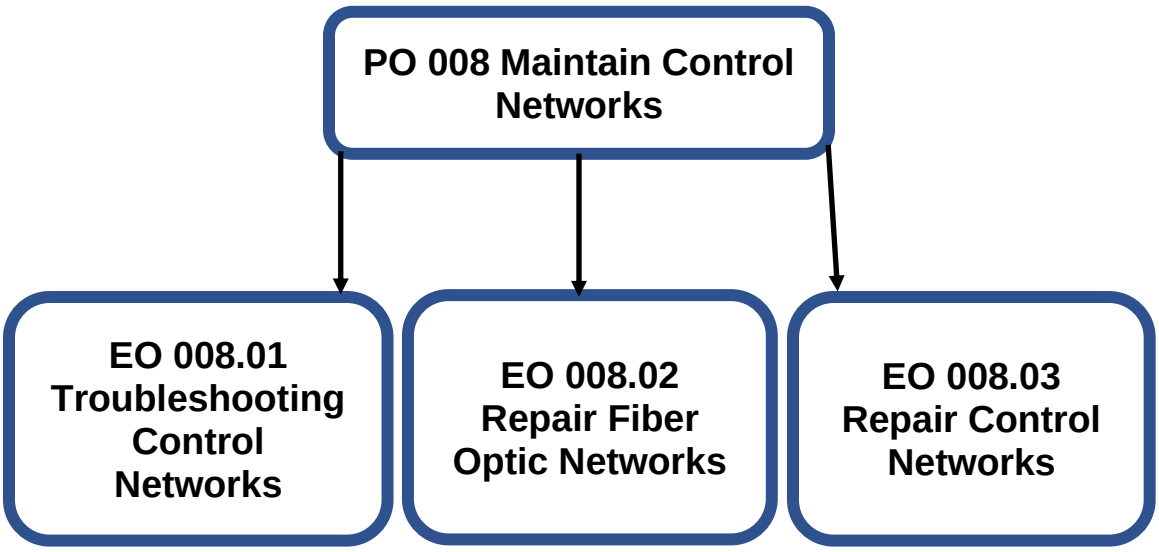

PO 003
Service Electronic Control Systems
EO 003.01 Apply a detailed knowledge of microprocessors
EO 003.02 Apply a detailed knowledge of theoretical concepts of control systems
EO 003.03 Troubleshoot control systems
EO 003.04 Repair control systems
EO 003.05 Repair engine control systems
EO 003.06 Apply a detailed knowledge of theoretical concepts of digital systems

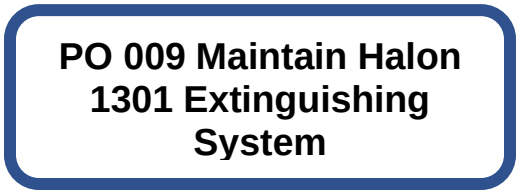

PO 004
Maintain Field Devices
EO 004.01
Apply a detailed knowledge of electronic circuit fundamentals
EO 004.02
Apply a detailed knowledge of electrical circuit fundamentals
EO 004.03
Troubleshoot field devices
EO 004.04
Repair Control Sensors

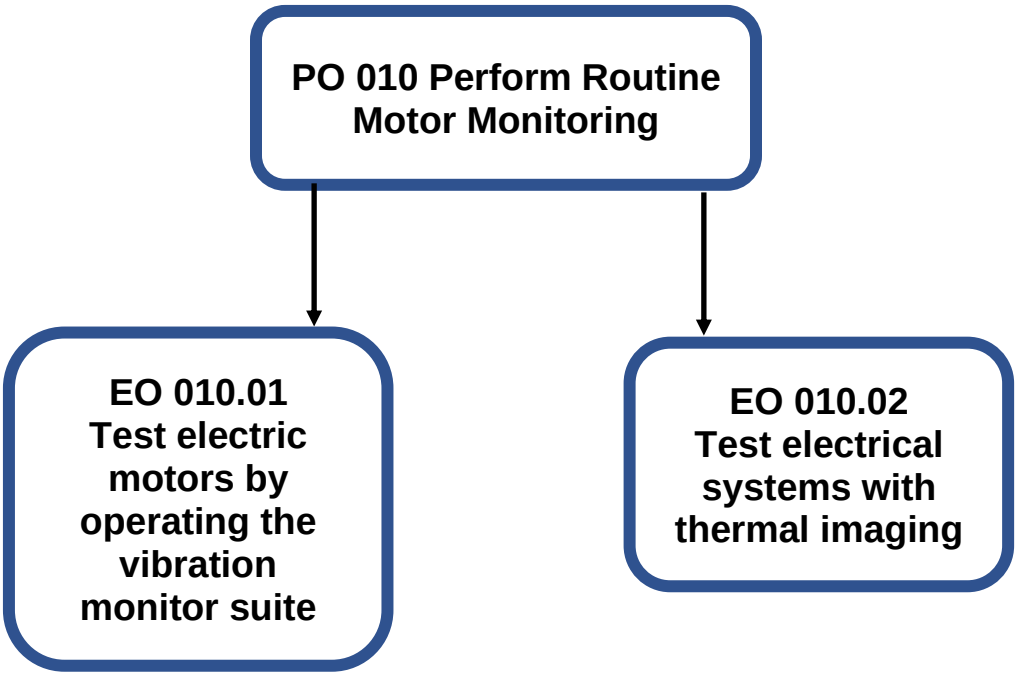

PO 005 Maintain Power Generation Systems
EO 005.01
Maintain Generators
EO 005.02
Maintain Motor Alternators
EO 005.03
Test Primary Power Generation Systems

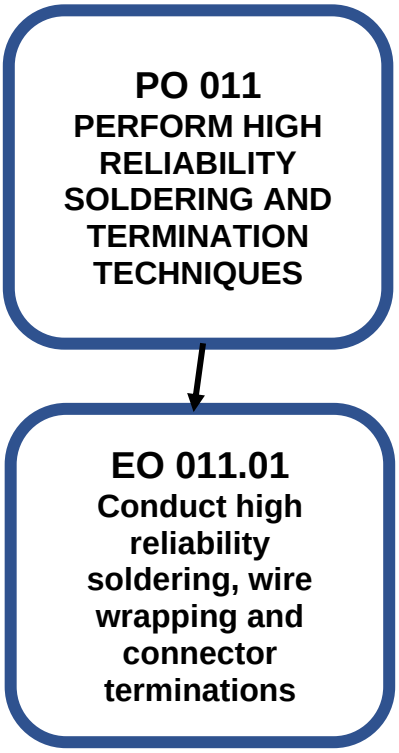

PO 006 Maintain Electrical Distribution Systems
EO 006.01 Troubleshoot electrical distribution systems
EO 006.02 Repair electrical distribution systems
EO 006.03 Test electrical distribution systems


PO 007 Service Fluid Power Systems
EO 007.01 Apply fluid power theory
EO 007.02 Apply a detailed knowledge of pneumatic systems and electrically operated and control fluid power
EO 007.03 Troubleshoot Electro-hydraulics
EO 007.04 Troubleshoot Electro-Pneumatics


PO 008 Maintain Control Networks
EO 008.01 Troubleshooting Control Networks
EO 008.02 Repair Fiber Optic Networks
EO 008.03 Repair Control Networks


PO 009 Maintain Halon 1301 Extinguishing System


PO 010 Perform Routine Motor Monitoring
EO 010.01 Test electric motors by operating the vibration monitor suite
EO 010.02 Test electrical systems with thermal imaging


PO 011
PERFORM HIGH RELIABILITY SOLDERING AND TERMINATION TECHNIQUES
EO 011.01
Conduct high reliability soldering, wire wrapping and connector terminations

ANNEX E – ON JOB PERFORMANCE RECORD (OJPR)

There is no OJPR for this training.

ANNEX F – ACRONYMS AND ABBREVIATIONS

Acronym and Abbreviation	Term or Definition
ACOS	Automatic Changeover Switch
AFFF	Aqueous Film Forming Foam
AVR	Automatic Voltage Regulator
BAM	Bus Access Module
CBRN	Chemical Biological Radiological Nuclear
CCT	Circuit
CCTV	Closed Circuit Television
CET	Certified Electrical Technologist
DG	Diesel Generator
DRMIS	Defence Resource Management Information System
EHS	Electronic Health Monitor
GLM	General Load Monitor
GD	Guided discussion
GS	Guided Study
GT	Gas Turbine
HFCP	Helo Fuel/De-fuel Control Panel
HMI	Human-Machine Interface
HRC	Halon Relay Controller
HVAC	Heating Ventilation Air Conditioning
IC	Integrated Circuit
IMO	International Maritime Organization
IPCS	Integrated Platform Control System
IPMS	Integrated Platform Monitoring System
ISSC	In-Service Support Contract
LCC	Local Control Console
LCP	Local Control Panel
LCU	Local Control Unit
LLMS	Liquid Level Management System
LOP	Local Operating Panel
LOTO	Lockout/Tagout
MCC	Machinery Control Console
MCOS	Manual Changeover Switch
MDE	Main Diesel Engine
MSI	Medium-Scale Integrated
NBCD	Nuclear Biological & Chemical Defence

NEM	Naval Engineering Manual
NPSH	Net Positive Suction Head
ODS	Ozone Depleting Substances
PCB	Printed Circuit Board
PCOT	Propulsion Control Operator Trainer
PG & D	Power Generation & Distribution
PLA	Power Level Actuators
PLC	Programmable Logic Controller
PPE	Personal Protective Equipment
PT	Power Turbine
PTIT	Power Turbine Inlet Temperature
QM	Quartermaster
RTU	Remote Terminal Unit
SCR	Silicon-Controlled Rectifier
SSS	Synchronous Self Shifting
TAU	Twin Agent Unit
TEL	Technology Enhanced Learning
TI	Technical Inspector
TTL	Transistor-Transistor Logic
VA	Vibration Analysis

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

Mod	PO / EO	PO / EO Title	EO	EC	CEC	PC	Scheduler Notes
1		**Academic**	**Academic**				
	002.02	Apply a Detailed Knowledge of essential mathematic skills	3650		250		Arithmetic
	002.01	Analyze electrical circuits	12800		500		Electronics
	001.01	Report Technical Information	1375				**NFS(A)** schedule 1375 Minutes in academic Phase.
	003.06	Apply a Detailed Knowledge of Theoretical Concepts of Digital Systems	4450		300		Digital Logic
	004.01	Apply a detailed knowledge of electronic circuit fundamentals	3300		300		Field Devices and Application
	007.01	Applying fluid power theory	4600		600		Fluid Power Theory
	004.02	Apply a detailed knowledge of electrical circuit fundamentals	3300		300		Electronics
	007.02	Apply a detailed knowledge of pneumatic systems and electrically operated and controlled fluid power	4500		1200		Fluid Power Theory
	003.01	Apply a Detailed Knowledge of Microprocessors	3700		500		Digital Logic
	003.02	Apply a Detailed Knowledge of theoretical Concepts of Control Systems	11950		100		Digital Logic
	008.02	Repair Fiber Optic Networks	1600		350		Fiber Optics
	011.01	Conduct high reliability soldering, wire wrapping and connector terminations	1300		100		
	Sub-total		55150		4500		
	Total		59650				
1		**Tech Comms**	**Application = 1375**				
	001.01	Report Technical Information	1375				**NFS(P)** schedule 1375 Minutes in Application

							Phase. Civ/ Slt-Lt(N)
2	**Maintain Field Devices**		**Application = 1075**				
	004.03	Troubleshoot Field Devices	650			50	
	004.04	Repair Control Sensors	375				
2	**Service Motor Controllers**		**Application = 4650**				
	002.03	Troubleshoot Motor Controllers	1900	50		600	FMF Eng Gas Free inst
	002.04	Maintain Motor Controllers	2050	50			
3	**Service Fluid Power Systems**		**Application = 3900**				
	003.03	Troubleshoot Control Systems	600	50			IPMS Inst
	003.04	Repair Control Systems	1450	50			IPMS Inst
	003.05	Repair Engine Control Systems	1450	300			IPMS Inst
4	**Networking**		**Application = 1650**				
	008.01	Troubleshoot Control Networks	200	50		600	IPMS Inst
	008.03	Repair Control Networks	700	100			IPMS Inst
4	**Halon**		**Application = 2650**				
	009.01	Maintain Halon 1301 Extinguishing System	1690			960	Requires: 2 days of bottle maint. from PO2 Mech.
5	**Perform Scheduled Motor Monitoring Routine**		**Application = 1450**				
	010.01	Test electric motors by Operating vibration monitor suite	1000	50		300	VA Lvl 1 Inst.
	010.02	Test electrical systems with thermal imaging equipment.	100				FMF Eng Inst.
2	**Maintain Primary Power Generation**		**Application = 2350**				
	005.01	Maintain Generators	900			300	
	005.02	Maintain Motor Alternators	100				
	005.03	Test Power Generation Systems	1050				
2	**Maintain Electrical Distribution System**		**Application = 1250**				
	006.01	Troubleshoot Electrical Distribution Systems	600				
	006.02	Repair Electrical Distribution Systems	600			50	FMF Eng Inst. 100 minutes
3	**Service Fluid Power Systems**		**Application = 3000**				
	007.03	Troubleshoot Electro-Hydraulics	850	400		300	Hyd Trainer
	007.04	Troubleshoot Electro-Pneumatics	550	600		300	Pneu Trainer

1	**DRMIS - Supervisor**		**Application = 285**				
	001.02	Produce Naval Maintenance Documentation within DRMIS	285				DRMIS SME
	Sub-total		18475	1700		3460	
	Total		23635				

All times are in minutes.
Academic timings for individual EO/EC/CEC/PC to be determined by academic subject matter experts. Includes review and debrief time.

ANNEX H – ADDITIONAL TRAINING RESOURCES

PO 002 SERVICE POWER CONTROLLERS	EO 002.01 Apply a Detailed Knowledge of Circuit Analysis
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Series resistive circuits	
a. Circuit analysis i. Current ii. Voltage drops iii. KVL and Analysis procedure for a series circuit i. Voltage divider (1) Voltage divider theorem (2) Voltage divider design (3) Potentiometer as a voltage divider i. Power in series circuits v. Voltage-dropping and current limiting resistors v. Open-circuit and short-circuit in series circuits	Tutorial introduction to software: https://www.youtube.com/watch?v=f52GV1IpwVk Browser-based circuit builder: https://www.circuitlab.com/circuit/g2qzxs/resistors-in-series-and-parallel/ Theoretical concepts: https://www.khanacademy.org/science/electrical-engineering/ee-circuit-analysis-topic
TP 2 Parallel resistive circuits	
a. Voltage and current in a parallel circuit. b. Kirchhoff's current law c. Parallel equivalent circuit d. Analysis procedure for a parallel circuit e. Conductance in parallel f. Current divider g. Power in parallel circuits h. Open-circuit and short-circuit in parallel circuits	See TP1.
TP 3 Series-parallel circuits	
a. Equivalent circuit b. Current and voltage droops in a series-parallel circuits: c. Open-circuit and short-circuit in series-parallel circuits d. Analysis of a series parallel circuit	See TP1.
TP4 Capacitance	
a. Reading assignment (1) Electric circuit storage (2) Electric field (3) Capacitance and capacitor dimensions (4) Capacitor types and characteristics 2) Parallel and series capacitors 3) Energy stored in a capacitor 4) Stray capacitance	See TP1.
TP5 Inductance	
a. Reading assignment a. Electro-magnetic induction b. Induced EMF and current c. Self-inductance d. Mutual-inductance e. Types of inductors	See TP1.

2) Energy stored in an inductive circuit 3) Inductors in series and parallel 4) Stray inductance	
TP6 Inductance and capacitance in DC circuits	
a. RL circuit operation b. Instantaneous current and voltage in LR circuits c. Open circuiting and inductive circuits d. RL circuit waveforms e. RC circuit operation f. Instantaneous current and voltage in a RC circuit g. Discharging a capacitor h. RC circuit waveforms	See TP1.
TP7 Alternating current and voltage	
a. Resistive load with AC supply b. Peak, average, and RMS values c. Oscilloscope control	See TP1.
TP8 Inductance and capacitance in AC circuits	
a. Alternating current and voltage in an inductive circuit b. Inductive reactance (XL) c. Alternating current and voltage in a capacitive circuit d. Capacitive reactance (XC) e. Series RL and RC circuits f. Series and parallel RLC circuits	See TP1.
TP9 Series-parallel AC circuits	
a. Series connected impedances b. AC voltage divider c. Impedance in parallel d. AC current divider e. Series-parallel impedances	See TP1.
TP 10 Power in AC circuits	
a. Power dissipated in resistance b. Power in an inductor c. Power in a capacitor d. True power and reactive power e. Power in RL and RC circuits f. Power factor	See TP1.
TP 11 AC circuit analysis	
a. Loop equations b. Nodal analysis c. Superposition d. Thevenin's and Norton's e. Maximum power transfer f. Delta-wye transformer	See TP1.
TP 15 Resonance	
a. Series resonance i. Q factor ii. Bandwidth 2) Parallel resonance i. Q factor ii. Resistance damping	See TP1.
TP 16 Filters	
a. Low pass b. High pass c. Resonance filters	See TP1.
TP 17 Electromechanical energy conversion	
a. Voltage induced in a conductor	See TP1.

b. Direction of induced voltage c. Lenz's law d. Induced voltage in series e. Commutation	
TP 18 DC generator characteristics	
a. Induced voltage, magnetisation curve b. Field resistance and the build-up process c. Critical resistance and the failure of the generator to build up d. Shunt-generator characteristics e. Cumulative-compound generator characteristics f. Differential-compound generator characteristics g. Series generator	See TP1. Video: https://www.youtube.com/watch?v=kbivA3INgYY
TP 19 DC motor characteristics	
a. Force on a conductor-magnitude and direction b. Generation of torque c. Counter EMF d. Power conversion e. Motor speed characteristics f. Effects of armature reaction g. Shunt motor characteristics h. Series motor characteristics i. Compound motor characteristics	https://www.youtube.com/watch?v=LAtPHANEfQo https://www.youtube.com/watch?v=cNl-G8gKsyU
TP 20 Dynamo efficiency	
a. Dynamo losses-coppers tray power and stray load b. Determining the stray power losses c. Method of duplicating flux and speed d. Dynamo efficiency	http://www.edisontechcenter.org/generators.html https://www.slideshare.net/MariaRominaAngustia/efficiency-rating-and-applications-of-dynamos
TP 21 Single phase transformers	
a. Transformer fundamentals b. No-load voltage relations c. Effects of frequency relations d. No-load phasor diagrams e. Current relations f. Phasor diagrams with load g. Equivalent transformer circuit h. Full load phasor diagram i. Short circuit test j. Full load copper loss k. Transformer regulation l. Transformer efficiency	http://www.electronics-tutorials.ws/transformer/transformer-basics.html https://www.youtube.com/watch?v=eE8XXiMbq6U https://www.youtube.com/watch?v=eolT3AqXy6E https://www.youtube.com/watch?v=tkcJ3VpVRwE
TP 22 Poly-phase transformer connections	
a. The Y-Y transformation b. The Δ-Δ transformation c. The Y-Δ transformation d. The Δ- Y transformation e. The open Δ connection f. Power supplied in the open Δ transformation	https://www.allaboutcircuits.com/textbook/alternating-current/chpt-10/three-phase-transformer-circuits/ http://www.electronics-tutorials.ws/transformer/three-phase-transformer.html https://www.youtube.com/watch?v=fAFPHbXBjmc https://www.youtube.com/watch?v=H2hYUu8lPY0
TP 23 Alternators	

a. Generated EMF b. Distributed winding and distribution factor c. Pitch factor d. Alternator characteristics e. Armature reaction i. Lagging power factor ii. Leading power factor 6) Alternator regulation 7) Alternator efficiency 8) Alternator ratings	https://www.youtube.com/watch?v=tiKH48EMgKE
TP 24 Parallel operation of alternators	
a. Requirements for parallel operation b. Synchronising procedure c. Variation of field current of alternators in parallel d. Dividing load of parallel alternators e. Hunting f. Torque at standstill g. Slip h. Rotor EMF, frequency and current i. Running torques and maximum torque j. Description of operation k. Wound rotor induction motor l. Current and power factor m. Losses and efficiency (theory) n. Equivalent circuit and slip	Woodward Group Power Generation Learning Module (desk-top based simulation): http://www.woodward.com/powergenlearningmodule.aspx https://www.youtube.com/watch?v=Qm8jAIl1xdw https://www.youtube.com/watch?v=5O86O4nfntA
TP 25 Poly-phase induction motor	
a. Introduction b. Effect of load c. Increased load d. Power and torque relationships e. Variation of field current f. Calculation of generated voltage and torque angle from load conditions g. Synchronous motor armature reaction h. V curves i. Motor ratings and speed control j. Power factor correction	Cutaway educational models: https://kandh.en.taiwantrade.com/product/cutaway-model-of-electrical-machine-for-educational-teaching-learning-demonstration-of-motors-permanent-magnet-induction-motor-shunt-compound-wound-synchronous-rotor-winding-squirrel-cage-497480.html https://www.youtube.com/watch?v=Qm8jAIl1xdw https://www.youtube.com/watch?v=5O86O4nfntA https://www.youtube.com/watch?v=X5OWCx5RKUc https://www.youtube.com/watch?v=S9DyIFIEb9E
TP 26 Single phase motors	
a. Direct current motors operated on alternating current b. Universal motor characteristics c. Single phase induction motor-cross field theory d. Split phase induction motor e. Capacitor start motor characteristics f. Permanent split capacitor motor g. Two value capacitor motor	Construction: https://www.youtube.com/watch?v=awrUxv7B-a8 How it works: https://www.youtube.com/watch?v=EX-71cLRp2Q How it's connected: https://www.youtube.com/watch?v=aHh0YxWUfH8

PO 002 SERVICE POWER CONTROLLERS	EO 002.02 Apply a Detailed Knowledge of Essential Mathematic Skills
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Basic mathematics:	
a. Review of basic mathematics: Numerical fractions; (1) Algebraic expressions; (2) Equations and equation manipulation; (3) Basic trigonometry; (4) Exponents and radicals; and (5) Factoring.	Review of basic math: For electrical apprentices (sample of PIP material): https://www.nic.bc.ca/PDF_docs/departments/Math_Review_for_Electricians.pdf Basic electrical theory exam (sample for DLN implementation): https://www.proprofs.com/quiz-school/story.php?title=master-electrician-prep-exam-part-1 40 Journeyman-level electrical practice exams which could be modeled in DLN as PIP/USQ content: http://electricianpracticetests.com/journeyman-electrician/journeyman-electrician-exam-practice-test-2/
TP 2 Quadratic equations:	
a. Quadratic equations: (1) Solution by factoring; (2) Completing the square; (3) The quadratic formula; (4) Quadratic function; and (5) Equations leading to quadratic equations.	Animated simulations for comprehension and manipulation of math: https://www.mathsisfun.com/algebra/quadratic-equation-graph.html http://animated-mathematics.net/quad.html
TP 3 Trigonometric formulae, equations, and graphing:	
a. Solving trigonometric formula, equations and graphing: (1) Basic identities; (2) Trigonometric equations; (3) Solution of oblique triangles law of sines and cosines; and (4) Applications of angles and angular movement. b. Trigonometric graphing.	Trig graphing: http://www.businessinsider.com/7-gifs-trigonometry-sine-cosine-2013-5 Introducing and proving basic trig identities: https://www.youtube.com/watch?v=zV6ziekbnrA
TP4 Analytic geometry:	
a. Solving analytic geometry: (1) The conic sections; (2) The distance formula; and	Theory: https://www.khanacademy.org/math/geometry-home/analytic-geometry-topic Animated understanding of concepts: http://www.mathwarehouse.com/animated-gifs/ Provide examples of why/how this can be used in actual employment or real-world applications: https://www.youtube.com/watch?v=JDghwEM2_Ys
TP 5 Functions and slope of the line:	
a. Solving functions and slope of the line:	Intuitive understanding of slope: https://www.khanacademy.org/math/cc-eighth-grade-math/cc-8th-linear-equations-functions/8th-slope/v/slope-intuition-example
TP 6 Differential calculus:	
a. Using differential calculus to: (1) limits;	Understanding the purpose, construction, and uses o calculus:

(2) Derivative – Delta process; (3) Rules for differentiation; (4) Product and quotient rules; and (5) Higher order derivatives.	https://www.youtube.com/watch?v=WUvTyaaNkzM
TP 7 Derivatives of transcendental functions:	
a. Learning derivatives of transcendental functions: (1) Trigonometric functions; (2) Logarithmic functions; and (3) Exponential functions.	Understanding the purpose, construction, and uses of calculus: https://www.youtube.com/watch?v=WUvTyaaNkzM
a. Use integral calculus: (1) The indefinite integral; (2) Basic differential equations; (3) Power rule for integration; (4) Integration applications (electrical); (5) Integration of trigonometric, exponential and logarithmic functions; (6) Definite integral; and (7) Mean and RMS values.	Understanding the purpose, construction, and uses of calculus: https://www.youtube.com/watch?v=WUvTyaaNkzM

PO 002 Service Power Controllers	**EO 002.03 Troubleshoot Motor Controllers**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Troubleshoot motor controller to include, but not limited to:	
a. Troubleshooting motor controllers by: (0) Gathering resources to include, but not limited to: a) Publications; b) Applicable technical schematics; c) Test equipment; d) LOTO equipment; e) PPE; and f) Assistance. (1) Visually inspecting for abnormalities to include: a) Fault lights; b) Trouble codes; and c) System configuration. (2) Confirming power to include: a) Voltage test; b) Amperage test; and c) Resistance test. (3) Conducting functional test; (4) Checking systematically individual components to include, but not limited to: a) Fuses; b) Relays; c) Switches; d) Contactors; e) Conductors; f) Leads; g) Test points; h) Overloads;	Multi-fault, configurable, industrial motor controller and panel reflective of ship's configuration for hands-on practice (2 of these controllers exist for reference. Located at NFS(A)—more modern HALIFAX-Class controllers need to be developed with help from FMF)

i) Field devices; and j) Circuit cards.	
TP 2 Inspect circuit cards by:	
a. Inspecting circuit cards for components to include, but not limited to: 1) Inspecting control relays; 2) Inspecting inductors; 3) Inspecting visually, motherboards; 4) Inspecting motherboards; and 5) Inspecting capacitors. b. Troubleshooting printed circuit boards for anomalies to include, but not limited to: i. Burn marks; ii. Broken traces; iii. Utilizing test points; iv. Damaged leads; v. Damaged components; and vi. Bent pins.	Sample circuit card that is envisioned: https://www.google.ca/imgres?imgurl=https%3A%2F%2Fi.stack.imgur.com%2FTLr2g.jpg&imgrefurl=https%3A%2F%2Felectronics.stackexchange.com%2Fquestions%2F14038%2Fis-the-damage-on-this-circuit-board-too-much-to-be-fixed-pics-inside&docid=QeB3rjWCi7CJfM&tbnid=4rRoWLpvCLs5OM%3A&vet=10ahUKEwihxZePxd_XAhVNON8KHfwzAjgQMwhBKAUwBQ..i&w=2048&h=1536&safe=active&bih=749&biw=1536&q=circuit%20card%20with%20burnt%20resistor&ved=0ahUKEwihxZePxd_XAhVNON8KHfwzAjgQMwhBKAUwBQ&iact=mrc&uact=8

PO 002 Service Power Controllers	EO 002.05 Repair Motor Controllers
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Repair and test motor controllers by:	
a. Repairing and testing motor controllers to include, but not limited to: 1) Disassembling/assembling component(s); 2) Repair fault and/or replace component(s), to include, but not limited to: i. Fuses; ii. Relays; iii. Switches; iv. Contactors by: 1. Dressing contacts; and 2. Replacing coil. v. Conductors; vi. Leads; vii. Test points; viii. Overloads by: 1. Testing parameters; 2. Replacing heaters; and 3. Adjusting settings. ix. Field devices; and x. Circuit cards. 3) Conducting bench test; and 4) Conducting system function test.	Multi-fault, configurable, industrial motor controller and panel reflective of ship's configuration for hands-on practice (2 of these controllers exist for reference. Located at NFS(A)—more modern HALIFAX-Class controllers need to be developed with help from FMF)
TP 2 Repair and test circuit cards:	
a. Repairing and testing circuit cards to include, but not limited to: a. Repairing circuit card cage assembly by:	Demonstration circuit card provided to trainees are to contain current protection devices.

1) Examining for bent pins; 2) Repairing bent pins; 3) Following ESD procedures; 4) Replacing fuses; and 5) Resetting overload protection. b. Testing circuit card cage assembly by: 1) Checking for correct seating.	
PO 002 Service Power Controllers	**EO 005 Troubleshoot Electric Motors**
Teaching Point	**TEL Examples covering course content or model styles to be developed**
TP1 Troubleshoot electric motors to include, but not limited to:	
a. Troubleshooting machinery space ventilation systems to include, but not limited to: (1) Resistance check; (2) Inspecting conn box; (3) Inspecting motor leads; (4) Meggering motor; (5) Checking motor bearings; (6) Checking mounting security; (7) Checking cable glands; (8) Conduct amp probe; (9) Checking air vents; (10) Checking air gap; (11) Checking flexible coupling; and (12) Troubleshooting various motor starters by: i. Conducting experiment in Lab Volt manual.	
TP 2 Repair electric motors to include, but not limited to:	
a. Repairing enclosure fans (AUT2320): (1) Connecting/disconnecting motor leads; and (2) Assembling/disassembling motor. b. Testing electric motors by: (1) Testing motor rotation; and (2) Checking motor alignment.	
TP 3 Utilize equipment health monitoring to troubleshoot motors, including but not limited to	
a. Training in conjunction with digital imaging system field operation course (MITE ID 115158); and b. Training in conjunction with thermal imaging.	

PO 003 SERVICE ELECTRONIC CONTROLLERS	EO 003.01 Apply a Detailed Knowledge of Microprocessors
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Explain introduction to microprocessors and microcomputers by:	
a. The digital computer by: (1) Brief history of computer development; and (2) Overview of integrate circuit technology. b. Mainframe computers, mini computers and microcomputers (1) Mainframe computers; (2) Mini computers; and (3) Microcomputers. c. Hardware elements of the digital computer system. d. General architecture of a microcomputer system. e. Types of microcomputers and single chip microcomputers.	http://www.softwareforeducation.com/sms32v50/ https://software.intel.com/en-us/articles/pre-release-license-agreement-for-intel-software-development-emulator-accept-end-user-license-agreement-and-download https://www.youtube.com/watch?v=8y1wfmAsNqY TEL Resources to include the microprocessor course under development at NFS(A)
TP2 Explaining number systems/ code/ architecture by:	
a. Number systems (1) examination of the decimal/hexadecimal and binary number systems; and (2) conversions between decimal, hexadecimal and binary.	
b. Codes (1) BCD code format and uses; (2) conversion between BCD and decimal; (3) the ASCII code format; and (4) twos complement signed number format.	
c. Arithmetic (1) adding/subtracting binary numbers; and (2) adding/subtracting hexadecimal numbers.	
TP3 Operating the INTEL™ microprocessor (Include most up to date system)	
TP4 Explaining operand addressing modes of the INTEL™ microprocessors	
TP5 Conducting programming techniques	
TP6 Examining semiconductor memories	
TP7 Explaining INTEL™ memory interface	
TP8 Explaining INTEL™ input/output interfaces	
TP9 Executing exception processing and the INTEL™ microprocessor	
TP10 Testing and troubleshooting	

PO 003 SERVICE ELECTRONIC CONTROLLERS	EO 003.02 Apply a Detailed Knowledge of Theoretical Concepts of Control Systems
Teaching Point	**Examples covering content or model styles to be developed**
TP1: Interact with AC switching controllers by:	
a. Examining bi-directional switches to include but not limited to: (1) AC switches using thyristors; and (2) AC switches using other devises.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system https://www.google.ca/url?sa=t&rct=j&q=&esrc=s&source=web&cd=8&cad=rja&uact=8&ved=0ahUKEwivzYnX3d_XAhXCnOAKHUr8DukQFghAMAc&url=https%3A%2F%2Fwww.labvolt.com%2Fdownloads

	%2F86356_f0.pdf&usg=AOvVaw31EvlyuO---LwmBqZKFQfU
b. Examining switching control of AC loads to include by not limited to: (1) Single phase resistance loads – integral half cycle control; (2) Phase control – resistive loads; (3) Single phase R-L loads; (4) Thyristor controlled inductors; (5) Static VAR compensation circuits; and (6) Three phase applications of switching control.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system
TP2: Classify choppers by:	
a. Examining voltage step-down choppers to include but not limited to: (1) Power circuit configuration and working principals; (2) Voltage relationships; (3) Duty cycle range of practical choppers; (4) Chopper operation when load presents a “back EMF”; (5) Use of filter on input side; (6) Use of filter on output side.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system
b. Examining step-up choppers including but not limited to: (1) Power circuit configuration and working principles; (2) Voltage relationships; (3) Duty cycle range of practical choppers; (4) Chopper operation when load presents a “back EMF”; (5) Use of filter on input side; (6) Use of filter on output side.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system
c. Examining two quadrant choppers including but not limited to: (1) Waveforms; (2) Switching circuitry.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system
d. Examining four quadrant choppers including but not limited to: (1) Waveforms; (2) Switching circuitry.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system
e. Examining switching control circuits for chopper converters including but not limited to: (1) Functional requirements of the switching control circuits.	Labvolt Power Electronics training module: https://www.labvolt.com/solutions/6_power_energy/59-8010-A0_power_electronics_training_system
TP3: Classify inverters by:	
a. Examining functions and features of inverters.	https://www.labvolt.com/solutions/6_power_energy/98-8013-00_2_kw_electromechanical_training_systems
b. Examining Inverter applications including but not limited to: (1) DC link inverters; (2) Adjustable speed AC drives; and (3) Uninterruptable power supplies.	https://www.labvolt.com/solutions/6_power_energy/98-8013-00_2_kw_electromechanical_training_systems

c. Examining types of inverters including, but not limited to: (1) Voltage source inverters; (2) Current source inverters; and (3) Current regulated inverters.	https://www.labvolt.com/solutions/6_power_energy/98-8013-00_2_kw_electromechanical_training_systems
d. Examining half-bridge rectifiers to include but not limited to: (1) Circuit configuration switching; (2) Analysis of half-bridge inverter with inductive load; (3) Adjustment of AC frequency and AC voltage; (4) Output waveform considerations; and sinusoidal pulse width modulation using half-bridge rectifiers.	Incorporate working animations : e.g. Bridge rectifier animation: https://www.youtube.com/watch?v=PfWmIQfYIRI Centre-tapped full wave rectifier : https://www.youtube.com/watch?v=qvcT2iC1N6k
e. Examining full-bridge configurations including but not limited to: (1) Single phase full wave bridge topology; (2) Control of the AC output voltage – pulse width modulation; (3) Shaping of the output waveform – sinusoidal pulse width modulation.	
f. Examining three phase inverters to include but not limited to: (1) Circuit configuration and switching sequence; (2) Three phase inverter feeding a delta connected load; (3) Three phase inverter feeding wye connected load; (4) Voltage control and waveform shaping in three phase converters; (5) Filtering requirements; and (6) Inverter operation with reverse power flow.	
TP4: Examine adjustable speed DC motor drives by	
a. Examining speed control methods of separately excited DC motors to include but not limited to: (1) Converter restrictions; (2) Speed control ranges with separate excitation; and (3) Regenerative braking and reversing,	An example of how to demonstrate PWM speed control with DC motors: https://www.youtube.com/watch?v=3To3Zgr1Jao
b. Examining chopper controlled DC motor devices to include but not limited to: (1) Driving and braking considerations; (2) Rotation reversal; and (3) Static four quadrant chopper.	
c. Examining DC motor drives using phase controlled thyristor converters including but not limited to: (1) Dual bridge configuration; (2) Speed control scheme for a DC motor drive using a dual	

converter with circulating current; and (3) Sequence of operations during a rotation reversal.	
d. Demonstrate control of series motors.	
TP5: Control solid state AC motors control by	
a. Examining voltage source inverters	
b. Examining current source inverters	
c. Examining phase controlled cycloconverters	
d. Examining load commutated inverters	
e. Examining adjustable speed drives using squirrel cage induction motors to include but not limited to: (1) Induction motor basics; (2) Open loop speed control of an induction motor fed from a voltage source inverter or cycloconverter; and (3) Closed loop control using a voltage source inverter.	Animation of VFD working principle: https://www.youtube.com/watch?v=DiKcKYbJ1A4&vl=en Squirrel cage machine animation: https://www.youtube.com/watch?v=0FCu7P5J2U8 Example of infographics demonstrating VFD applications: http://electrical-engineering-portal.com/why-do-we-need-variable-speed-drives-vsd
TP6: Utilizing power supply systems by	
a. Examining power supply components to include but not limited to: (1) Linear regulators; (2) Pre-regulators; (3) Off-line switchers; (4) SMPS converter circuit topologies; (5) Controllers for SPMS; and (6) UPS construction	https://www.labvolt.com/solutions/2_electronics/40-91009-20_power_supply_regulation_circuits
TP7: Analyzing synchro and servo motors by	
a. Examining synchro transmitters construction and basic operation	Field device and synchro training modules: http://www.kitektechnologies.com/sensor-trainer-kit.html http://www.sigmatrainers.com/Htm/Industrial%20Drives%20and%20Control%20Lab.htm Animations of operation: https://www.youtube.com/watch?v=FO-05cyGRCE https://www.youtube.com/watch?v=Voq6nEdNwGo
b. Examining synchro receivers to include but not limited to: (1) Construction and basic operation; and (2) Differences between transmitters and receivers.	https://www.vimicrosystems.com/vi/Web_Catalog/5_Control%20System%20Trainer.pdf
c. Examining control systems including but not limited to: (1) Closed loop system (TX/RX); (2) Analysis of systems – controlled variable response to set-point changed	https://www.khanacademy.org/science/electrical-engineering/ee-amplifiers/ee-opamp/v/ee-feedback
d. Examining construction and basic operation of differential transmitters	
e. Examining construction and basic operation of differential receivers	https://www.labvolt.com/solutions/2_electronics/40-91012-20_operational_amplifier_fundamentals

f. Examining construction and basic operation of servo amplifiers.	
g. Examining AC and DC servo motors	
PO 003 SERVICE ELECTRONIC CONTROLLERS	**EO 003.05 Repair Engine Control Systems**
Teaching Point	**Examples covering content or model styles to be developed**
TP1: Examine Governors by	
a. Repairing main diesel engine electronic governor by: (1) Describing the governor box; (2) Identifies governor components; (3) Describe governor functions including: i. Speed measuring system control; ii. Fuel limited system; iii. Speed droop system; iv. Speed sensor; and v. Frequency divider.	Labvolt Synchronous Generator Control Function Set 9069-A0 Animated video demonstrating how a governor speed circuit operates in diesel controller, such as: https://www.youtube.com/watch?v=xCGHHCe1bFc
b. Testing propulsion diesel engine controls by: (1) Describing principles of speed measuring control to include, but not limited to: i. Electronic mode; ii. Module 721 (digital control); iii. PLC; iv. PGA-EG; and v. Pneumatic mode. (2) Testing diesel engine electronic governor by: i. Identifying and repairing: 1. Types of faults; 2. Remote control mode; 3. Local control mode; and 4. Mechanical mode.	
TP2: Repair Engine Control systems by:	
a. Installing propulsion control system components.	
b. Assembling/disassembling diesel engine electric starting system components; (1) Components of an electrical starter; (2) Operation of the starter-mounted solenoids; and (3) Operation of the overrunning clutch and drive pinion.	
c. Calibrating electrical propulsion control system components by: (1) Testing power lever angle actuators.	
d. Testing momentum modules	

PO 003 SERVICE ELECTRONIC CONTROLLERS	EO 003.06 Apply a Detailed Knowledge of Theoretical Concepts of Digital Systems
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Explain concepts of digital logic	
a. Numerical representation; b. Digital and analog systems; c. Digital number systems; d. Representing binary quantities; e. Digital circuits; f. Parallel and serial transmission; g. Memory; h. Digital computers; and i. Switch representation of logic functions.	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/
TP2 Comparing number systems and codes	
a. Binary to decimal conversions; b. Decimal to binary conversions; c. Octal number system; d. Hexadecimal number system; e. BCD code; f. Excess-3 code; g. Grey code; h. Alphanumeric code; and i. Parity method of error detection.	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/
TP3 Analyze logic gate sand Boolean algebra	
a. Boolean constants and variables; b. Truth tables; c. OR operation; d. AND operation; e. NOT operation; f. Describing logic circuits algebraically; g. Evaluating logic circuits outputs; h. Implementing circuits from Boolean expressions; i. NOR gates and NAND gates; j. Boolean and DeMorgan theorems; k. Boolean identities and proofs; l. Universality of NAND gates and NOR gates; m. Alternate logic gate representations; and n. IEEE/ANSI stand logic symbols.	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/
TP4 Compare combinational logic circuits	
a. Sum of products form b. Simplifying logic circuits c. Algebraic simplifications d. Proof by perfect induction e. Combinational logic, min-terms and max-terms f. Designing combinational logic circuits g. Karnaugh map method h. Exclusive OR and exclusive NOR circuits i. Parity generator and checker j. Inhibit circuits k. Basic characteristics of digital ICs l. Troubleshooting digital systems m. Internal digital IC faults n. External faults o. Troubleshooting case study	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/
TP5 Explain flip flops	
a. Introduction to flip flops b. NAND gate latch	Web-based platform for design and visualization of digital logic:

c. NOR gate latch d. Troubleshooting case study e. Clock signals and clocked flip flops f. Clocked SET CLEAR flip flops g. Clocked J-K flip flop h. Clocked D flip flops i. D latch j. Asynchronous inputs k. IEEE/ANSI symbols l. Flip flop timing considerations m. Detecting an input sequence n. Data storage and transfer o. Serial data transfer, shift registers p. Frequency division and counting q. Schmitt trigger devices r. One shot (monostable Multivibrator) s. Analysing sequential circuits t. Astable multivibrators u. Troubleshooting flip flops circuits	http://www.dld-visu.com/
TP6 Conduct digital arithmetic operations and circuits	
a. Binary addition b. Repressing signed numbers c. Addition in the 2s complement d. Subtraction in the 2s complement e. Multiplication of binary numbers f. Binary division g. BCD additions h. Hexadecimal arithmetic i. Arithmetic circuits j. Parallel binary adder k. Design of a full adder l. Complete parallel adders with registers m. Carry propagation n. Integrated circuits parallel adder o. 2s complement system p. BDR adder q. Binary multipliers r. Complex arithmetic ICs s. Comparators circuit t. Parity circuits u. IEEE/ANSI symbols v. Troubleshooting case study	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/
TP7 Analyze counters and registers	
a. Asynchronous (ripple) counters b. Counters with MOD numbers greater than 2n c. IC asynchronous counters d. Asynchronous down counter e. Propagation delay in ripple counters f. Synchronous (parallel) counters g. Synchronous down and up/down counters h. Pre-settable counters i. The 74193 (LS193/HC 192) counter j. IEEE/ANSI dependency notation k. Decoding a counter l. Decoding glitches m. Cascading BCD counters n. Synchronous counter design o. Shift-register counters p. Counter applications, frequency counter and digital clock q. Integrated circuit registers	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/

r. Parallel in/parallel out registers s. Serial in/serial out registers t. Parallel in/serial out registers u. Serial in /parallel out registers v. IEEE/ANSI register symbols w. Troubleshooting counters and registers	
TP8 Analyze MSI logic	
a. Encoders and decoders, terminology b. Decoders c. BCD to 7 segment decoders/drivers d. Liquid crystal displays e. Encoders f. Decoder and encoder IEEE/ANSI symbols g. Troubleshooting decoders and encoders h. Multiplexers i. Multiplexer application j. Multiplexers to replace random logic k. De-multiplexers l. Multiplexer/ de-multiplexer IEEE/ANSI symbols m. Troubleshooting multiplexer/de-multiplexers circuits n. Magnitude comparator o. Code converters p. Digital IC terminology q. Standard TTL series characteristics r. TTL open collector outputs s. Tristate (3-stage) TTL t. Tristate registers u. Data bussing	Web-based platform for design and visualization of digital logic: http://www.dld-visu.com/

PO 004 MAINTAIN FIELD DEVICES	**EO 004.01 Apply a detailed knowledge of electronic circuit fundamentals**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Define semiconductors	
a. Semiconductor theory b. Doping a semiconductor c. Unbiased diode d. Forward bias e. Reverse bias f. Breakdown g. Energy levels and the energy hill h. Barrier potential and temperature i. The reversed biased diode	Uses of, construction and operation: https://www.youtube.com/watch?v=Xi18hI1LqAA https://www.youtube.com/watch?v=JBtEckh3L9Q https://www.youtube.com/watch?v=WT1bwFDSq9I https://www.youtube.com/watch?v=W6QUEq0nUH8
TP 2 Applying diode theory	
a. Schematic symbol: b. Diode curve c. Forward and reverse regions d. Ideal diode e. Second and third approximation f. Approximation g. Data sheet h. Bulk/diode resistance i. Load line	https://www.labvolt.com/solutions/2_electronics/40-91005-20_semiconductor_devices
TP 3 Recognize diode circuits	
a. Input transformer b. Half wave rectifier	

c. Full wave rectifier d. Bridge rectifier e. Capacitor input filter f. Calculating other quantities g. Troubleshoot h. Read data sheet to include (1) fuses (2) real transformers (3) design guidelines; and (4) surge current. i. RC and LC filters j. Voltage multipliers k. Clippers and clampers l. Peak to peak detector and m. DC return	
TP4 Identify special purpose diodes	
a. Zener diode b. Loaded zener regulator c. Optoelectronic devices d. Schottky diode e. Varactor f. Varistors g. Data sheet h. Troubleshooting i. Load lines j. Second approximation k. Ripple zener drop out point l. LED design guidelines m. reverse recovery time n. varactor characteristics o. other diodes	
TP5 Identify bipolar transistors	
a. unbiased transistor b. biased transistor c. CE connection d. Transistor characteristics e. Transistor approximations f. Data sheets g. Energy viewpoint h. Base curves, cut off and breakdown i. Third approximation j. Base spreading resistance k. Ebers-Moll model l. CB connection	
TP6 Apply transistor fundamentals	
a. Variations in current gain b. Load line c. Operating point d. Proof of the load line e. Saturation f. Transistor switch g. Transistor current source h. Optoelectronic devices	
TP7 Recognize transistor biasing	
a. Voltage-divider bias b. Voltage-divider analysis c. Voltage-divider load line and Q-point d. Two supply emitter bias e. PNP transistor f. Other types of bias i. Emitter feedback bias	

ii. Collector feedback bias iii. Collector and emitter feedback bias iv. Voltage divider bias g. Troubleshooting	
TP8 Utilize AC models	
a. Coupling capacitor b. Bypass capacitor c. Superposition in amplifiers d. Small signal operation e. AC resistance of the emitter diode f. AC beta g. Common emitter amplifier h. AC model of a CE amplifier i. AC quantities on the data sheet j. H parameters	
TP9 Utilize voltage amplifiers	
a. Highlights of a CE amplifier b. Voltage gain c. Predicting voltage gain d. Simplified analysis e. Swamped amplifier f. Cascaded stages g. Output impedance h. Negative feedback i. Cascade stages j. H parameters k. Common base amplifiers l. Common collector amplifier m. Maximum unclipped output n. Cascading CE and CC	
TP10 Utilize power amplifiers	
a. AC load line b. Limits on signal swing c. Class A operation d. Transistor power rating e. AC saturation and cut off f. AC output compliance g. Thermal resistance h. Class B operation i. Class B power formulae j. Biasing class B amplifiers k. Class B driver	

PO 004 MAINTAIN FIELD DEVICES	**EO 004.02 Apply a detailed knowledge of electrical circuit fundamentals**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Identify field effect transistors	
a. JFET b. Biased JFET c. Drain curves d. Transconductance curve e. JFET approximation f. MOSFET depletion mode g. MOSFET enhancement mode h. Data sheets i. Saturation j. Mathematical derivation	JFET working animation: https://www.youtube.com/watch?v=uea-BxQR71A MOSFET working animation: https://www.youtube.com/watch?v=ItOV1nkTlPU https://www.labvolt.com/solutions/2_electronics/40-91010-20_fet_fundamentals
TP 2 Utilizing FET circuits	
a. Self-bias JFETs	

b. Graphical solution of self-bias c. Solution with universal JFET curve d. Transconductance e. JFET amplifiers f. JFET analog switch g. Depletion mode MOSFET amplifiers h. Enhancement mode MOSFET amplifiers i. Other JFET biasing 1) Voltage divider biasing 2) Source bias 3) Current source bias j. Output impedance of the source follower k. Other JFET applications	
TP 3 Recognizing frequency effects	
a. Frequency response of an amplifier 1) Input coupling capacitor 2) Output coupling capacitor b. Emitter bypass capacitor c. Collector bypass circuit d. Miller's theorem e. High frequency bipolar analysis f. Total frequency response g. Decibels h. Decibels voltage gain 1) Voltage gain outside the midband i. Proof of Miller's theorem j. High frequency FET analysis k. dBM l. Power and voltage gain m. Bode plots n. Risetime bandwidth relation o. Stray effects p. Several critical frequencies	
TP 4 Applying op-amp theory	
a. Integrated circuits (ICs) b. Differential amplifier c. Two input characteristics d. AC analysis of a differential amplifier e. Output offset voltage f. Common mode gain g. Current mirror h. Differential amplifier with current mirrors i. Small signal frequency response j. Large signal frequency response k. Power bandwidth l. Operational amplifier (op amp) m. Op amp characteristics n. Popular op amps o. Other linear ICs	
TP 5 Recognizing negative feedback	
a. Non-inverting voltage feedback b. Open loop and closed loop voltage gains c. Input and output impedance	

d. Other benefits of negative feedback e. Inverting voltage feedback f. Bandwidth g. Exact value of feedback fraction h. More on inverting voltage feedback i. Non-inverting current feedback j. Inverting current feedback k. More on closed loop bandwidth	
TP 6 Utilizing linear op-amp circuits	
a. Voltage amplifiers 1) Non-inverting amplifiers 2) Inverting amplifiers b. Op amp inverting circuits c. The summing amplifier d. Current boosters e. Voltage controlled current sources f. Differential and instrument amplifier g. Active filters 1) Automatic gain control 2) Howland current course 3) Other active filters	
TP 7 Utilizing non-linear op-amp circuits	
a. Active diode circuits i. Peak detector ii. Active clipper and clamper b. Comparator c. Schmitt trigger i. Moving the trip points d. The integrator e. The differentiator f. Relaxation oscillator g. Window comparator	
TP 8 Designing regulated power supplies	
a. Voltage feedback operation b. Current limiting c. Power supply characteristics d. Three terminal IC regulators e. Current boosters f. DC to DC converter g. Switching regulators i. Foldback limiting	
TP 9 Applying oscillatory theory	
a. Theory of sinusoidal oscillation b. The Wien bridge oscillator c. Other RC oscillators d. The Colpitts oscillator e. Other LC oscillators i. Crystal oscillators f. Unwanted oscillators g. Other ways to reduce loop-gain Aβ to unity	

PO 004 MAINTAIN FIELD DEVICES	**EO 004.03 Troubleshoot Field Devices**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Troubleshoot control sensors to include, but not limited to:	

a.	Discriminating operation of field devices to include but not limited to: (1) Temperature sensors; (2) Pressure sensors; (3) Level sensors; (4) Reed switches; (5) Limit switches; (6) Thyristors; (7) Tension transducers; (8) Synchro-servos; (9) Thermistor; (10) Explosive gas detectors; (11) Optical sensors; (12) Photo sensors; (13) Speed sensors; (14) Carbon monoxide sensor; (15) Vacuum sensors; (16) Pyrometer; (17) Resistive temperature devices; (18) Smoke detector; (19) Heat detectors; (20) Thermocouples; (21) Flow sensor; and (22) Leak detector.	Siemans provides working animations of multiple field devices. An example is as follows: http://instrumentationtools.com/pressure-transmitter-animation/#.Whyc1E2Wypo Working principle can be demonstrated using animation: https://www.youtube.com/watch?v=TiKWBSzOk7M To be instructed using a 3D cutaway model (virtual and otherwise) demonstrating flow and signal through the system. An example of the type of models can be found as follows: https://www.google.ca/imgres?imgurl=https%3A%2F%2Fwww.brooksinstrument.com%2F~%2Fmedia%2Fbrooks%2Fimages%2Fproducts%2Fpressure%2520and%2520vacuum%2Fpressure%2520gauges-regulators-switches-transmitters%2Fpressure%2520transducers%2520and%2520transmitters%2520-%2520electronic%2Fpr-ssii-cutaway-blue-callouts.png%3Fas%3D0%26h%3D609%26la%3Den%26w%3D600%26hash%3DCBC3C61D064D5D21ECA5CB3E9042E96E8D012408&imgrefurl=https%3A%2F%2Fwww.brooksinstrument.com%2Fen%2Fproducts%2Fpressure-vacuum&docid=NQJ7JlfJmpGCfM&tbnid=ICclZ2pkGvJ01M%3A&vet=10ahUKEwiDzbyl8d_XAhVvct8KHZZvApsQMwg-KAIwAg..i&w=600&h=609&safe=active&bih=749&biw=1536&q=pressure%20sensor%20%2B%20cutaway%20models&ved=0ahUKEwiDzbyl8d_XAhVvct8KHZZvApsQMwg-KAIwAg&iact=mrc&uact=8 https://www.gemssensors.com/fluid-power Testbed & Calibrator: http://www.gunt.de/index.php?option=com_gunt&task=gunt.list.category&product_id=932&lang=en&Itemid=148
b.	Inspecting field devices as to include but not limited to: (1) Inspecting explosive gas detector; (2) Inspecting speed sensors; (3) Inspecting tachogenerator; (4) Inspecting optical sensors; (5) Inspecting photo sensors; (6) Troubleshooting propulsion diesel engine enclosure carbon monoxide sensor; (7) Inspecting thermistors; (8) Troubleshooting tension transducer; (9) Inspecting thermistors; (10) Inspecting rudder angle indicator; (11) Inspecting feedback transmitter; and	

(12) Troubleshooting shaft knot indicators.	
PO 004 MAINTAIN FIELD DEVICES	**EO 004.04 Repair Field Devices**
Teaching Point	**Examples covering content or model styles to be developed**
a. Repairing speed sensors by: (1) Adhering to proper lock-out/tag-out procedures; (2) Mustering tools required; (3) Mustering appropriate PPE; (4) Utilizing appropriate documentation; and (5) Reviewing all Safety and Environmental concerns with regards to repairing control sensors.	
TP 2 Repair rudder angle indicator:	
a. Repairing feedback transmitter; and	
b. Repairing steering feedback electrical control unit.	

PO 004 MAINTAIN FIELD DEVICES	**EO 004.05 Test Field Devices**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Test control sensors to include, but not limited to:	
a. Testing pressure switches by: (1) Reviewing manufacturing specifications; and (2) Test IAW manufacture's specifications.	

PO 005 Maintain Primary Power Generation Systems	**EO 005.01 Troubleshoot Primary Power Generation Systems**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Troubleshoot Power Generation	
a. Inspecting automatic voltage regulators to include, but not limited to: (1) Rotating rectifier; (2) Main exciter; (3) Pilot exciter; (4) PCB regulator to include: i. Power supply; ii. Under-speed protection; and iii. Regulator circuit. (5) PCB firing unit; (6) Fault logic system; (7) Active and reactive load sharing; (8) No load and short circuit conditions; and (9) Limitation of output voltage.	SIM - interactive exploded view of generator and AVR To be built in a fashion similar to the model seen here: http://www.sun-a.com/img/sp/3969/cover.jpg
b. Troubleshooting generators to include, but not limited to: (1) Generator excitation;	Voltage regulation through Lab Volt experiments

(2) Generator under load; (3) Regulation of output voltage; (4) De-excitation; and (5) Function of the control loop of AVR.	SIM - interactive exploded view of generator and AVR (each component should have the ability to be individually selected, removed, operated in isolation of the other elements
c. Troubleshooting motor alternator sets to include, but not limited to: (1) Review motor theory; (2) Review alternator theory; and (3) Purpose of MA	Working animation to be designed in the following style: https://www.youtube.com/watch?v=OOeFhL92vC8 https://www.youtube.com/watch?v=KHUiAs8NElQ

PO 005 Maintain Primary Power Generation Systems	EO 005.02 Repair Primary Power Generation Systems
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Repair generators:	
a. Repairing generators to include, but not limited to: (1) Enclosure fans (2) Enclosure heaters (3) Left-hand terminal block components (4) Leak detector (5) AVR location b. Assembling/disassembling generator components	SIM - interactive exploded view of generator and AVR (each component should have the ability to be individually selected, removed, operated in isolation of the other elements
TP2 Repair AVRs:	
a. Applying voltage regulation theory, to include, but not limited to: (1) Rotating rectifier (2) Main exciter (3) Pilot exciter (4) PCB regulator (5) Power supply (6) Under-speed protection (7) Regulator circuit (8) PCB firing unit (9) Fault logic system (10) Active and reactive load sharing (11) No load and short circuit conditions (12) Limitation of output voltage (13) Function of a control loop b. Repairing automatic voltage regulators	SIM - interactive exploded view of generator and AVR (each component should have the ability to be individually selected, removed, operated in isolation of the other elements. Instructor to provide trainees with homework assignment on AVR control loop theory. e.g. http://www.sanfoundry.com/avr-microcontroller-mcqs-interrupt-programming/ http://www.sanfoundry.com/avr-microcontroller-mcqs-architecture/
c. Adjusting automatic voltage regulators	
d. Assemble/disassemble AVRs	

PO 005 Maintain Primary Power Generation Systems	EO 005.03 Test Power Generation Systems
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Test generator to include, but not limited to:	
a. Measuring electrical parameters on generator windings by:	

(1) Testing generator field windings to include: i. Testing continuity; ii. Testing resistance; and iii. Testing thermistors.	
TP 2 Test voltage regulators:	
a. Testing modes of operation to include: (1) Main; and (2) Standby.	
b. Testing no-load and full-load voltage droop	

PO 006 MAINTAIN ELECTRICITY DISTRIBUTION SYSTEM	EO 006.01 Troubleshoot Electricity Distribution Systems
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Troubleshoot power generation and distribution system to include:	
a. Troubleshooting electric switchboard control (ECS) by: (1) Examining switchboard control components to include but not limited to: i. RLC cards; ii. Error check cards; and iii. Surge protectors.	SIM – virtual environment consisting of different switchboards from different platforms. As per the switchboard simulator seen here: http://www.arisimulation.com/brochures/ARI%20Marine%20Simulation%20%282%29/index.html Alternatively, a PC-based electronic-control simulator could be utilized: http://www.pcmaritime.com/maritime-training/simulators/ SIM – virtual environment consisting of different switchboards from different platforms. Assembly of CCT breakers to be included in EO 006.02. To include high voltage (1000V+). Circuit breaker simulators to be modeled after: http://www.unitest.pl/?page_id=1720 Field trip to various classes of ship, to show differences in switchgear components and construction.
b. Inspecting synchroscope by: (1) Examining synchroscope components and operation; and (2) Utilizing synchroscopes to troubleshoot paralleling abnormalities.	
c. Inspecting electrical switchboard transducers by: (1) Examining power transducers; (2) Examining linear transducers; (3) Examining rotary transducers; and (4) Analyzing application of switchgear protection.	
d. Inspecting switchboard tangle box by: (1) Examining connections between generators and switchboards.	
e. Inspecting electrical switchboard breakers by: (1) Disassemble moulded circuit breakers; (2) Disassemble air circuit breakers to include: i. Inspecting 440 VAC distribution breakers.	
f. Examining switchgear components to include but not limited to: (1) Inspecting backplane; (2) Inspecting bus-bars; and	

(3) Inspecting electrical disconnect links.	
TP2 Troubleshoot bus transfer switches by:	
a. Dissembling components of bus transfer switches.	Training products available in consultation with L3-M&PS. To include but not limited to ACOS, MCOS, & SABT
b. Inspecting switching unit auto (SABT) by: (1) Reviewing SABT technical safety bulletin.	

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.01 Apply fluid power theory
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Define units	
a. SI units b. English gravitational system c. Metric prefixes d. Consistency in units e. Force f. Acceleration due to gravity g. Weight h. Cancellation procedures	
TP 2 List basic fluid properties	
a. Compressibility b. Mass, weight and density c. Specific weight d. Class examples e. Viscosity of fluids	
TP 3 Differentiate viscosity of fluid	
a. Dynamic and kinematics viscosity b. Viscosity index c. Viscosity measurements methods d. Rotating drum viscometer e. Capillary tube f. Saybolt universal viscometer g. SAE viscosity measurement h. Viscosity tables	
TP4 Compute pressure measurements	
a. Definition of pressure b. Pascal's law c. How pressure is created d. Hydraulic power transmission e. Absolute and gauge pressure in SI and English units	https://www.youtube.com/watch?v=wHMHxO9Ys_0 https://www.youtube.com/watch?v=bzaCFkptIII
TP 5 Relate the relationship between pressure and elevation	
a. Fluid flow rate i. Q=Av ii. W=Y(gamma) Q iii. M= (rho) Q b. Continuity equation	https://www.youtube.com/watch?v=wykn-JTnacE
TP 6 Compare flow of fluids	
TP 7 Analyze conservation of energy	
a. Forms of energy in a field b. Bernoulli's equation c. Restrictions on Bernoulli's equations d. Application of Bernoulli's equation e. Torricelli's theorem	Bernouilli applied: https://www.youtube.com/watch?v=bC8v6hlXnSk Torricelli's: https://www.youtube.com/watch?v=3qrUjQilvgE
TP 8 Apply general energy equations	

a. Nomenclature of energy losses and additions b. General energy equation	https://www.youtube.com/watch?v=JoGAuD_pcRE
TP 9 Classify pumps and motors	
a. Positive and non-positive displacement b. Construction and accessories c. Vacuum d. Gear, vane and piston pumps e. Pressure compensation and load sensing f. High/low volume circuit and closed loop circuits g. Mechanical efficiency of pumps h. Power delivered to motors i. Mechanical efficiency of motors	Pump animations and operating principles to be produced in the following styles: https://www.youtube.com/watch?v=AWwYRYr21nY https://www.youtube.com/watch?v=stjvbAO_6JQ https://www.youtube.com/watch?v=F3oWC_9R1Nc
TP 10 Explain Reynolds number, laminar flow and turbulent flow	Laminar vs. turbulent flow: https://www.youtube.com/watch?v=p08_KlTKP50 https://www.youtube.com/watch?v=WG-YCpAGgQQ https://www.youtube.com/watch?v=cOGDfdR8FjE
a. Reynolds number b. Critical Reynolds number c. Velocity profiles	Reynold's number demonstration: https://www.youtube.com/watch?v=kmjFdBxbV08 Velocity profile applications: https://www.youtube.com/watch?v=bHXGJ1fx_Qk https://www.youtube.com/watch?v=CAQ9_reH0Zg
TP 11 Introduce fluid power	
a. Basic hydraulic theory b. Basic hydraulic circuits c. ISO and ANSI symbols d. Series and parallel circuits e. Forces/pressure relationship f. Speed flow relationship g. Advantages and disadvantages of fluid power Laboratory experiments to support knowledge	Hydraulic system circuits and hydraulic manifold design: https://www.youtube.com/watch?v=mBF4r6fdWE8 https://www.youtube.com/watch?v=zZ0WwZcRrV4 https://www.youtube.com/watch?v=KgphO-u7MlQ Volume/speed/flow relationship: https://www.youtube.com/watch?v=Cz4sHr2Mhxw Example of homework/quiz that could be placed on DLN for ANSI/ISO symbol knowledge: https://www.proprofs.com/quiz-school/story.php?title=ntu3mtk2
TP 12 Describe hydraulic fluids	
a. Fluid types b. Fluid properties c. Fluid conductors i. Types ii. Advantages iii. Fittings d. Pressure and flow measurement e. Contamination detection and control f. Demonstrate Millipore sampling kit g. Operational of pressure gauge and flow meter h. Seals and leakage i. Fluid conditioners	

i. Fliters ii. Heaters iii. Coolers j. reservoirs	
TP 13 Utilize directional control valves	
a. check valves b. pilot pressure and pilot operated check valves c. shuttle valves d. DCV cneters and control methods e. Rotary and spool type DCVs f. Solenoid controlled, pilot operated DCV g. Proportional DCVs Laboratory experiments to support knowledge	Working principles and components of types of vales should be presented in cut-away live-operation, as per: https://www.youtube.com/watch?v=mvWRYXFBjYw https://www.youtube.com/watch?v=uqmVLxRPHC0 https://www.youtube.com/watch?v=XAItnsUcES0 Hydraulic schematic animation: https://www.youtube.com/watch?v=4VdU9U5YgNQ
TP 14 Understand flow control	
a. Relationship between i. Flow ii. Speed iii. Pressure drop b. Basic restrictor type of flow control valve(FCV) c. Meter in and meter out d. Pressure compensated FCV e. Deceleration valves f. Proportional flow control valve Laboratory experiments to support knowledge	
TP 15 Understand pressure control	
a. Introduction to the six main pressure control valves b. Relief valves i.Pressure override ii.Compound relief valve iii.Venting c. Unloading valve d. Sequence valve e. Counterbalance valve f. Reducing valve g. Break valve h. Proportional pressure control valves Laboratory experiments to support knowledge	See details as identified in TP13
TP 16 Describe the function of accumulators	
a. Accumulator types i. Weight ii. Spring iii. Gas b. Gas laws c. Accumulator circuits i. Leakage compensation ii. Emergency supply iii. Shock absorber Laboratory experiments to support knowledge	Accumulator animation: https://www.youtube.com/watch?v=RMDE5v3ouN8 Gas law animation: https://www.youtube.com/watch?v=5fsQ7YxsAHc
TP 17 Describe the function of actuators	
a. Linear actuators b. Rotary actuators including motors c. Actuator accessories	Actuators vs. hydraulics: https://www.youtube.com/watch?v=nDjDUcTEPBI

i. Cushioning ii. Mounts d. Intensifier e. Limited rotary actuator(torque generator) f. Pressure compensation	Hydraulic actuator working principle: https://www.youtube.com/watch?v=nv6Ym2QtOto Fluid power software simulation package: https://www.labvolt.com/solutions/3_fluid_power/98-6387-00_hydraulics_and_pneumatics_simulation_software_lvsim_hyd_pneu
TP 18 Indicate hydraulic colouring	Hydraulic colour code: https://www.youtube.com/watch?v=az-EFZ9e9UA Example of a basic hydraulic circuit and colour-code interactive exercise which could be DLN course material: https://www.wisc-online.com/learn/career-clusters/stem/hyp2506/hydraulic-color-code-circuit-example
TP 19 Interpret hydraulic circuits	
a. Hydraulic circuit design b. Troubleshoot various hydraulic circuits using drawings	Basic Hydaulic Schematic Interpretation: https://www.youtube.com/watch?v=xxoAm3X4iw0 Electrical control of hydraulic circuits: https://www.youtube.com/watch?v=JsFcfudj3rE Using flowmeters and other devices to assist troubleshooting: https://www.youtube.com/watch?v=5POaSPGI2yY
TP 20 Troubleshoot hydraulics	
a. Introduction to hydraulic troubleshooting	

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.06 Troubleshoot Pneumatics
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Describe pneumatics	
a. Differences/ advantages/ disadvantages of pneumatic systems compared to hydraulic systems b. Air compression c. Gas laws d. Compressor types e. Moisture in air f. Cooling methods g. Filters, regulators and lubricators h. Quick air exhaust circuit.	Laboratory experiment to support knowledge
TP 2 Define electro-hydraulic/pneumatic systems	
a. Ladder diagrams b. Relays, limit switches c. Fluid power circuits controlled by ladder logic/ diagrams	Fluid power and ladder logic: https://www.youtube.com/watch?v=ZOQ82Nn2OZo Interactive design can be practiced or taught utilizing: Fluid Power - Pneumatics Fundamentals, Model 6081 - Lab-Volt
TP 3 Troubleshoot electro-hydraulic systems	
a. Introduction of the correct method of troubshooting electro-hydraulic circuits using schematics and simulated faults.	
TP 4 Differentiate proportional valves	

a. Remote control of hydraulic valves b. Associated electronic devices c. Control amplifier functions d. Basic principle of proportional valves e. Proportional pressure control valves f. Proportional flow control valves g. Proportional directional control valves	
TP5 Arrange electro-hydraulic servo systems	
TP6 Troubleshoot electro- hydraulic servo systems	
a. Introduction of correct method of troubleshooting electro- hydraulic servo circuits using schematics and simulated faults	Laboratory experiments to support knowledge
TP7 Recall electro-hydraulic servo colouring	
l. Allow trainees to use the colour code they learned in the previous semester on more advanced circuits	
TP 8 Troubleshoot circuit design	Laboratory experiments to support knowledge

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.03 Troubleshoot hydraulics
Teaching Point	**Examples covering content or model styles to be developed**
TP 1 Gather resources to include	
d. Adhering to proper lock-out/tag-out procedures; e. Mustering tools required; f. Mustering appropriate PPE; g. Utilizing appropriate documentation; and h. Reviewing all Safety and Environmental concerns with regards to hydraulic systems.	Use knowledge of hydraulics, manufacturers' publications, and sound troubleshooting practices to inspect. TEL resources in the form of working animations (3-D cutaways with real-time flow) are to be used in support of TPs 1-4. e.g. pump animations, valve operation, DCV spool operation, solenoid-activation, and piggy-back valve interrelationships
TP 2 Inspect hydraulic controls to include, but not limited to	
a. Valves: (1) Recognizing flow-control markings on physical valves b. Pumps; c. Motors; d. Cylinders; e. Piping; f. Actuators; g. Accumulators; h. Filters; i. Coolers; j. Reliefs; k. Safety controls and interlocks; l. Remote controls; m. Pilot-operated piggyback valves; and n. Recognizing symptoms of hydraulic faults.	Use knowledge of hydraulics, manufacturers' publications, and sound troubleshooting practices to inspect. Instructor to describe operating characteristics, benefits and disadvantages of different hydraulic component types in different operations.
TP 3 Inspect hydraulic directional control valves and hydraulic actuating-control, to include, but not limited to	

a. Multi-position, multi-direction control valves; b. Manually-operated; c. Remotely-operated, to include, but not limited to: a. Electrically; b. Hydraulically; and c. Electro-hydraulic.	Instructor to describe operating characteristics, benefits and disadvantages of different valve types in different operations.

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.04 Repair hydraulics
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Gather resources	
a. Adhering to proper lock-out/tag-out procedures; b. Mustering tools required; c. Mustering appropriate PPE; d. Utilizing appropriate documentation; and e. Reviewing all Safety and Environmental concerns with regards to hydraulic systems.	Use knowledge of hydraulics, manufacturers' publications, and sound troubleshooting practices to repair. Instructor to introduce the concept of hydraulic feedback.
TP 2 Repair hydraulic steering controls	
a. Discussing unique characteristics of the steering controls.	
TP 3 Repair hydraulic power unit	
a. Discussing auto-centering	Instructors to emphasize the differences between the hydraulic power unit, focussing on the rudder centering unit. Field trip consists of demonstration of components of auto-centering unit, to be done in conjunction with another field trip.

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.05 Test Hydraulics
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Gather resources	
a. Adhering to proper lock-out/tag-out procedures; b. Mustering tools required; c. Mustering appropriate PPE; d. Utilizing appropriate documentation; and e. Reviewing all Safety and Environmental concerns with regards to hydraulic systems.	
TP 2 Test hydraulic steering control	

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.06 Troubleshoot Pneumatics
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Gather resources to include	
a. Adhering to proper lock-out/tag-out procedures; b. Mustering tools required; c. Mustering appropriate PPE;	Use knowledge of pneumatics, manufacturers' publications and sound troubleshooting practices.

d. Utilizing appropriate documentation; and e. Reviewing all Safety and Environmental concerns with regards to pneumatic systems.	
TP 2 Troubleshooting HVAC system	
(1) Inspecting smoke evacuation system components (AOT0630); and (2) Inspecting electro-pneumatic valves (ATT0370).	Instructor to discuss the differences with regards to working fluids (hydraulic vs. pneumatic). Requires a pneumatic trainer.
TP 3 Troubleshooting propulsion diesel engine low pressure control air, to include, but not limited to	
a. Troubleshooting gearbox pneumatic control system to include, but not limited to: (1) Clutch; and (2) Governor.	Trainees to learn about pneumatic actuators.

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.07 Repair Pneumatics
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Gather resources to include	
a. Adhering to proper lock-out/tag-out procedures; b. Mustering tools required; c. Mustering appropriate PPE; d. Utilizing appropriate documentation; and e. Reviewing all Safety and Environmental concerns with regards to pneumatic systems.	Use knowledge of pneumatics, manufactures publications, and sound troubleshooting practices to repair.
TP 2 Assembling and disassembling electro pneumatic valves.	
a. Installing HVAC/NBCD air distribution components to include, but not limited to: (1) Indications; (2) Pilot valves; (3) Actuators; and (4) Louvres. b. Adjusting smoke evacuation system components.	
TP3 Assembling and disassembling pneumatic control system components	
a. Installing electro-pneumatic valves; b. Repairing pneumatic-operated valves; c. Repairing electro-pneumatic valves; d. Repairing variable-position pneumatic-controlled valves; and e. Repairing pneumatic actuators. f. Repairing pneumatic motors.	Trainees to participate in training lab to confirm knowledge.

PO 007 SERVICE FLUID POWER SYSTEMS	EO 007.08 Test Pneumatics
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Gather resources to include	

a. Adhering to proper lock-out/tag-out procedures; b. Mustering tools required; c. Mustering appropriate PPE; d. Utilizing appropriate documentation; and e. Reviewing all Safety and Environmental concerns with regards to pneumatic systems.	Use knowledge of pneumatics, manufactures publications, and sound troubleshooting practices to test.
TP 2 Trial HVAC/NBCD air distribution systems by	
a. Testing pneumatic-control components to include, but not limited to: (1) Actuators; (2) Valves; (3) Dampers; b. Testing electro-pneumatic valves to include, but not limited to: (1) Smoke-control.	Field trip will be conducted in concert with hydraulic field trip.

PO 008 MAINTAIN CONTROL NETWORKS	**EO 008.01 Troubleshoot Control Networks**
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Investigate Communication Networks by	
a. Troubleshooting control access network buses by: (1) Identifying network types to include but not limited to: i. Stars; ii. Rings; iii. Peer to Peer vs. Client servers; iv. Mesh; v. LAN/WAN; vi. Ethernet; vii. Point to point and point to multipoint. (2) Identifying network components to include but not limited to: i. Network interface cards; ii. Hubs; iii. Switches; iv. Bridges; v. Routers; vi. Gateways; and vii. Severs.	Instructor to provide quizzes to support instructed material focus on identifying network components.
b. Troubleshooting CCTV Servers by: (1) Identifying video display components including but not limited to: i. Video display types; ii. Video creation and signal analysis; and iii. Video network interface.	Trainees to install CCTV into an existing network
c. Inspecting Network Communication Cables by: (1) Troubleshooting Cat 5e cable networks by:	Trainees to build a working Cat 5 network.

i. Identifying network cable types to include but not limited to: 1. Co-Axial; 2. Twisted pair (CAT5e/CAT 6); and (2) Troubleshooting Fibre Optics networks by: i. Identifying Fibre cable types.	

PO 008 MAINTAIN CONTROL NETWORKS	EO 008.02 Repair Control Networks
Teaching Point	**Examples covering content or model styles to be developed**
TP1 Repair Communication Networks	
a. Repairing Cat 5e cable networks, to include but not limited to: (1) Analyzing the OSI Model: i. Layer 1: Physical; ii. Layer 2: Data Link; iii. Layer 3: Network; iv. Layer 4: Transport; v. Layer 5: Session; vi. Layer 6: Presentation; and vii. Layer 7: Application. (2) Examining Ethernet including: i. Varieties of Ethernet; ii. Frame structure; and iii. Error conditions. (3) Repairing twisted pairs including: i. Cat 5; ii. Cat 5e; and iii. Cat 6. (4) Testing Cat 5e networks; and (5) Testing network communication cables.	Practical to consist of trainee building a small functional communication network. TEL trainer will be used to allow trainees to construct a virtual communication network.
b. Repairing fibre optic networks to include but not limited to: (1) Connecting/disconnecting CCTV networks and equipment by: i. Investigating TCP/IP protocols to include but not limited to: 1. Key architectural principles; 2. Abstraction layers; 3. Comparison of TCP/IP & OSI layering; 4. Addressing including: a. IPv4; and b. IPv6. 5. Error checking tools for	Fibre optic splicing to be covered under b.3.iv using a fusion splicer.

connectivity problems. (2) Testing CCTV network and equipment. (3) Analyzing fibre optic cabling to include but not limited to: i. Uses and principal of operation; ii. Mechanisms of operation; iii. Manufacturing processes; and iv. Troubleshooting and repair. (4) Analyzing fibre optic communications to include but not limited to: i. Transmitters; ii. Receivers; iii. Digital predistortion; iv. Amplifiers; v. Wavelength-division multiplexing; vi. Parameters; and vii. Comparison with electrical transmissions.	

ANNEX I – COSTING INFORMATION

Course:	***LS Electrical USQ***					
TD	No. Of Trainees/	Cost per	# Days	Initial TE	Each successive	5 Year
	Instructors	each	(if applicable)	Set-up	course	total
Not Applicable						
Material	Quantity	Cost per each	# Days (enter 1	Initial TE	Each successive	5 Year
			if not applicable)	Set-up	course	total
Classroom/Stationary Supplies						
DWAN Connected Laptops	12	$1,800.00		$21,600.00		$21,600.00
Pens/Pencils/Erasers/Paper/etc.	1	$400.00		$400.00	$400.00	$4,000.00
			Sub-Total:	########	**$400.00**	**$25,600.00**
Instructor Stationary Supplies						
DWAN Connected Laptops	2	$1,800.00		$3,600.00		$3,600.00
Photo copier	1	$1,968.00		$1,968.00		$1,968.00
Pens/Pencils/Erasers/Paper/etc.	1	$150.00		$150.00	$150.00	$750.00
Printer for 11’ x 17’ paper	1	$2,500.00		$2,500.00		$2,500.00
Paper 11x17 per pkg of 500	10	$15.00		$150.00	$30.00	$750.00
			Sub-Total:	**$8,368.00**	**$180.00**	**$9,568.00**
References						

Math Textbooks	12	$150.00		$1,800.00		$1,800.00
Physics Textbooks	12	$150.00		$1,800.00		$1,800.00
Chemistry Textbooks	12	$150.00		$1,800.00		$1,800.00
Hydraulics Textbooks	12	$150.00		$1,800.00		$1,800.00
Electric Circuit Analysis Textbooks	12	$200.00		$2,400.00		$2,400.00
Field Device/Controls Textbooks	12	$200.00		$2,400.00		$2,400.00
Digital Logic/Microprocessor Textbooks	12	$200.00		$2,400.00		$2,400.00
Miscellaneous References	12	$100.00		$1,200.00		$1,200.00
			Sub-Total:	########	**$0.00**	**$15,600.00**
Electrical Testing Equipment						
Portable Multi-meters	12	$ 400.00		$4,800.00		$4,800.00
Oscilloscope	12	$5,000.00		$60,000.00		$60,000.00
			Sub-Total:	########	**$0.00**	**$64,800.00**
Motor Controllers						
Multi-fault configurable motor controller	6					
			Sub-Total:	**$0.00**	**$0.00**	**$0.00**
AC & DC Electrical Machines						
Low power synchronous generator	12	$350.00		$4,200.00		$4,200.00
Automatic voltage regulator	12	$500.00		$6,000.00		$6,000.00
Electric governor	12	$120.00		$1,440.00		$1,440.00
AC Motor starter with variable speed control system	12	$350.00		$4,200.00		$4,200.00
DC Motor starter with variable speed control system	12	$350.00		$4,200.00		$4,200.00

Generators	12	$ 500.00		$6,000.00		$6,000.00
Transformers	12	$ 1,400.00		$16,800.00		$16,800.00
			Sub-Total:	########	**$0.00**	**$42,840.00**
Switch Gear						
Low-voltage main switchboard	3	$8,000.00		$24,000.00		$24,000.00
Switchboard meters	12	$450.00		$5,400.00		$5,400.00
Switchgear Simulator	1					
			Sub-Total:	########	**$0.00**	**$29,400.00**
Control Systems						
PID Control system or simulator						
PLC control and monitoring network with associated computer system						
Lab volt trainer						
-			Sub-Total:	**$0.00**	**$0.00**	**$0.00**
-						
Electrical control casements/equipment						
contactors	12	$137.95		$1,655.40		$1,655.40
relays	12	$31.50		$378.00		$378.00
time delay relays	12	$33.50		$402.00		$402.00
thermal relays	12	$31.50		$378.00		$378.00
over voltage relays	12	$ 25.50		$306.00		$306.00

under voltage relays	12	$ 25.50		$ 306.00		$ 306.00
switches	12					
circuit breakers	12	$61.99		$743.88		$743.88
push buttons	12	$12.00		$144.00		$144.00
control lamps	12	$6.00		$72.00		$72.00
fuses	12	$ 12.75		$153.00		$153.00
			Sub-Total:	**$4,538.28**	**$0.00**	**$4,538.28**
Field Devices						
temperature sensors	12	$154.00		$1,848.00		$1,848.00
pressure sensors	12	$175.00		$2,100.00		$2,100.00
level sensors	12	$80.00		$960.00		$960.00
reed switches	12	$20.00		$240.00		$240.00
limit switches	12	$96.00		$1,152.00		$1,152.00
overload trips	12	$46.00		$552.00		$552.00
thyristor	12			$0.00		$0.00
rudder angle indicator	12			$0.00		$0.00
tension transducer	12	$137.00		$1,644.00		$1,644.00
thermistor	12	27.5		$330.00		$330.00
explosive gas detector				$0.00		$0.00
optical sensors	12	$ 152.00		$1,824.00		$1,824.00
photo sensors	12	$ 121.00		$1,452.00		$1,452.00
output speed sensors	12	$ 140.00		$1,680.00		$1,680.00

electronic speed sensors	12	$ 120.00		$1,440.00		$1,440.00
magnetic speed sensors	12	$ 90.00		$1,080.00		$1,080.00
carbon monoxide detector	12	$ 1,050.00		$12,600.00		$12,600.00
salinity cell	12	$ 87.95		$1,055.40		$1,055.40
manual pull station solenoid	12	$ 51.00		$612.00		$612.00
shaft knot indicator	12	$ 400.00		$4,800.00		$4,800.00
vacuum sensors	12	$ 830.00		$9,960.00		$9,960.00
electric thermostatic control valve	12	$ 123.00		$1,476.00		$1,476.00
temperature control valve	12	$ 128.00		$1,536.00		$1,536.00
pyrometer	12	$ 140.00		$1,680.00		$1,680.00
			Sub-Total:	########		**$50,021.40**
Hydraulics						
Hydraulic Trainer with Manifold	6	$ -		$0.00		$0.00
			Sub-Total:	**$0.00**	**$ -**	**$0.00**

Pneumatics						
Pneumatic Trainer with Manifold	6	$ -		$0.00		$0.00
Hydraulics and Pneumatics Simulation Software (LVSIM®-HYD & PNEU)	12	$ -		$0.00		
			Sub-Total:	**$0.00**	**$0.00**	**$0.00**
Network Systems						
Network interface cards	6	$ 400.00		$2,400.00		$2,400.00
Hub	6	$ 80.00		$480.00		$480.00
Switch	6	$ 20.00		$120.00		$120.00
Ethernet Bridge	6	$ 1,000.00		$6,000.00		$6,000.00
Monitor	6	$ 96.00		$576.00		$576.00
CCTV Camera	6	$23.00		$138.00		$138.00
802.11 Router	6	$100.00		$600.00		$600.00
Gateway	6	$140.00		$840.00		$840.00
Servers	6	$400.00		$2,400.00		$2,400.00
Cat 5e Cable	1	$ 80.00		$80.00		$400.00
Professional networking tool kit	12	$120.00		$1,440.00		$1,440.00
Fusion splicing kit	2	$7,300.00		$14,600.00		$14,600.00
Fibre termination tool	6	$30.00		$180.00		$180.00
Virtual communicaton network building software (Open source program)	12	$0.00		$0.00		$0.00

			Sub-Total:	########	**$0.00**	**$29,854.00**
Contingent Consumables						
Miscellaneous	1	$ 5,000.00		$5,000.00		$25,000.00
			Sub-Total:	**$5,000.00**	**$1,160.00**	**$25,000.00**
PYs (only if course includes incremental on SWE, or course to be delivered off-site)	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year Total
Total Estimated Costs				########	########	#########

ANNEX J – COURSE HISTORY

1. The Leading Seaman Marine Technician Electrical Specialist course has its roots in the legacy QL5A Electrical Technician course. It incorporates elements of the IPMS Maintainer course, as well as Gas Free Certification course taken from the legacy Hull Technician QL5 course. It was decided to not incorporate the Unit Technical Inspector training.

Amendment table

Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below.

Reason for amendment (with date)	Details on course history
26 Aug 24	**TM - Prerequisites changed to align with the USQ-S1 Mechanical Specialist course. Additionally, a pre-req was added for PCCs to confirm with CoCs that the member has relevant experience prior to be loaded on their technician's course.** **Rationale** **1. The USQ-S1 Electrical and Mechanical specialist QSPs had differing prerequisites, but are at the same level within the MAR TECH OCC. Aligning them will create the same standard and reduce confusion.** **2. In the absence of an OJPR between the USQS2 and the USQS1 levels, adding in CoC concurrence of experience @ pre-req C will help encourage units to ensure their members are ready for their next level of training. With the MAR TECH split, OJPRs will be re-introduced at between this level of coursing.**

ANNEX K – ASSESSMENT PLAN

Mod	EC / PC	Scope Covered	Purpose	How	Pass / Fail Criteria	Impact
6	EC 001.01	EO 001.01	To assess the trainees' ability to report technical information as part of the Naval Technical Communication course Mod B. This assessment is embedded within another QSP	IAW Naval Technical Communication MOD B QSP (MITE ID 118129)	IAW Naval Technical Communication MOD B QSP (MITE ID 118129)	IAW Naval Technical Communication MOD B QSP (MITE ID 118129)
7	EC 001.02	EC 001.02	To assess the trainees' ability to use DRMIS to produce Naval maintenance documentation. This assessment is embedded with another QSP	IAW DRMIS Maintenance Supervisor MOD 2 QSP (MITE ID119097)	IAW DRMIS Maintenance Supervisor MOD 2 QSP (MITE ID119097)	IAW DRMIS Maintenance Supervisor MOD 2 QSP (MITE ID 119097)
1	CEC 002.01	EO 002.01	To assess the trainees' ability to apply a detailed knowledge of circuit analysis.	500 min Written Test consisting of 2 parts and administered by academic instructors	60% part 1 and 60% part 2	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
1	CEC 002.02	EO 002.02	To assess the trainees' ability to apply a detailed knowledge of essential Mathematic skills.	250 min Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
2	EC 002.03	EO 002.03	To assess the trainees' ability to troubleshoot motor controllers	50 min Written Test administered upon completion of EO 002.03	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

2	EC 002.04	EO 002.04	To assess the trainees' ability to maintain motor controllers	50 min Written Test administered upon completion of EO 002.04	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
2	PC 002	PO 002	To assess the trainees ability to maintain motor controller through performance.	50 min Practical Test administered upon completion of PO 002	60% IAW with PC Checklist 002	TED / Remedial Training / Re-test / TRB after second unsuccessful attempt
1	CEC 003.01	EO 003.01	To assess the trainees' ability to apply a detailed knowledge of microprocessors.	500 min Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
1	CEC 003.02	EO 003.02	To assess the trainees' ability to apply a detailed knowledge of theoretical concepts of control systems.	100 Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
4	EC 003.03	EO 003.03	To assess the trainees' ability to troubleshoot control systems	50 min Written Test administered upon completion of EO 003.03	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
4	EC 003.04	EO 003.04	To assess the trainees' ability to repair control systems	50 min Written assessment administered upon completion of EO 003.04	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.

2	EC 003.05	EO 003.05	To assess the trainees' ability to repair engine control systems	25 min Practical Test administered upon completion of EO 003.05	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
1	CEC 003.06	EO 003.06	To assess the trainees' ability to apply a detailed knowledge of theoretical concepts of digital systems.	300 min Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

2	PC 004	PO 004	To assess the trainees' ability to identify faults on field devices	50 min written assessment administered upon completion of PO 004	60%	TED / Remedial Training / Re-test / TRB after second unsuccessful attempt
1	CEC 004.01	EO 004.01	To assess the trainees' ability to apply a detailed knowledge of electronic circuit fundamentals.	Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
1	CEC 004.02	EO 004.02	To assess the trainees' ability to apply a detailed knowledge of electrical circuit fundamentals.	Written and/or Practical Test	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
2	EC 004.03-004.04	EO 004.03-004.04	ECs are conducted via in class activities with feedback provided by instructors	Accounted for in instructional timings	IAW MLP	Additional activities could be assigned at the instructor's discretion to support student achievement

2	PC 005	PO 005	To assess the trainees' ability to maintain power generation systems.	25min /trainee minute Practical assessment administered upon completion of PO 005	60% IAW with PC Checklist 005	TED / Remedial Training / Re-test / TRB after a second unsuccessful attempt
2	EC 005.01-03	EO 005.01-03	ECs are conducted via in class activities with feedback provided by instructors	Accounted for in instructional timings	IAW MLP	Additional activities could be assigned at the instructor's discretion to support student achievement

2	PC 006	PO 006	To assess the trainees' ability to maintain electrical distribution systems.	50 min Written Test administered upon completion of PO 006	60%	TED / Remedial Training / Re-test / TRB after a second unsuccessful attempt
2	EC 006.01-02	EO 006.01-02	ECs are conducted via in class activities with feedback provided by instructors	Accounted for in instructional timings	IAW MLP	Additional activities could be assigned at the instructor's discretion to support student achievement
3	PC 007	PO 007	To assess the trainees' ability to service fluid power system	50 min 2 part Practical assessment administered upon completion of PO 007	Part 1 60% Part 2 60% IAW with PC Checklist 007	TED / Remedial Training / Re-test / TRB after a second unsuccessful attempt
1	CEC 007.01	EO 007.01	To assess the trainees' ability to applying fluid power theory.	600 min Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
1	CEC 007.02	EO 007.02	To assess the trainees' ability to apply a detailed knowledge of pneumatic systems and electrically operated and controlled fluid power.	1200 Written Test administered by academic instructors	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial

						training/counselling as required.
3	EC 007.03 EC 007.04	EO 007.03 EO 007.04	ECs are conducted via in class activities with feedback provided by instructors	Fault practice on Hydraulic/Pneumatic Trainer.	IAW MLP	Additional activities could be assigned at the instructor's discretion to support student achievement
2	PC 008	PO 008	To assess the trainees' ability to maintain control networks.	50 min Practical assessment administered upon completion of PO 008	60% IAW with PC Checklist 008	TED / Remedial Training / Re-test / TRB after a second unsuccessful attempt
1	CEC 008.02	EO 008.02	To assess the trainee's ability to repair a fiber optic network	Administered by Academic Instructor	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.
2	EC 008.01 EC 008.03	EO 008.01 EC 008.03	ECs are conducted via in class activities with feedback provided by instructors.	Accounted for in instructional timings	IAW MLP	Additional activities could be assigned at the instructor's discretion to support student achievement
4	PC 009	PO 009	To assess the trainees' ability to maintain the Halon 1301 Extinguishing System. This assessment is embedded within another QSP	IAW Halon 1301 Extinguishing System Maintainer QSP (MITE ID 109149)	IAW Halon 1301 Extinguishing System Maintainer QSP (MITE ID 109149)	IAW Halon 1301 Extinguishing System Maintainer QSP (MITE ID 109149)

5	EC 010.01	EO 010.01	To assess the trainee's ability to perform Vibration analysis	Accounted for in instructional timings	IAW Vibration Monitor QSP (MITE ID 108503)	Additional activities could be assigned at the instructor's discretion to support student achievement
5	EC 010.02	EO 010.02	ECs are conducted via in class activities with feedback provided by instructors.	Accounted for in instructional timings	IAW MLP	Additional activities could be assigned at the instructor's discretion to support student achievement
5	PC 010	PO 010	To assess the trainee's ability to Test electrical systems using the vibration monitoring suite.	50 min Practical assessment administered upon completion of EC 010.01	60%	Trainee issued Trainee Experiencing Difficulty (TED) and provided additional training/instruction.
5	CEC 011	PO 011	To assess the trainee's ability to use soldering and termination techniques.	100 min practical assessment administered on completion of EO 11.01	60%	First CEC Fail results in a retest. Second CEC failure results in a TRB IAW NPTORD. Remedial training/counselling as required.

ANNEX L – TECHNOLOGY ENABLED LEARNING PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact on Trg/Assessment if TEL not implemented - Realized benefits - E/W Coast Delta
002.02	Creation of comprehensive pre-instruction package to be delivered through DLN.	Delivery of threshold knowledge exams. Review time to be based on results.	Receive 1200 minutes of review of academic material received in-house.
003.01 003.02 003.06	To be determined by the academic SME.	To be determined by the academic SME.	TBD.
004.01 004.03 004.04	3D virtual cutaway models demonstrating the function of the following field devices (to supplement physical versions): 1) Temperature sensors; 2) Pressure sensors; 3) Level sensors; 4) Reed switches; 5) Limit switches; 6) Thyristors; 7) Tension transducers; 8) Synchro-servos; 9) Thermistor; 10) Explosive gas detectors; 11) Optical sensors; 12) Photo sensors; 13) Speed sensors; 14) Carbon monoxide sensor; 15) Vacuum sensors; 16) Pyrometer;	Physical demonstration copies of the following field devices: (1) Temperature sensors; (2) Pressure sensors; (3) Level sensors; (4) Reed switches; (5) Limit switches; (6) Thyristors; (7) Tension transducers; (8) Synchro-servos; (9) Thermistor; (10) Explosive gas detectors; (11) Optical sensors; (12) Photo sensors; (13) Speed sensors; (14) Carbon monoxide sensor; (15) Vacuum sensors; (16) Pyrometer;	Trainee will be unable to perform IAW the standard, as the first day on the job will be their initial exposure to these devices.

	17) Resistive temperature devices; 18) Smoke detector; 19) Heat detectors; 20) Thermocouples; 21) Flow sensor; and 22) Leak detector.	(17) Resistive temperature devices; (18) Smoke detector; (19) Heat detectors; (20) Thermocouples; (21) Flow sensor; and (22) Leak detector.	
004.02 002.01	Creation of comprehensive pre-instruction package to be delivered through DLN.	Delivery of threshold knowledge exams. Review time to be based on results.	Receive 1200 minutes of review of academic material received in-house.
007.01 007.02	To be determined by the academic SME.	To be determined by the academic SME.	TBD.
007.03 007.04	Virtual (Amatrol) and physical hydraulics and pneumatics trainers (6 of each).	Use existing trainer at NFS.	Trainees will not be exposed to these systems prior to course completion, requiring OJT to meet the standard. Will significantly increase assessment time, resulting in extended course length.
005.01 005.02 005.03 002.03 002.04 002.05 006.01 006.02	Industrial multi fault configurable motor controller, that mimics switchboard.	Use existing Lab Volt trainer for instruction and testing.	Trainees will be required to use Lab Volt equipment, which does not properly mimic shipboard environment, which leads to techs unable to perform the standard expected of a LS tech upon posting to the fleet.
003.03 003.04 003.05	Basic steering system model.	Existing PLC trainer and existing Lab Volt trainer	Trainees will be required to use Lab Volt equipment, which does not properly mimic shipboard

	Models of rectifier firing board, voltage phase monitoring cards, EMI filters, protection, and low voltage power supplies. Universal engine starting/control system available on computer. Virtual PLCs and micro-controllers. OBTS (IPMS trainer).			environment, which leads to techs unable to perform the standard expected of a LS tech upon posting to the fleet.
008.01 008.02	Six of the following: . Network interface cards; . Hubs; . Switches; . Bridges; . Monitors; . CCTV Cameras; . Routers; . Gateways; and . Severs Cat 5e cabling Professional networking tool kit for Cat 5e cable Fusion splicing equipment Fibre cable Fibre termination tools	West coast currently does not have the capability to instruct this material at NFS(P).	East coast capabilities currently unknown	Current status of NFS does not support training of this content. Trainees will be unable to support the fleet with regards to properly maintaining control networks.

	Virtual communication network building software.			
001.01	To be determined by the academic SME.	To be determined by the academic SME.		TBD
001.02	As per MITE code 119097			

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